PakEmis v1.0: an evidence-graded public-data reconstruction of Pakistan's non-LULUCF greenhouse-gas emissions, FY2015–16 to FY2023–24

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Abstract. National greenhouse-gas inventories are usually documents, their consistency editorial. We present Pakistan's inventory built as software: ten sector modules satisfy a common contract, and its cross-cutting properties are executable assertions on every commit.

The inventory reports $444.96\,\mathrm{Mt}$ CO_2e (AR5, $438.19\,\mathrm{Mt}$ on AR6) over twenty-four IPCC categories, **excluding LULUCF**, for one common reporting period: 1 July 2023 to 30 June 2024—the total of the *estimated PakEmis scope*, not a complete national account: comparator-bounded absences (F-gases, the cement kiln-fuel excess) are priced beside it. Every figure is conditional on a named structural configuration, `REF_2023_24`, whose twenty-two registered axes are published as a scenario registry; dung fuel follows the household survey, its diverted remainder unresolved; the destination assumption L gives $450.20\,\mathrm{Mt}$.

Each annual figure resolves into twelve months closing to it exactly; $54.65\,\%$ of the national total sits on a flat profile—a declared assumption, not a finding. 199 of 202 streams are gridded at five hundredths of a degree; the 3 cruise-aviation streams, $0.05\,\%$, stay unallocated, and closure is raster plus unallocated.

A nine-year, ten-sector reconstruction ships complete.

Uncertainty is carried at parameter level: the national interval is $\pm 11.72\,\%$, honestly asymmetric at $-10.3\,\%/+13.1\,\%$ about the median, the waste and fugitives engines' bounded draws undiluted. Named structural alternatives still open move the total by up to $9.07\,\mathrm{Mt}$—outside that interval by construction—and are priced by a bridge. The public-power coal basis led that list until this release and is now decided on evidence; the control it replaced ships as a legacy configuration worth $17.78\,\mathrm{Mt}$. Completeness is reported with notation keys over 138 category–gas combinations, 54 estimated and 71 explicitly not, with EDGAR coverage as three numbers rather than one ratio. The engine, activity data and every output quoted here are in one repository under release `1.0.0rc67`. The dataset DOI is assigned on deposit; until then the release is identified by version and manifest digest.

1 Introduction

A national greenhouse-gas inventory is a large arithmetic object whose correctness depends almost entirely on properties that span its parts. Fuel supplied to a country must be allocated among the sectors that burn it without any of it being counted twice

or dropped. Nitrogen removed from one balance must arrive in another. A category reported under one IPCC code must not also appear under a second. Each of these is a statement about the relationship between two sectors, and in the conventional production model each is enforced by a compiler reading two spreadsheets side by side and satisfying themselves that the numbers agree.

That model has a failure mode that is difficult to detect from the outside and easy to introduce from the inside: a change to one sector silently invalidates a property that was true when a different sector was written. Nothing breaks. The total still sums. The inconsistency is discoverable only by repeating the original reading, which nobody does, because the reading was tedious the first time and there is no signal that it needs repeating.

1.1 The argument of this paper

We built Pakistan's inventory so that those cross-cutting properties are executable. Each sector is a Python module satisfying a common contract; the contract is enforced by 32 tests that run against every registered sector; the properties that span sectors are assertions with == in the sector code and in the test suite, not sentences in a methods document. When a change to the agriculture sector's manure-management shares alters how much dung is diverted from the manure balance as fuel, the residential sector's dung ledger changes with it, because it is derived from the same herd through the same volatile-solids rates rather than estimated separately. The two cannot drift apart, because there is only one number.

This is a software-engineering claim about inventory compilation, and it is testable in the ordinary way: the repository is public, the test suite runs in continuous integration, and every figure in this paper is regenerated from the engine's own output files by a build step that fails if a figure is stale.

1.2 Why Pakistan, and why now

Pakistan is the fifth most populous country in the world and among the most exposed to climate impacts, yet its inventory infrastructure is thin. Four official filings are in circulation and they are not interchangeable, so this paper distinguishes them throughout. The First Biennial Update Report (Ministry of Climate Change, Government of Pakistan, 2022) carries the most recent full narrative inventory with a published sectoral breakdown, for FY2017–18, and is the comparator for categories where only a narrative figure exists. The First Biennial Transparency Report (Government of Pakistan, 2024) supersedes it as the current submission and carries Pakistan's first Common Reporting Tables (Government of Pakistan, 2026), for inventory year 2022; those tables are machine-readable at cell level and are the comparator used wherever a category cell exists, because a cell can be audited and a narrative total cannot. The Third National Communication (Government of Pakistan, 2025) and the Initial National Communication complete the set; the latter is used once, as a labelled magnitude check in the railway benchmark register (Section 6), and nowhere else. Where this paper says a category is compared against Pakistan's own reporting, it names which of these it means, and Section 6 prefers the CRT cells over the BUR-1 narrative wherever both exist.

That distinction is not bookkeeping. The CRT's own notation keys record that residential biomass—dung cake and fire-wood, the country's two largest household fuels—is NE in the national filing, so a comparison against its residential total is a comparison against the metered-fossil sliver of the sector rather than against the sector.

International datasets fill the remaining gap—EDGAR (Crippa et al., 2025), FAOSTAT (FAO, 2025), Climate TRACE (Climate TRACE, 2026)—but each is a global product applying globally uniform assumptions to nationally reported activity, and their Pakistani sector estimates disagree with each other by amounts that are large relative to the mitigation decisions they inform.

There is also a specific structural reason a national inventory for Pakistan is worth building as a modular system rather than a document. Its emissions are unusually concentrated in agriculture—enteric fermentation alone is $34.1\,\%$ of the national total, larger than every energy category combined except road transport and electricity—which means an inventory dominated by one sector must nevertheless get the other nine right to be useful for anything except a headline. Sectors that are small in CO_2e can be large in policy: residential biomass is $5.4\,\%$ of national emissions and the largest estimated source of gross primary $PM_{2.5}$ emissions among the sectors quantified here. Whether it is also the dominant source of *exposure* is a separate question this inventory cannot answer: exposure depends on ventilation, proximity and time-activity patterns that are outside its scope.

1.3 Contributions

1. **A ten-module national inventory covering the principal currently reconstructable anthropogenic sources** for Pakistan at $444.96\,\mathrm{Mt}$ CO_2e under reference scenario `REF_2023_24`—$87\,\%$ of the corresponding EDGAR aggregate magnitude, with every mapped IPCC category estimated over $11\,\%$ of that scope and the unestimated categories named over the rest—in which every input carries an explicit provenance and evidence-status record, Pakistani official sources are used where available and proxy, survey-derived or assumed values are identified separately, every sector is allocated to provinces, and IPCC notation keys are reported over all 138 category–gas combinations considered.

2. **A sector-as-module architecture** in which 32 contract invariants are enforced on every sector, so that adding a sector cannot silently weaken the inventory (Section 2).

3. **Four enforced accounting closures** asserted by construction rather than described, each of which corrects a double-count or an orphaned flow that the prose version of the same inventory had already documented as a known defect. *Two of the four cross a module*—dung nitrogen from agriculture into residential, and generator petrol carved out of transport's control by residential—and two are between categories inside one module, which is why they are called accounting closures here rather than cross-sector ones (Section 4.8).

4. **A methodological finding about the IPCC default tables**: four instances in which the plausible wrong table leaves CO_2 exactly correct and is therefore invisible to every external comparator, moving the affected category by $8\,\%$ where the category is small enough for a trace gas to matter and by less than $0.5\,\%$ where it is CO_2-dominated (Section S3).

5. **Uncertainty carried at the level of parameters rather than of emission rows**, so that one row may depend on an activity, a conversion and a factor at once, each shared with a different set of other rows (Section 4.9). *This overturns a finding of the previous edition.* Under the earlier model, in which a row carried a single correlation group, sharing the fuel-carbon factors widened the road-transport category by more than half; at parameter level it does not move it at all, because the carbon content of diesel carries about $2\,\%$ and the road rows carry thirty to sixty, and correlating a small part of a large variance

cannot widen the sum. What the parameter graph now finds is that $2.5\,\%$ of the national variance has no named parameter, after agriculture's IPCC factor uncertainties (composed inside each row's declared width, invisible to the registry) were given names and the remaining sectors' factor, conversion and activity groups were scoped to their source streams the same way, with no central number changed in either step. Where the variance sits is read from the released ensemble: enteric fermentation carries $59.8\,\%$ of it. Parameter-level importance is not claimed.

6. **A national key-category analysis ranked by variance as well as by level**, identifying where measurement effort reduces national uncertainty rather than where the emissions are (Section 6.7).

The agriculture sector's methods, its geospatial allocation and its air-pollutant layer are documented separately (Ahmad, 2026a), as is the nowcasting system that extends it beyond the census year (Ahmad, 2026b). This paper treats agriculture as one of ten sectors and does not repeat that material. Eight of the nine non-agricultural sectors also have a full technical report—data provenance, methods, factors, validation, uncertainty, air pollutants and limitations—maintained in the repository, compiled on every release and held to the same generated-from-the-engine discipline as this manuscript; industrial fuel combustion was added after those reports were written and is documented in Section 2.5 and the completeness table rather than in a standalone report. Those reports are versioned documentation of the release rather than companion submissions: this is the one article, and it carries what a reader needs in order to judge the dataset, while they carry what a reader needs in order to audit and maintain it.

1.4 Related Work

1.4.1 Global gridded inventories

EDGAR (Crippa et al., 2025, 2020) is the most complete source of Pakistani sector estimates and the principal external comparator used here. It applies a consistent global methodology to nationally reported activity data, which is its strength for cross-country comparison and its limitation for any single country: where a national statistic exists that EDGAR's global pipeline does not ingest—a railway operator's metered fuel purchase, a regulator's fuel-consumption table, a provincial production series—EDGAR necessarily models what a national compiler can read.

FAOSTAT (FAO, 2025; Tubiello et al., 2013) covers agriculture and land use with a Tier 1 method applied to FAO's own activity database. Climate TRACE (Climate TRACE, 2026) takes a different approach again, inferring emissions from remote sensing and asset-level data. All three disagree with each other for Pakistan by margins that are material: their agriculture totals span $233\,\mathrm{Mt}$ to $261\,\mathrm{Mt}$.

The disagreement is not a defect of any of them. It reflects that a global product must choose defaults where a national compiler can consult a source, and this inventory's value proposition rests on that difference: it is built from Pakistani statistics—the Pakistan Statistical Year Book, NEPRA's State of Industry Report, OGRA's petroleum tables, Pakistan Railways Year Books, provincial statistical bureaux—rather than from a global model's Pakistani slice.

1.4.2 National inventory compilation as software

Most national inventories are compiled in spreadsheets, often in the IPCC Inventory Software or a national derivative. The literature on this is thin, which is itself notable: the software engineering of inventory compilation is largely undiscussed in the emissions literature, while the substantive question—how do you know a large multi-source arithmetic artefact is internally consistent?—is well studied in software engineering under the heading of invariant checking.

Reproducibility infrastructure has reached adjacent fields. Emissions research increasingly publishes code alongside results, and gridded products ship their processing chains. What is less common is treating the *cross-sector properties* as testable artefacts: the double-counting checks, the mass balances between sectors, the notation-key bookkeeping. Those are usually documented in a methods report and verified by review. This paper's contribution is to argue, with a worked ten-sector instance, that they are better expressed as executable assertions.

1.4.3 Pakistani inventories

Pakistan's First Biennial Transparency Report (Government of Pakistan, 2024) is the country's current submission and publishes a full IPCC-category inventory for FY2020–21; the First Biennial Update Report (Ministry of Climate Change, Government of Pakistan, 2022) precedes it and reports for FY2017–18. The Global Change Impact Studies Centre has published national inventory work (Global Change Impact Studies Centre, 2019), and sectoral studies exist for brick kilns (Hamdani et al., 2025), mangroves (Hussain and Rahman, 2022) and crop residue burning (Mir et al., 2021). None is simultaneously current, reproducible and accompanied by per-sector uncertainty, which is the gap this work addresses. Completeness is not the axis on which this inventory can claim an advantage, and it would be unfair to grade others on it while exempting ourselves: of the 138 detailed categories tracked here, 71 remain NE and 8 IE, fluorinated gases and the non-gas captive fleet among them, so PakEmis is a more reproducible inventory rather than a more complete one. Measured against the current national submission for a matched fiscal year (Section 6.5) the shortfall is $6.04\,\mathrm{Mt}$ across seven categories, and that is nearly the whole of the national difference between the two accounts. Section 2.5 reports its own notation keys on the same terms.

Where this inventory disagrees with BUR-1 it says so and argues the case; the railways disagreement in Section 6 is by a factor of ten and is resolved on the operator's own published fuel account.

1.4.4 Uncertainty treatment

IPCC's Volume 1 Chapter 3 defines Approach 1 error propagation and Approach 2 Monte-Carlo simulation, and the 2019 Refinement (IPCC, 2019) restates both. Approach 2 permits correlated inputs; in practice, published national inventories rarely state whether shared emission factors were correlated across sectors, and the Approach 1 formula makes independence the default reading. Section 4.9 quantifies what that undeclared assumption is worth for a country with six fuel-burning categories drawing on the same IPCC defaults.

2 Dataset scope and architecture

This section states what the dataset is and is not. PakEmis is a public-data reconstruction of Pakistan's anthropogenic greenhouse-gas emissions outside land use, land-use change and forestry, reported for fiscal years running from July to June and carried on one reference configuration whose alternatives are registered rather than implied. The architecture that produces it is described first, because the properties a reader can rely on—mass conservation from category to gas to total, one reporting period across every sector, and an explicit notation key wherever a category is not estimated—are properties of that architecture and are asserted by it on every build. The completeness statement that closes the section is the boundary of the estimated scope: what is estimated, what is not, and what a comparator implies the absent part is worth.

The inventory is a Python package in which each of the ten sectors is a module registered in a single list. Registration is not a formality: it is the point at which a sector becomes subject to 32 invariants that are checked against every registered sector on every commit. A sector that cannot satisfy them does not enter the inventory.

2.1 What a sector must provide

The contract is deliberately small. A sector module declares a name, a reference year and the set of IPCC categories it emits, and exposes two functions: one that loads its activity data and emission factors, and one that turns that data into a list of component rows. A component row is the atomic unit of the whole inventory—one gas, from one sub-item, within one category—carrying the emission in gigagrammes, its CO_2e value on both AR5 and AR6 metrics, a relative uncertainty, and its evidence, temporal and spatial classes. The uncertainty is resolved to *parameters* rather than carried on the row: a separate registry (`data/parameter_scope.csv`) names the physical quantities behind each row and which other rows share them, so one component may depend on an activity, a conversion and a factor with three different dependency structures. That resolution is used by the Approach-2 simulation; the Approach-1 quadrature remains row-level, which is one reason the two are worth running against each other. Everything downstream in the system—the national rollup, the uncertainty analysis, the key-category screen, the completeness report, the provincial allocation, the figures—consumes component rows and nothing else. No downstream module knows what a buffalo or a locomotive is.

The cross-sector GWP contract is explicit. On the AR5 basis used for headline totals, $CO_2 = 1$, biogenic $CH_4 = 28$, fossil $CH_4 = 30$ and $N_2O = 265$. Each component row carries its gas origin, so agriculture and waste methane use the biogenic value while fossil-fuel sectors use the fossil value; the national aggregator never infers origin from the sector name. The corresponding AR6 values are stored alongside AR5 rather than substituted silently.

The consequence is that adding a sector requires no change to any analysis. Fugitive emissions and cement were added late in the project's life; neither required a line of change in the uncertainty, key-category or benchmark code, and both appeared in the national outputs on first registration.

2.2 The invariants

Table 1 groups the 32 contract tests by what they protect. They are not unit tests of any sector's arithmetic—each sector has those separately—but properties that must hold of every sector, present and future.

Several deserve comment because they encode a mistake that was actually made or narrowly avoided. *Category names are globally unique* exists because two sectors independently wanted the category name "Other"; had both been allowed, the national rollup would have summed them into one row and the key-category screen would have ranked a chimera. *Correlation groups stay within one category* exists because a correlation group that spans categories makes the Approach-1 quadrature in the key-category analysis ill-defined: the analysis apportions variance to categories, and a group straddling two of them has no category to be apportioned to. *CO_2e is exactly the GWP conversion* pins the relationship between the gigagramme figure and the two CO_2e figures to an exact multiplication, so that a sector cannot report a CO_2e value that its own gas mass does not support—the single most common transcription error in spreadsheet inventories. *Intermediates are not cached* asserts that two consecutive builds of a sector return equal but distinct objects, which is what makes the sensitivity analyses in several sectors safe: they rebuild a sector under a changed assumption, and a cached intermediate would leak the changed assumption into the baseline.

What it protects	Invariants
Shape and declaration	Declared attributes present; row shape complete; every declared category is actually emitted; category–sub-item pairs unique within a sector
Global namespace	Sector names unique; category names globally unique across all sectors
Metric integrity	Every gas known to the GWP table; CO_2e exactly equals mass $\times$ GWP on both AR5 and AR6; all methane correctly flagged fossil or biogenic
Sinks and signs	Signs match the declared sink flag; sinks keep their sign through aggregation; an unflagged sink is dropped rather than silently summed; the sign helper is a no-op on positive values
Uncertainty structure	Correlation groups do not span categories
Spatial allocation	An allocation proxy resolves for every emitted category; an unknown category raises rather than falling back to a default proxy
Provenance metadata	Evidence, temporal and spatial classes are drawn from the shared vocabulary where present; a sector adopts them for all of its rows or none, so a partially classified sector cannot be read as a fully classified one
Reference year	Every sector either returns a value for a requested year or raises; no sector may return a plausible wrong number for a year its sources do not cover, and a sector whose magnitude is pinned to one year is declared to be so
Reproducibility	Building a sector twice is deterministic; intermediates are not cached between builds
Cross-sector flow	Managed soils receives manure nitrogen (the agriculture-internal case of the closures in Section 4.8)
Declared accessors	Every optional accessor a sector exports is callable; the three gas-consuming sectors export their live gas throughput by one name; power's plant-side closure channel stays off the emissions path and behind the registry's diagnostics hook; no report module imports sector code at import time

Table 1. The 32 contract invariants enforced against every registered sector. A sector is admitted to the national inventory only if all of them hold. Grouped by the failure each is designed to catch; several encode an error that was made in an earlier version of this inventory and is now unrepeatable.

2.3 Why a registry rather than a spreadsheet

The proximate argument is the one above: a registry gives cross-cutting checks somewhere to live. The deeper argument concerns what happens when the inventory is revised.

A spreadsheet inventory is revised by editing cells. The provenance of a cell—which source, which year, which conversion—lives in the compiler's memory or in a note beside it, and the previous value is gone. In this system every activity figure is a row in a version-controlled CSV with a source column, every emission factor cites its IPCC table, and every revision is a commit with a message. When the residential sector's dung-cake stream changed because agriculture's manure-management shares changed, the resulting $0.4\,\mathrm{Mt}$ movement in the national total is attributable to a specific commit rather than being a difference between two spreadsheet files.

That is also what makes the *sensitivity* results in this paper cheap to produce. A sensitivity analysis in a spreadsheet means copying the workbook. Here it means calling the sector's build function with one parameter changed, which is why the residential sector can report its total with firewood excluded, at $-35\,\%$ and at $+35\,\%$ on every run, and why the waste sector can publish both readings of the disputed sewer methane correction factor rather than choosing one (Section 6).

2.4 Verification and reproducibility

The engine ships 3882 defined test functions. Beyond the 32 contract invariants and the per-sector arithmetic tests, three categories are worth naming because they guard the manuscript rather than the code. *Provenance tests* assert that every emission factor used anywhere in the inventory cites a document in the bibliography, and that any source declared informal is explicitly listed as such. *Number tests* parse this paper's LaTeX source and check each quoted figure against the generated CSV it came from, so a number cannot drift between the engine and the text. *Figure-currency tests* assert that every image in a manuscript directory is byte-identical to the file the engine most recently produced.

Continuous integration runs the full suite, regenerates every output a build phase writes, rebuilds every figure and compiles all manuscripts with `tectonic` on every push. A manuscript that does not compile, or that quotes a number the engine does not produce, is a build failure.

That qualification is load-bearing and this release states it rather than leaving it implied. 96 of the shipped products are *held*: no build phase writes them, because their producers need network access, a GIS extra, an archive too large to commit, or simply hours. Each is pinned by the digest it carried at release, so a reader can prove the file they have is the file that was published — but that digest was taken from the file itself, so it cannot detect a producer whose output has since moved. Only 12 of the 96 are re-derived and compared against the file on disk by a mode that writes nothing; for 3 the only recorded instruction is the producer itself, which would rewrite the product rather than check it, and the remaining 81 record no verification at all. Those counts are a declared property of each row, not an inference from its command, and the twelve are re-derived on every test run — each announcing which products it compared, so a product added to a producer cannot inherit a verification that does not read it. The first run found a held provenance record whose generator disagreed with it, and the file, not the generator, was the

correct one: the stale-file warning had been inviting a regeneration that would have replaced a true provenance claim with a false one.

2.5 Completeness: What Is Estimated and What Is Not

IPCC requires that an inventory account for every source category it might contain, using notation keys where a number is absent: NE for not estimated, NO for not occurring, IE for included elsewhere, NA for not applicable. The requirement is easy to satisfy formally and rarely satisfied usefully, because a table of NE entries with no magnitudes tells a reader that something is missing without telling them whether it matters.

We report notation keys over 138 category–gas combinations spanning 72 IPCC codes, and where a source exists that can bound a gap, the completeness table carries the magnitude. Table 2 summarises; the full table is generated by the engine and versioned with it.

Key	Count	Meaning in this inventory
E	54	Estimated and reported
NE	71	Not estimated. Each carries a justification, and those a comparator can bound carry a magnitude
IE	8	Included elsewhere: the cross-sector transfers of Section 4.8, household electricity counted at the power station, biogenic CO_2 held as a memo, and the transport boundary's declared inclusions—stationary agricultural plant, captive diesel and construction machinery reported inside the road and off-road controls at those categories' factors. (Railway non-traction fuel left this key: it is an estimated row at its own 1A2/1A4c-unresolved category rather than a declaration inside 1A3c)
NO	4	Not occurring: managed savanna burning, other agricultural sources, direct combustion on urban electric rail, and electric road vehicles
NA	1	Not applicable: soda ash production (2.B.7), where the category exists in the reporting hierarchy and the process does not arise in Pakistan—distinct from NO, which records an activity that does not occur, and from NE, which records one that does and is unquantified

Table 2. Completeness across 138 category–gas combinations and 72 IPCC codes. The generated table carries a justification for every non-E entry.

2.5.1 The declared holes, in order of size

Manufacturing industries and construction (1A2), roughly $15.85\,\mathrm{Mt}$ remaining — halved in this release by the solid-fuel activation. EDGAR puts Pakistani industrial fuel combustion near $44\,\mathrm{Mt}$. This inventory reports $5.55\,\mathrm{Mt}$ of captive self-generation (screened by configuration and fuel class, with a bracket and a status per row, in `outputs/industry_captive_screening_2024.csv`) and $5.61\,\mathrm{Mt}$ of industrial process gas, both from OGRA's sector-wise gas table, $3.40\,\mathrm{Mt}$ of industrial liquid fuels from OCAC's Total Industry customer class, and—new in this release, through the category's written activation gate—$2.78\,\mathrm{Mt}$ of brick-kiln and $10.80\,\mathrm{Mt}$ of cement-sector coal from the HDIP coal-by-consumer series

republished in the Finance Division's Statistical Supplement. About 15.85 Mt is outstanding against 29 in the previous edition, 33 before that and 38 before that.

Both of the recent additions closed an error rather than a data gap, and they closed the same error twice. This inventory first recorded that Pakistan publishes no sector-disaggregated industrial fuel consumption. It does: OGRA reports gas by sector, in the table that was already the source of this inventory's residential control total, its LPG total, its kerosene total and the captive series. The register was then rewritten to say the remainder was specifically non-gas and required a paid source. That was also wrong: OCAC's Total Industry customer class—furnace oil, diesel, LDO, kerosene and gasoline—had been in the repository since the transport sector first ingested the same annual panel for its road control total, and no module read those rows. What was missing was not the statistic but the reading of it, twice, in the same category—which is the general finding of Section 7.2's account of the gap register, and the reason the register now distinguishes a datum that was never acquired from one that was acquired and never extracted.

The activation is the graduation the previous edition named. Its screenings had read the old 29 Mt as largely interpreted—kiln fuels, mostly—and said what would graduate them: the HDIP industrial coal breakdown. That series arrived (the Finance Division's Statistical Supplement republishes it, Table 14.1, with categories that exhaust supply), and the two large claimants passed through the category's written gate as reference streams priced on each year's own coal. Two findings came with them. The measured cement-sector coal (10.80 Mt on 4391 kt) sits *below* the superseded transfer model's 11.99 Mt to 15.65 Mt bracket—either the transferred heat intensity overstates Pakistani kilns or part of the kiln heat is not coal; the model is demoted to a comparator with that finding attached, not reconciled away. And the brick column is a residual-style bucket in HDIP's own accounting—supply exhausted means whatever industrial coal is not called power, cement or household lands in it—which is priced as the stream's 30 % activity band rather than footnoted.

The codes in this table are IPCC 2006 throughout, from Volume 1 Chapter 8 Table 8.2. An earlier release coded sector 3 to the UNFCCC common reporting tables and sector 4 to the 2006 Guidelines under one column, which made a join on the code string unsafe; the scheme and its crosswalk to the reporting tables, including the two categories that have no counterpart in either direction, are published as `outputs/pakemis_category_crosswalk.csv`.

That characterisation is now measured rather than asserted, and the measurement says the band is too narrow. `outputs/pakemis_coal_allocation_identity.csv` puts the national coal supply beside the consumer columns summed: over the 14 years both tables cover they close to within 1 kt in 13, far closer than two independent statistics would agree, which is the signature of a constructed identity rather than a balance. Inside it the split is a redistribution, and over FY2019–20 to FY2023–24 the constraint binds: the total moves 438 kt while the power column gains 8299 kt and brick kilns plus cement lose 8739 kt. Every tonne the Thar ramp added to the power column came out of another column, so the brick series' 86.2 % fall over those years is not independent evidence about brick activity. Over the whole held span the same total grew by 17 001 kt, so the redistribution reading belongs to the recent window, and the product publishes both.

The consequence for this inventory is that the priced band does not reach the only independent bound the repository holds. At 1129 kt the booked column spans 790 kt to 1468 kt under its 30 % band, while the brick fleet—counted once in late 2020 at 10 394 kilns on one reading and 9486 on the other—bounds its coal at 3301 kt and 3095 kt even on the shortest firing season

either source declares. The band's upper end is a factor of 2.25 and 2.11 short of those. Read in kilns at the survey's own 328.2 t per kiln-year, the booked column supports 3440 firing kiln-years at the reference year against 24 936 in FY2019–20—from more than twice the counted fleet to about a third of it, for a fleet counted once. `outputs/pakemis_coal_sector_reallocation.csv` gives that shortfall a candidate source, from a direction with no connection to kiln counting. Because HDIP's consumption is its supply, replacing the power column with the stations' own metered coal changes the industrial residual by the same tonnage the other way—at the reference year by 3321 kt, worth 8.14 Mt CO_2e against the power lane's −18.03 Mt. Booked, the brick column sits below the fleet's own floor in 3 of the 9 years; with that residual it is inside the bound in all 9. The claim this supports is modest, because six years were already inside and the bound is wider than the quantity in dispute: the reallocation never worsens the brick position and repairs it where it was broken. Which non-power consumer took the coal is not resolved.

The fleet estimate is still not booked and should not be: it has no time axis, which is precisely why it was published beside the column rather than in place of it. What changes is that the brick stream's uncertainty is now known to be understated, with the factor stated.

What remains outstanding is narrower again: petroleum coke (never in the coal table; still the Comtrade trade-screen lane), captive coal (a 0.28 Mt to 1.31 Mt configuration, deliberately *not* activated because it sits inside the brick residual and would double count), and process bagasse, still bounded rather than estimated: the entire cane crop's bagasse reaches only 0.39 Mt of countable CH_4 and N_2O (*bagasse_ceiling_in_scope*). Given that every narrowing so far came from reading the repository rather than from acquiring anything, the remaining 15.85 Mt is better read as an upper bound on what is absent than as a measurement of it.

One row of that same table is deliberately *not* taken, and the reason is a method, not a materiality judgement. Fertiliser draws 756 MMCFD, more than twice general industry and second only to power. The IPCC ammonia method allocates a plant's *total* gas requirement—feedstock and process fuel together—to the ammonia category, 2B1, and removes that gas from Energy so the same molecules are never counted twice; only separately metered auxiliary combustion outside the ammonia process boundary stays in 1A2. NFDC's monthly plant annex puts the split at 83.5 % feedstock and 16.5 % process fuel in 2023–24, the reference year itself, so a sixth of the fertiliser row is burned—and the burned sixth belongs to 2B1 anyway, because the method draws the boundary around the process rather than around the flame. The same annex puts the sector at 751.9 MMCFD against OGRA's 756, two independent Pakistani records of one fiscal year 0.5 % apart: an industry body's plant meters against a regulator's utility returns. An earlier edition of this section said almost none of the row was combustion, which was wrong twice over: a sixth of it is, and that was never why the row is excluded. The exclusion stands on the method: booking the row as industrial fuel would inflate the category three and a half fold and double-count against the ammonia accounting.

That the estimated part is *measured* rather than modelled is worth noting against the usual pattern for a gap this size: OGRA publishes the gas annually, OCAC publishes the liquid products monthly and annually, and a 1,105-plant register assembled from the two gas utilities' captive fleets and the NEPRA licence list reconciles against the national figure to within 0.01 MMCFD. The dominant uncertainty on the gas half is not the activity but the heating value, which nothing read here pins: 900 against 1000 Btu per standard cubic foot moves the gas rows from 5.05 Mt to 5.61 Mt and the industrial category as a

whole from 22.02 Mt to 22.58 Mt, and both are reported. The liquid rows do not move with it: they are metered in tonnes and converted by calorific value, so the difference between the two figures is a direct measure of how much of this category still rests on an unpinned gas assumption.

Fluorinated gases (2F), 11.54 Mt on EDGAR's estimate. Refrigeration and air-conditioning substitutes for ozone-depleting substances. If EDGAR is approximately right, Pakistani F-gases are larger than the waste sector's solid-waste disposal category and a third the size of waste overall. They are absent from the totals—but the absence is a *machine-declared state*, not a sentence: an HFC equipment-bank model is implemented (`pkinv.sectors.ippu.fgases`) and gated off because only an aggregate consumption control is held, with no bank-resolving import series. The same is true across the thirty-five non-cement IPPU category rows: every one carries an NE state in the register, and the chemical categories additionally publish `outputs/ippu_chemical_activation_status_2024.csv` (which estimator exists, what activity it holds, and the named evidence that would activate it—ammonia's carbon-ledger estimator waits on complete plant feedstock and CO_2-recovery data, nitric acid's abatement estimator on national acid output) and `outputs/ippu_chemical_tier_gate_2024.csv` (why each tier is not ready). The completeness ask is answered by those shipped, tested states; what none of them can supply is the species- and application-resolved activity behind them. The customs half is now processed (rc30): UN Comtrade's HS-2022 HFC lines give net substance imports through customs as a fourth typed screening lane, corroborating the UNEP supply aggregate without resolving the mixture split—which is exactly what the National Ozone Unit request remains for.

Captive and industrial self-generation (1A2 partial). IPCC places self-generation in 1A2 on Table 2.3 factors. NEPRA's regulatory scope excludes it, which is most of the reason Ember's Pakistani electricity total exceeds NEPRA's. It is inside the 44 Mt 1A2 hole rather than additional to it.

Fugitive activity detail (1B2, partial). Transmission, distribution, processing, refining and post-meter sources are now included under the 2019 Refinement. The remaining gaps are exploration, abandoned wells, episodic releases and the facility characteristics needed to replace higher/default practice factors: pipeline material and pressure, LDAR, compressor seals, plant throughput and well status.

Industrial wastewater (4D2). Needs industrial output by sector and organic loading per unit of output, neither available nationally. Domestic wastewater is reported; industrial is not, and given that domestic alone is 5.5 % of the inventory, this gap is not small.

Water-borne navigation (1A3d), domestic and international. The domestic activity figure now exists in this repository—OCAC's sectoral table carries a Domestic Bunkers row (6.9 kt of diesel in FY2021–22, read by the transport fuel ledger)—so what remains is a small reconciliation, not an absent statistic. The gap is still worth describing precisely because it contains a trap. The national petroleum statistics carry an "Overseas/Export" furnace-oil line of 1.01 Mt that reads like bunker fuel and is not: it is exported cargo, matched independently by commercial trade data showing 1.11 Mt of furnace-oil exports in the same period. Karachi's bunkering operation began only in 2026. Booking that line as bunkers would create a memo item of roughly 3 Mt CO_2 out of a commodity export.

Iron and steel (2C1), lime (2A2), ammonia (2B1), soda ash (2B7). Four IPPU categories where the emission factor is uncontroversial and the activity data are the obstacle. The repository holds production figures near two of them—9100 kt of *total fertiliser product*, a sum over urea, DAP, CAN and the rest from the Punjab Pocket Book, and 785 kt of soda ash from the PBS quantum index, which supersedes the 651 kt an earlier edition of this section still carried—and it is worth being precise about why neither is the process activity the method needs.

The held figures are not the process activity the method needs. No fertiliser tonnage is ammonia: the IPCC method needs ammonia output itself, the feedstock carbon entering the process, and the CO_2 recovered downstream for urea synthesis, and a total-product tonnage supplies none of the three. Nitric acid needs acid output, the abatement technology installed and its operating factor. *No magnitude is asserted for these categories here.* A previous edition of this section said the four were together "of order 1 Mt", which had no category-by-category derivation behind it and is withdrawn; Pakistan's own filing puts the chemical industries near 19.4 Mt for 2022, which is the right order of magnitude for the question and is not a substitute for an independent estimate either. They stay NE because the available production quantities do not match the physical process boundaries the IPCC methods require, which is a statement about evidence rather than about size. For ammonia the evidence has now been assembled far enough to test the official figure: the NFDC review supplies production by product and manufacturer and the plant-wise gas split, and the generated reconciliation table compares four independent activity pathways—official filing, plant census, product stoichiometry and gas balance. They do not agree. The filing's 11 467 kt of ammonia would need roughly half again more gas than the entire fertiliser sector draws, while embodied ammonia in the products made sits near 4500 kt; the official activity reads like a total-product or capacity figure, and until the pathways reconcile the category stays NE.

2.5.2 Notation keys as a first-class output

Every NE entry above carries a justification in the generated completeness table, and where a comparator can bound the gap, its magnitude. That is a deliberate inversion of the usual practice, in which the estimated categories are documented in detail and the unestimated ones receive a two-word note.

The reason is that a reader's most important question about an inventory reporting 445 Mt is not how that total was computed but what is not in it. Section 6 gives the aggregate answer—this inventory is 87 % of EDGAR's Pakistani sectors summed—and the completeness table is what makes that 13 % attributable to specific named categories rather than being a diffuse disagreement.

Four IE entries deserve their own note because they are the only cases where a real emission is deliberately reported at zero in the place a reader might look for it. Household electricity—39 728 GW h, nearly half of national consumption—is not in the residential sector, because those emissions occur at the power station and are counted in 1A1a; adding them would double-count roughly a quarter of the inventory. Biogenic CO_2 from dung, firewood and landfilled organic matter is a memo item outside the total, per IPCC's biomass convention and Pakistan's own stated practice. And the two cross-sector transfers of Section 4.8 are IE by construction. Each is asserted by a test: the residential sector fails its build if an electricity stream is ever added.

3 Source data and processing

Every activity quantity in the dataset is traced to a document in the public record: the regulators' annual reports and the statistics bureau's yearbooks for energy, the economic survey and the fertiliser development centre for agriculture, census and survey tables for population and households, and the international defaults where the country publishes no factor of its own. The source hierarchy—a national document over an international default, a period-matched edition over a borrowed one, a transcribed table over an inferred one—is applied per stream and recorded per stream, so that the evidence grade of a number is a column beside it rather than a sentence about it. Two released products are the register of record for this section and are described with the other data records in Section 5: `outputs/pakemis_source_register.csv` names every source document the release read, its hash, its custody status and where its publisher provides it; `outputs/pakemis_source_streams.csv` joins each source to the emission streams it feeds and carries the temporal and spatial evidence classes that the monthly estimate and the spatial allocation inherit from it. Transcriptions are held to their documents by a re-extraction check, and the transformations between a document's units and the inventory's are code in the repository, not steps in a methods note. The redistribution terms that follow from this custody model are stated in the Data availability section: the derived products are open, and the source documents are named and located rather than re-published.

4 Dataset generation

The dataset is generated in four stages that this section follows in order. The annual reconstruction computes every category–gas cell for the reference configuration and for each fiscal year of the panel, and closes it exactly to the national total. The monthly estimate resolves each annual cell into twelve months on a declared temporal profile, and records for every stream whether that profile was observed, borrowed or flat. The spatial allocation places the national mass on administrative units and on a regular grid, and keeps the part it cannot place as an explicit unallocated remainder, so that the closure identity is raster plus unallocated rather than raster alone. The uncertainty stage carries parameter-level draws through the same engine and prices the structural alternatives that a parametric interval cannot contain. The ledgers between sectors—the quantities estimated once and consumed twice across an accounting boundary—are part of the generation rather than of its validation, because they are what makes the total a single sum instead of a collection of sector totals.

The inventory reports $444.96\,\mathrm{Mt}$ CO_2e on AR5 100-year global warming potentials, or $438.19\,\mathrm{Mt}$ on AR6, under reference scenario `REF_2023_24`—every figure in this section is conditional on that configuration, whose twenty-two registered axes are published as a scenario registry. Its priced configurations run from the fugitives envelope's joint floor at 431.06 to the retained legacy power account at $462.74\,\mathrm{Mt}$; those are alternatives of different kinds and are not one range (Section 4.9). Table 3 gives the sector breakdown and Figure 1 its composition.

Module	Activity-vintage qualification	Mt CO_2e (AR5)	Share (%)	Uncert. (±%, 95%)
Agriculture	census herd + FY crop/fertiliser flows	223.45	50.2	19.1
Power module	grid + captive self-generation	45.59	10.2	9.7
Transport module	provincial direct, fleet borrowed	41.93	9.4	11.4
Waste	population projected to the midpoint	37.45	8.4	51.1
Residential fuel	—	24.22	5.4	23.2
Oil and gas fugitives[e]	—	23.50	5.3	37.7
Cement (IPPU)	—	25.54	5.7	8.1
Industrial fuel (1A2)	gas, liquids and HDIP solid fuels	22.58	5.1	18.7
Railways	—	0.39	0.1	14.6
Domestic aviation	—	0.30	0.1	58.8
Total	FY2023–24	**444.96**	**100.0**	**10.9**

Table 3. National inventory by module, all on the common reporting period **1 July 2023 to 30 June 2024**. The year column this table used to carry is gone deliberately: it printed 2023 for power and aviation and 2024 for everything else, for the same twelve months, because those sectors number a fiscal year by the calendar year it opens in and the others by the year it closes in. That is the ambiguity the period contract removed, and a table is the last place it should have survived. Three rows are MODULES rather than single IPCC categories, and are named so. The power module is main-activity generation (1A1a, 40.04 Mt—the selected member of the `power_grid_control` structural family, whose other members are a fuel-property alternative at 38.45, a coverage alternative at 44.05 (the K-Electric system's fuel added) and the retained legacy control at 57.82 Mt, three kinds that do not form one range; the selection rule is declared on the family's rows in `outputs/national_configurations.csv` and in Section 4.9) plus captive self-generation (1A2, 5.55 Mt); the transport module is road transport (1A3b, 40.19 Mt) plus mixed off-road machinery (1A4c/1A2, 1.73 Mt). The category-level split is Table 4. The industrial-fuel row is metered gas (5.61 Mt) plus the liquid fuels of OCAC's Total Industry customer class (3.40 Mt). Since the re-based reference the railways row is a module too: strict locomotive traction (1A3c, 0.389 39 Mt) plus the measured non-traction claimant (1A2/1A4c unresolved, 0.004 98 Mt). [e] The fugitives reference sits one membership step below the top of its own 9.601 Mt to 24.694 Mt structural envelope: conservative default practice and full coverage, with fertiliser feedstock outside the post-meter leakage base. The top is the feedstock-inside membership of every release through rc66, retained and not defended; the priced members, including a geometric-mean central reading of 17.68 Mt, are the `fugitives_structural_envelope` family rows of the same configurations file. The qualification column names what a common period cost. Agriculture's herd is a census count, not a flow over the period; transport's magnitude and provincial shares are the direct FY2023–24 OCAC editions and only its fleet/class composition is borrowed from the registration panel's last year; waste's population is projected one year past its census anchor. Uncertainty is IPCC Approach 1 over each sector's own components; the national figure is the quadrature of all components and is smaller than most of its parts, which is the expected behaviour of independent errors in a sum. Memo items—international aviation bunkers and biogenic CO_2—are excluded by construction.

4.1 One reference period

All ten sectors report the same twelve months: FY2023–24, July 2023 to June 2024. Every sector declares that period as explicit dates rather than as an integer year, and every generated output carries them, so a reader meeting a CSV without this manuscript can see which twelve months a row describes.

Writing the dates down was itself the finding. The integer year carried three incompatible meanings across the inventory—the fiscal year a sector's period opened in, the one it closed in, and a calendar year—with nothing enforcing any of them, and the collision had already put a contradiction into an earlier edition of this paper about which year the power sector reported. It exposed three sectors adrift, one of which nobody suspected.

Transport was reporting FY2021–22 because that was the year its provincial fuel-sales panel was complete for, which put the national total two years behind eight other sectors in order to protect an allocation the total does not depend on. It now takes its magnitude *and* its provincial shares from the reference year's own twelve direct OCAC area pages—nothing provincial is

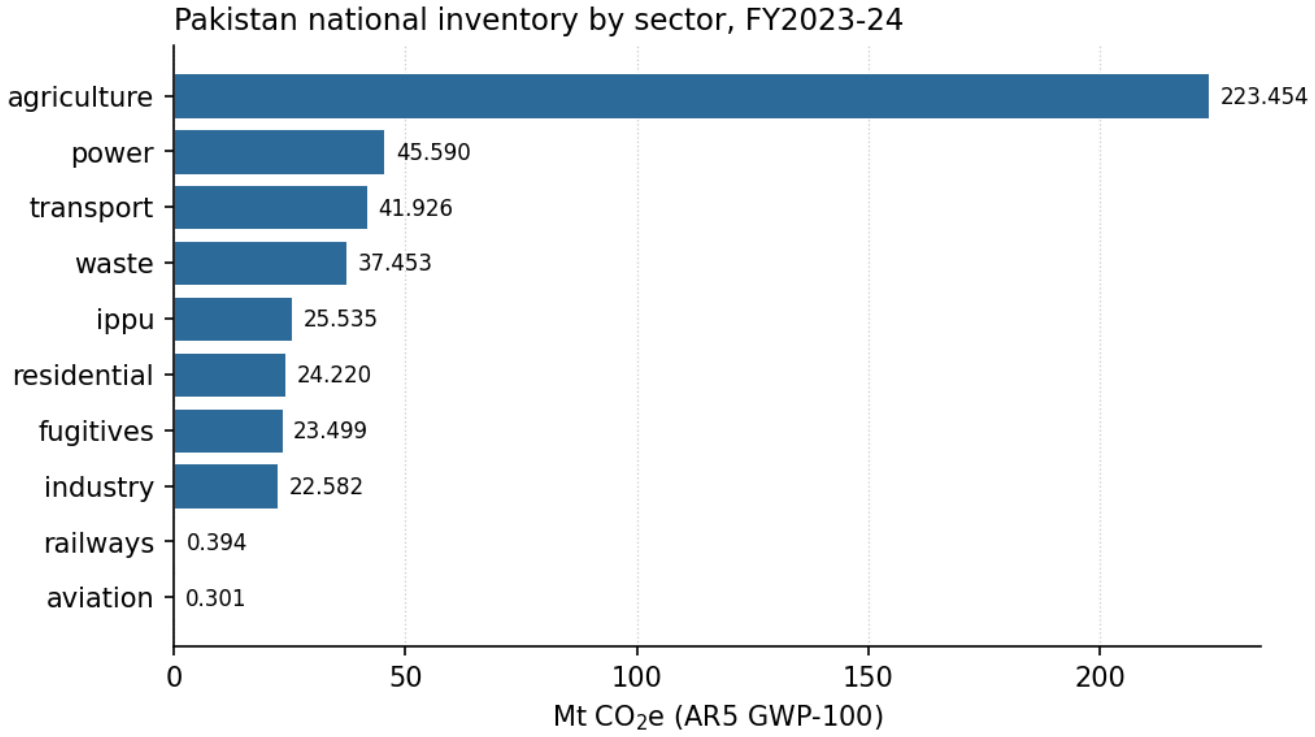


Figure 1. Sector composition of the national inventory. Agriculture is $50.2\,\%$ of the total, and the three energy sectors—road transport, electricity generation and residential fuel—together are $23.5\,\%$.

borrowed since those editions were ingested—while its fleet/class *composition* still comes from the registration panel's last year (FY2022–23), declared as borrowed and bracketed: a share set sums to one, so a borrowed shape moves emissions between classes and cannot change a total. The same publication's *industry* pages had been sitting unread in the same file. Industrial liquids had one provincial vector for six products, and it gave furnace oil the retail geography of fuels it does not burn—Punjab 0.657 against furnace oil's own 0.784—while industrial diesel, which is very nearly half Sindh, was pulled the other way. The two largest liquids now carry their own product's area pages, with the four small ones blended because their shares swing on a few thousand tonnes; the pages also supply a year axis for the two periods before the reference year. The national total is unchanged. Waste was reporting the calendar year of its census anchor; its district population panel is now projected one year to the fiscal midpoint at the panel's own intercensal growth rate—an extrapolation where every other year is an interpolation, and its rows carry evidence class E3 to say so.

Agriculture, at half the inventory, was a fiscal year *ahead*: its crop quantities were the Pakistan Economic Survey 2024–25 tables and its nitrogen offtake the NFDC Review 2024–25. Both now read the 2023–24 editions. $77.37\,\%$ of the sector—enteric fermentation and manure management—rides a livestock census count rather than a flow over any period, which the component rows record as an enumeration rather than a span, and which is why those two categories did not move when the rest of the

sector did. The Economic Survey publishes livestock estimates for the same year; this inventory does not use them, because a count is evidence and a projection is not.

Closing that offset exposed something larger than the offset. The two publications agriculture cites were not in this project's source archive at all—nor was any Economic Survey or NFDC Review—while nine other sectors archived their primary documents. No test could see it, because the release manifest hashed the activity files and not the documents behind them: the guarantee was "these are the exact numbers", never "and here is what they were read from". Both documents are archived now, and the class of defect is closed rather than the instance: `data/source_documents.csv` classifies every activity file as archived, derived, or not-archived; a not-archived file must carry a gap-register row saying so; and the release manifest hashes the archived documents beside the data. An audit at rc65 found 12 of 217 activity files citing at least one document this project does not hold. The register had read twenty-seven through rc64, and neither count had been what it seemed. A reviewer's question prompted a test of every claim in both directions, against both document trees and inside every archive. Of the twenty sources then listed as not kept, three were on disk the whole time and seventeen had never been saved: the scripts that fetched them parsed each response in memory, and pages were read without a snapshot. Those documents were fetched again, copied in from the fetchers' caches or restored from the repository's history, and each is compared with the table that was read from it. The same audit found eight files marked archived whose directory did not contain a document they cite; they are among the twelve. What remains unheld is a priced energy yearbook, a railway year book its publisher no longer lists, a superseded edition of a facility tracker, the district tables of the 2017 census, the satellite fire responses behind the burning proxy, and web pages that could not be captured with their content. `data/catalog/not_retained_register.csv` names each with the reason and, where one exists, a held reprint of the same figures. A fetch script now writes what it receives before it parses it, and a ledger of held documents that only grows fails the test suite if one is removed. They are declared now, which is the difference between a limitation and a silence.

4.2 The national result by IPCC category

The inventory is computed in ten software modules, and it is reported under twenty-two IPCC categories. Those are different partitions, and Table 4 is the one an inventory reader needs: it is the level at which this work can be compared against Pakistan's own filings, against EDGAR, or against any other country's inventory. The module view is kept in Table 3 because it explains where each number is computed and therefore where an error would live, but it is secondary.

The two partitions differ in one place, and it is the consequential one. Manufacturing industries and construction (1.A.2) is computed in two modules: the captive self-generation that the power module reconstructs from metered gas, and the process heat, boilers and industrial liquid fuels the industry module reads from the petroleum customer classes. Together they are $28.13\,\mathrm{Mt}$, which is the number to compare against EDGAR's $43.98\,\mathrm{Mt}$ for the same category—and neither module's own total is.

IPCC	Category	CO_2	CH_4	N_2O	Total	Computed in
3.A.1	Enteric fermentation	—	151.658	—	151.658	agriculture
1.A.3.b	Road transportation	39.285	0.398	0.511	40.194	transport
1.A.1.a	Public electricity and heat production	39.907	0.027	0.106	40.040	power
1.A.2	Manufacturing industries and construction	28.009	0.053	0.069	28.131	industry + power
4.D.1	Domestic wastewater	—	21.886	2.572	24.458	waste
1.A.4.b	Residential	18.388	5.176	0.656	24.220	residential
1.B.2	Fugitive emissions, oil and natural gas[e]	0.916	22.583	0.001	23.499	fugitives
2.A.1	Cement production	21.816	—	—	21.816	ippu
3.A.2	Manure management	—	13.182	6.148	19.330	agriculture
3.C.7	Rice cultivation	—	16.548	—	16.548	agriculture
3.C.4	Direct N2O emissions from managed soils	—	—	16.434	16.434	agriculture
3.C.5	Indirect N2O emissions from managed soils	—	—	12.091	12.091	agriculture
4.A	Solid waste disposal	—	9.169	—	9.169	waste
3.C.3	Urea application	4.824	—	—	4.824	agriculture
4.C.2	Open burning of waste	1.853	1.711	0.262	3.826	waste
2.B.1	Ammonia production	3.243	—	—	3.243	ippu
3.C.6	Indirect N2O emissions from manure management	—	—	1.898	1.898	agriculture
1.A.4.c	Agriculture/forestry/fishing (off-road)[†]	1.569	0.003	0.160	1.732	transport
3.C.1.b	Biomass burning in croplands	—	0.538	0.132	0.670	agriculture
2.C.1	Iron and steel production	0.393	—	—	0.393	ippu
1.A.3.c	Railways	0.353	0.001	0.036	0.389	railways
1.A.3.a	Civil aviation	0.299	0.000	0.002	0.301	aviation
2.D.1	Lubricant use	0.083	—	—	0.083	ippu
1.A.2/1.A.4.c	Railway stationary/off-road (unresolved)[‡]	0.005	0.000	0.000	0.005	railways
Total		**160.944**	**242.932**	**41.079**	**444.95542**	

Table 4. The 1.A.1.a grid figure is the `power_grid_control` family reference; the family's other members read 38.45 Mt (fuel property), 44.05 Mt (coverage) and 57.82 Mt (legacy account), and none is a sensitivity. [e] The 1.B.2 reference sits one membership step below the top of its own 9.601–24.694 Mt structural envelope, fertiliser feedstock being outside the post-meter leakage base; the priced `fugitives_structural_envelope` members, including a 17.68 Mt geometric-mean central reading, sit beside it in `outputs/national_configurations.csv`. The national inventory by IPCC category and gas, Mt CO_2e on AR5, reference scenario `REF_2023_24`, whose residential dung axis is the household survey reading. This is the primary national result; Table 3 reports the same components by software module, which is a different partition. The final column is why both tables exist: 1.A.2 is computed in two modules—captive self-generation inside the power module and process heat and boilers inside the industry module—so a module table cannot show it, and 1.A.2 is the category the largest coverage gap sits in. Industry is the one covered sector with no paper of its own, so its headline tables are named here: its result by category in both vintages (`outputs/industry_totals_2024.csv`), the stream ledger with each row's fossil flag, gigagrams and uncertainty (`outputs/industry_by_stream_2024.csv`), the province-level AR5 total (`outputs/industry_provincial_2024.csv`) and the twelve-month layer with each month's share and the basis of the shape (`outputs/industry_monthly_2024.csv`). The table is a stored query against the semantic release (`paper_national/queries/paper_artefacts.csv`, `tab_categorygas`): every cell is the contract's display value—the exact decimal sum over the same atoms `outputs/national_category_gas.csv` reports, rounded half-even at the contract's precision—and the receipt on the generated file's first line replays it. A dash is a gas with no component in that category, not an unestimated one, which Section 2.5 reports separately. Cells are rounded independently, so rows and columns are additive only to within a trailing digit. [†]1.A.4.c's IPCC placement is unresolved: the source's residual class mixes agricultural machinery with construction and industrial plant that the Guidelines book under 1.A.2, and the split is not published. [‡]The rail re-base's receiving category: measured railway workshop, power-house, pump and track-machinery fuel; the Year Book publishes one non-locomotive aggregate, without the per-use split a single-code booking would need.

4.3 Gas composition and the metric choice

On AR5 the inventory is $160.94\,\mathrm{Mt}$ CO_2, $242.93\,\mathrm{Mt}$ of methane and $41.08\,\mathrm{Mt}$ of nitrous oxide, expressed as CO_2e. Methane is $54.6\,\%$ of the national total, which is the single most consequential fact about Pakistan's emissions profile: it is a country whose inventory is dominated by a short-lived gas, and therefore one where the choice of metric and time horizon carries unusual policy weight.

This paragraph is now read from a generated ledger, and it is worth saying why. It was typed, because until this release no output reported the inventory by gas—every national output was organised by module. It went stale in place: through three releases it printed a decomposition whose three parts summed to $455.14\,\mathrm{Mt}$ against a headline of 448.00, with CO_2 wrong by nearly $10\,\mathrm{Mt}$, while every headline number around it was re-anchored. It was recorded as a known gap in the coverage test's declared-debt list, and the two tests that bound that list both passed throughout, because neither compares a listed quantity against anything. *A declared debt reads as tracked and behaves as unchecked.* The debt mechanism has been retired for numeric claims (Section 2), the decomposition comes from `outputs/national_category_gas.csv`, and a test asserts the three gases sum to the national total under both configurations and both metrics.

The AR5-to-AR6 difference is $6.76\,\mathrm{Mt}$, or $1.5\,\%$ lower. It decomposes cleanly: methane falls by $8.00\,\mathrm{Mt}$ as its GWP moves from 28 to 27 for biogenic sources and 30 to 29.8 for fossil ones, and nitrous oxide rises by $1.24\,\mathrm{Mt}$ (265 to 273), partially offsetting. CO_2 is unchanged by definition. Both metrics are carried through every component row rather than one being derived from the other at the end, which is what allows the AR5/AR6 pair to be reported at every level of aggregation.

The fossil/biogenic distinction is enforced rather than assumed. Every methane-emitting component declares which it is, and a contract test asserts that the declaration is present and that the CO_2e conversion used the corresponding GWP. Only $23.11\,\mathrm{Mt}$—$9.5\,\%$ of the inventory's methane—is fossil, and it matters most in the one category that emits both: residential fuel combustion, where dung and firewood methane is biogenic and natural gas and LPG methane is fossil, within the same IPCC category and the same output rows.

4.4 Provincial allocation

Every sector is allocated to Pakistan's provinces using a category-specific proxy—cropland for off-road machinery, population for wastewater, provincial fuel sales for transport, plant-wise generation for electricity, plant capacity for cement—and every allocation is checked to reconcile exactly against the national figure it distributes. The contract requires that a proxy resolve for every emitted category and that an unknown category raise an error rather than fall back to a default, because a silent default proxy is how a category ends up allocated by population when population has nothing to do with it.

Two allocations have unusually strong controls. Fugitives use separate provincial production, pipeline, consumption, refinery and terminal controls rather than one generic proxy; captive generation rides a 1,105-plant register whose metered gas sums by province, reconciling with the national OGRA total to within 0.01 MMCFD.

A third is worth reporting because it shows the value of writing a proxy's weakness down. Electricity generation was allocated on a capacity database that this inventory's own parameter file described as provisional and the weakest element of the

sector, and which named its likely failure: assigning provinces by a coordinate rule "likely overstates Khyber Pakhtunkhwa". Replacing it with plant-wise generation confirmed exactly that. On generation, Khyber Pakhtunkhwa carries $0.59\,\%$ of the sector's emissions rather than the $14.24\,\%$ of installed capacity the retired proxy gave it, because its electricity is overwhelmingly hydro, which burns nothing and belongs nowhere in a combustion sector. The correction moved roughly $8.8\,\mathrm{Mt}$ out of that province at an unchanged national total. A documented weakness is a specification for the work that fixes it, which is the same pattern the accounting closures follow in Section 4.8.

4.5 The nine-year series, in two views

The reference year is the most complete year, not the only one. The engine reconstructs each sector back to FY2015–16 where its activity record supports it and publishes the result as `outputs/pakemis_national_series.csv`, with the sector-by-category-by-gas expansion in `outputs/pakemis_annual_fy2015_16_fy2023_24.csv`. The product is a nine-fiscal-year panel in which all ten sector modules report in every year—a complete 90-sector-year rectangle since the FY2015–16 product splits were recovered from the public record. What is complete is the set of MODULES, not the set of sources: which categories inside them are estimated is a separate question with its own register, and Section 2.5 answers it. Two things about that product matter more than its magnitudes.

First, coverage is constant at last: every year carries all ten sectors, the national total runs from $393.84\,\mathrm{Mt}$ at FY2015–16 to a maximum of $483.45\,\mathrm{Mt}$ at FY2021–22 (cement was larger that year than in FY2023–24), and `coverage_share_pct_of_reference` reads $100\,\%$ in every row. Until this release the first year carried eight of ten sectors and the product warned that differencing across it measured the record filling in as much as emissions changing; that caveat is retired because its premise is, not because it was wrong.

How the last two sector-years closed is worth recording in full, because the closure is an argument about method. For four releases FY2015–16 road transport and industrial combustion were NE with their observed totals held in `outputs/pakemis_unallocated_activity_controls.csv`—Road 12440195 tonnes and Total Industry 2023377 tonnes, from the FY2016–17 edition's prior-year column—because the split by product existed in no held document and a sixteen-source search was logged negative. The search's negative was true of what it crawled and false of the record: the *Pakistan Oil Report 2015–2016* is publicly served on ocac.org.pk as a 242-page image flipbook, and its page-50 sectoral table carries the split. The page images are 602 pixels wide—below reliable OCR—so the table was transcribed and is accepted only under a four-way identity gate (`sectors/transport/ocac_image_edition.py`): every row sums to its printed total, every block to its class total, the six classes to the printed national total, and—the two-document gate—the recovered Road and Total Industry totals equal the independently held prior-year column exactly. The controls that recorded the absence now anchor the recovery. Cells below the glyph floor are committed only where the closing arithmetic fixes them uniquely, and each carries its read basis (`glyph_identity` or `identity`) in the committed transcription.

The structural configuration that would have MODELLED the year (`FY2015_16_FULL_SCOPE_BRIDGE`) stays declared and permanently unpopulated: the observed split made every permitted alternative moot, which is the strongest argument its contract will ever receive. The validator ships; the bridge never will.

The other absences closed the same way, and none by acquiring anything unheld. Cement was reported for the reference year alone until its clinker-trade panel was fetched back to July 2014 and Table 13.2 of the *Pakistan Statistical Year Book 2020*—held throughout—supplied the four earliest years. Industrial combustion FY2020–21 was an image scan whose table is legible once rendered. Declared gaps that turn out to be unread or unfound documents are the common kind, which is the fifteenth review's point.

Second, the product's *two views* now coincide. The HARMONIZED COMMON SCOPE view is the trend-safe one: the sectors reconstructed on a consistent method in every year of the window. Since the FY2015–16 recovery the window opens at the panel's first year, all ten sectors qualify, and the view runs $393.84\,\mathrm{Mt}$ to $444.96\,\mathrm{Mt}$ across the nine years—identical to the full-scope view, which is what a complete rectangle means. Membership is still derived from the rows rather than declared: for one release the list was a hand-written tuple that excluded cement and industry for reasons the data had already overtaken, and a test that checks only that members are present cannot notice a sector that has become eligible. Both views continue to ship, because the next partial year would split them again and the machinery that tells them apart is the guard. Figure 2 draws them coinciding.

4.5.1 Waste: a stateful series, not a scaled one

One property of the panel is easy to miss and easy to under-claim, and an external review graded it as absent, so it is stated here as a product fact: the waste rows are *per-year first-order decay*, not a reference-year figure scaled along population. The deposit history behind the FOD model is rebuilt to each requested year and the decay recursion run to that year, so every year's solid-waste figure carries the stock the years before it laid down. Solid-waste disposal runs 7.593 to $9.169\,\mathrm{Mt}$ of $\mathrm{CH_4}$ across the nine years ($20.8\,\%$ growth against population growth of roughly two per cent a year)—the convexity of a decaying stock under a growing deposit stream, which a scaled series cannot produce—and the waste module total runs 29.33 to $37.45\,\mathrm{Mt}$. The honest boundary of the claim is equally a product fact: across those years the district population panel varies and, since this release, so do the sanitation pathway shares, each year reading its own row of the JMP series. Generation and collection rates and the waste composition are held at their reference values—the panel's `held_constant` register says so on every row—so the series resolves demographic growth through a stateful model and the recorded shift between sanitation pathways, and does not resolve other behavioural or infrastructural change. Year-varying generation and collection rates await sources (provincial service statistics) recorded as acquisition targets, not modelling.

4.5.2 Agriculture: what a fiscal reconstruction changed, and what it revealed

Two agricultural histories exist and they answer different questions, so which is authoritative is stated rather than inferred. The *fiscal panel* runs FY2015–16 to FY2023–24 on Pakistani fiscal crop and nitrogen activity, and is the series the national trend, the nine-year product and the four-product release are built from. The *calendar diagnostic* runs 1990–2023 on FAOSTAT with a reference-year census anchor, and exists for long context only: it changes basis at the anchor, so it must not be differenced across that seam. The join between them is published rather than described, in `outputs/pakemis_series_`

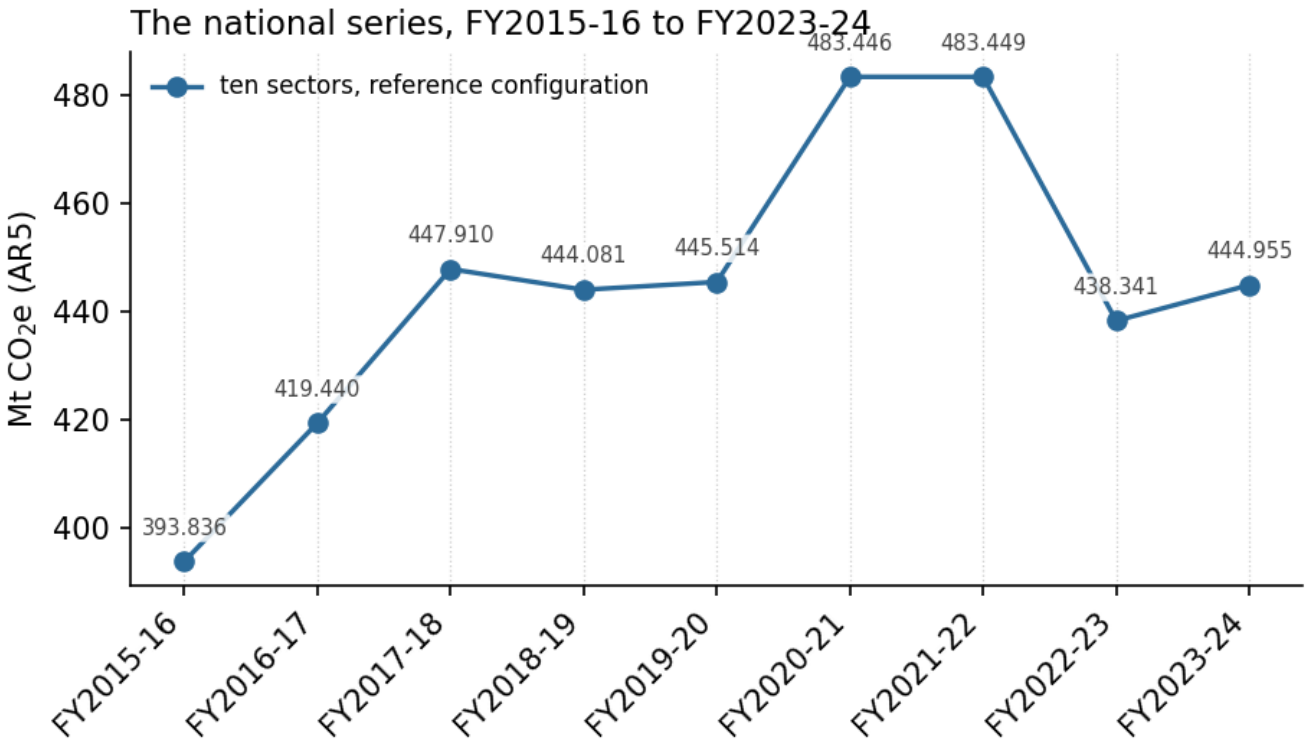


Figure 2. The national series, with its two views coinciding: since the FY2015–16 recovery the full-scope and constant-scope lines are the same 10 sectors in every year. **Section 4.9.9 shows that differencing the series still does not resolve a year-on-year change at 95 %.** Whether a series may be differenced and whether the difference is distinguishable from zero are separate questions. Agriculture sits inside the common scope on a reconstruction from Pakistani *fiscal* activity: crop and nitrogen drivers from the Economic Survey's own fiscal tables, livestock stocks still on the FAOSTAT projection because no fiscal enumeration of the national herd exists. The basis is therefore carried per category rather than per sector, in `outputs/pakemis_agriculture_fiscal_series.csv`, and the residual six-month phase error on the two categories that keep a calendar driver is measured at under 2 % of each rather than asserted. All ten sectors are in the view in every year: cement, industrial combustion and transport, each excluded for as long as an early year was declared absent, report the whole window now.

splice_agriculture.csv, which prices what the change of basis cost per year and per category. Where this paper reports agriculture, it reports the fiscal panel.

Agriculture is half the inventory, so its history decides whether the common-scope view can be differenced at all. Earlier releases drove that history from FAOSTAT and indexed it by fiscal-end year. Rebuilding it on the Economic Survey's own fiscal tables did not simply substitute one source for another: it exposed a misalignment of a full calendar year in 3 of the four crops the former history carried.

FAOSTAT labels a crop by its own harvest calendar year. A Pakistani fiscal year running July to June contains the *kharif* crop harvested in the preceding calendar year and the *rabi* crop harvested in its own. Reading the whole history at the fiscal-end year was therefore correct for wheat and a year late for rice, maize and sugarcane; cotton was absent from the extract altogether. The evidence is not an appeal to the cropping calendar but a comparison against fiscal tables, in outputs/pakemis_agriculture_crop_calendar.csv: those three agree with the fiscal record to within $0.13\,\%$ one year back and disagree by $7\,\%$ to $11\,\%$ where they were read, while wheat behaves in the opposite direction. The concrete cost was that FY2022–23, the year of the floods, was published with the *following* year's recovered rice area.

The rebuild is declared per driver rather than per sector, because the two are not the same. Four categories—rice cultivation, managed soils, urea application and field burning—now run on observed fiscal crop and nitrogen activity. Enteric fermentation and manure management, $77.37\,\%$ of the sector's reference-year mass, keep a calendar driver, because Pakistan publishes no fiscal-year *enumeration* of the national herd: the Economic Survey's fiscal head counts are, by their own footnote, the inter-census projection FAOSTAT already carries, and adopting them would buy a label and no observation. The residual six-month phase error those two categories inherit is measured rather than argued, at under $2\,\%$ of each category in every year, and it is small for a structural reason—the series it acts on is a smooth geometric projection, so a phase shift moves the level and barely touches the trend. The volatility that the seam genuinely distorted lived in the crop drivers, which are precisely the ones the rebuild fixes.

Across the eight backcast years the change was worth $-7.6969\,\mathrm{Mt}$ of basis and $1.055\,\mathrm{Mt}$ of cotton that the former history did not carry, decomposed by year and category in outputs/pakemis_agriculture_basis_change.csv. The two terms are never summed into one column: a source correction and a missing category are different claims with different standing. The reference year does not move, and neither does the national total that anchors it.

4.6 Monthly resolution

The same twelve months are also reported one at a time. For every source stream s the layer asserts

$$E(s,m) = E(s) \cdot p(s,m), \qquad \sum_{m} p(s,m) = 1, \tag{1}$$

so the twelve months reproduce the annual component exactly and no profile can create or destroy a gramme. The direction matters: the annual figure remains the control and the monthly layer only decides *when* within the year, which is what makes a bad seasonal profile a presentational error rather than an inventory error. All ten modules now carry a layer, the national

concatenation closes against the same component rows every other output in this paper is built from, and a module that shipped without one would fail rather than contribute silently.

Resolution is not observation, and the layer is built so that the two cannot be confused. Every month row carries a temporal class on the same four-level scale as the evidence and spatial axes: TE1 observed monthly activity for this source and year, TE2 reconstructed from monthly administrative records, TE3 a process-based seasonal profile, TE4 flat or weakly informed. Weighted by emissions, the national inventory is $0.9\,\%$ TE1, $25.29\,\%$ TE2, $19.12\,\%$ TE3 and $54.65\,\%$ TE4 (Figure 3). That last figure is the honest headline of this section. A flat profile is a claim—that a source is aseasonal—and for most of this inventory it is a claim made because no monthly activity statistic exists, not because the claim was tested.

Where the months are real, two publishers carry most of them. NEPRA's State-of-Industry Detail-of-Generation listings give the grid its plant-month generation—coal TE2, stated plant by plant, and gas TE3, where the annual control itself forces the mixed-labelled combined-cycle group onto gas—so the largest single block of newly timed mass ($38.09\,\mathrm{Mt}$) follows the fleet's own dispatch. OCAC issues petroleum sales by customer class monthly, and four modules—road transport, railways, domestic aviation and industrial combustion—take their shape from it, differencing the cumulative year-to-date columns that the FY2023–24 editions print. That differencing is an arithmetic identity on two published numbers rather than a model, which is why those months close to the annual control exactly; only July, whose cumulative column *is* the month, is read directly, and it is the inventory's only TE1. Agriculture's synthetic-fertiliser nitrogen rides NFDC's consolidated offtake by month, published on this inventory's own July–June fiscal year, so all twelve months are observed in every year and nothing is borrowed (an earlier edition rode the April–March Annex-7 and borrowed three months; that path survives only as the declared fallback for a year the consolidated table does not cover). Rice methane follows cultivation area-days across the provincial calendars; residential fuel follows province-weighted degree days; cement rides its own fiscal months in every year—all nine PBS QIM editions are parsed, so no cement shape is borrowed from anywhere.

That identity is now measured rather than argued. OCAC also publishes a single-month edition of the same months it later folds into the cumulative columns, and the two are independent statements of one quantity, so the recovery can be tested against a document that never entered it. Across four customer classes and three fiscal years, 144 months were compared: 132 agree to the tonne, none disagrees by more than $0.14\,\%$, and the two largest deviations sit either side of a single month OCAC restated after the panel was assembled (`outputs/ocac_direct_edition_crosscheck.csv`). The check can fail—a one-month shift injected into the published side, the off-by-one this project has twice found at a fiscal boundary, is caught by 141 of the 144 comparisons, and that positive control is published with the result rather than asserted about it (`ocac_parse_summary.csv`). Two further products are shipped beside them: `ocac_edition_currency.csv`, which records for every month whether the shipped panel is built on the edition OCAC currently serves and flags the one that is superseded, and `ocac_direct_edition_quarantine.csv`, which holds the two rows out of 6055 where the source's own products do not sum to the total printed beside them, unrepaired.

Cement is the one stream whose months were rebuilt this release, and the rebuild is worth a sentence because the shape it replaced was wrong in a measurable direction. All nine fiscal editions of PBS Table 2 are now parsed rather than the single hand-transcribed calendar-2022 window held before: 1017 item-years close on the annual total each row prints for itself, a

rate of 99.12 %, and the 9 that do not are kept unrepaired in `outputs/pbs_qim_quarantine.csv` because PBS's own arithmetic does not close on them — one prints twelve identical monthly values against a stated annual total that is not twelve times any of them. The check has a positive control published beside it: 932 of the 1017 series are distinguishable from a rotation of their own months, which is the property a seasonal profile actually depends on, and the remainder are flat series where no parse could tell (`pbs_qim_parse_summary.csv`). Two independent things agree with the result — every month of the earlier hand transcription reproduces exactly, and the borrowed shape it replaces was overstating cement's seasonality by roughly half, peaking at 2.27 times its trough where FY2023–24 peaks at 1.48 times its own.

Two decisions in the layer are worth stating because the alternative would have closed just as exactly. The livestock categories—38.9 % of the national inventory—ride a census, which counts a stock once and carries no month at all; they are flat, declared TE4, and a test asserts that no row whose activity is an enumeration ever receives a shaped profile. The same rule reaches inside agriculture's largest category: more than half of the nitrogen in managed soils is deposited or applied off that same census rather than bought from a dealer, so it is flat too, and putting all of managed soils on the fertiliser shape would have claimed an observed monthly basis for 15.65 Mt of emissions that has none.

The national year peaks in August at 41.50 Mt and troughs in April at 32.13 Mt, a ratio of 1.29 against a flat 37.08 Mt. That curvature now has two authors: agriculture—rice's flooded window and the fertiliser calendar—and, since the grid's plant-month generation arrived, the thermal fleet's summer dispatch, which deepens the August peak and the spring trough in the same direction. A reader should take the shape of Figure 3 as a statement about a handful of profiles and an assumption about the rest.

The whole of that is now published month by month. `outputs/pakemis_monthly_coverage_summary.csv` carries exactly 108 rows, one per month, giving the mass under each temporal class and naming any sector the month does not have at all; `pakemis_monthly_coverage.csv` backs it at the month × sector × category grain. Read strictly—TE1 and TE2 only, a monthly *record* of the activity rather than a modelled shape—18.79 % of the nine-year mass is observed and 60.84 % sits on a declared-flat profile. The single figure is the least useful thing in the table, because the strong classes are not spread across the window: the six years before OCAC's panel opens average 13.79 % observed and the three after it 28.42 %—a gap the SIR record first widened (the listings cover the panel years) and the same archive's earlier sheets then narrowed, labelling the pre-panel fleet in its own years. No sector-year is missing its months any more: the two that were—industry and transport in FY2015–16—gained their twelve rows when the year was recovered, and the `missing` class waits empty for the next partial year.

The second axis is reported beside it for the first time. Every priced row in the atomic annual table now declares how well its *quantity* is supported, where 41.2 % of the nine-year mass previously carried no class at all—unpoliced by any test and absent from every output. Derived from the basis each category already declares rather than graded afresh, the split is 11.1 % E1, 33.0 % E2, 54.8 % E3 and 1.1 % E4. That E3 majority is the honest consequence of agriculture: half the inventory, and eight of its nine years rest on a livestock projection rather than on a count taken in the period reported.

Unshaped no longer means equal twelfths. A stream that declares no seasonality is declaring something about the SOURCE, and that is not a licence to divide the calendar wrongly: a continuous emission is proportional to elapsed time, and February

is $10.7\,\%$ shorter than July. Every declared-flat profile is now weighted by its month's own length, leap days included, which is what widened the peak-to-trough ratio from 1.26 to 1.288 without moving a gramme of the annual total. The change is an arithmetic correction rather than new evidence, and the class of these rows is unchanged. The ratio has moved again, each time for the opposite kind of reason: to 1.32 when cement stopped borrowing a calendar-2022 shape and took its own fiscal months, and onward when NEPRA's plant-month generation record put grid coal and gas on generation-dispatch timing—coal stated plant by plant (TE2), gas assigned by the annual control's own bound and therefore capped at TE3, never called observed—new evidence about $38.09\,\mathrm{Mt}$ that had sat flat, and not a gramme moved either time.

4.6.1 Two cross-checks on the timing

A seasonal profile cannot be validated against the truth: nobody meters when Pakistan's nitrogen is taken up or when its furnace oil is burned. What can be done is to build the same profile a second way and publish the disagreement.

The fertiliser profile is the inventory's largest use of an observed monthly series, and the agronomic alternative is a crop calendar—each crop's share of national nitrogen against its own application months. The two answer different questions, one recording when urea left the depot and the other when the crop needs it, and they disagree by 11.5 percentage points in the worst month and 51.5 points summed over the year, which is roughly a quarter of the annual nitrogen changing month (Figure 4). The offtake series is markedly flatter than agronomy: it shows no November spike for the wheat basal dressing, because dealers stock and farmers buy on price, credit and availability weeks before the field needs it. Neither series is uptake, and the gap between them is the honest width of the monthly claim for the largest category in the inventory.

The second check tests the differencing method rather than a profile. Summing the twelve recovered months back to a year is not circular when the annual edition is a separate publication, and OCAC issues one. For FY2023–24 every customer class agrees to the tonne, which confirms the ingest and the arithmetic and has no power to detect revision, because that annual edition reprints the June cumulative column unchanged. FY2021–22 is where the check bites: its months are twelve separately printed direct editions, reassembling them into a year is a real test, and they return within $0.34\,\%$ across every class the four OCAC-driven modules consume.

4.6.2 What the layer found in the annual rows

The component schema has carried a declared temporal class since before there was a layer to use it, which made it a claim about evidence that nothing checked. Reconciling the two moved 34 of 202 component rows (`outputs/national_uncertainty_decomposition.csv`), in both directions, and the direction is the interesting part. Aviation's TE1 became TE2: FY2023–24's OCAC editions are cumulative, so only July is read directly and the sector had been claiming an observed monthly basis for the whole year. Road CNG, bagasse in the power sector, crop-residue burning and the livestock-origin soil nitrogen all fell to TE4 at that reconciliation, because the shapes their declarations implied were not then assembled. Two have since climbed back on real evidence—crop-residue burning on the nine-year fire record, waste disposal on the FOD state engine—while the sugar crushing season remains documentable and undocumented. Residential charcoal rose to TE3 with the

rest of its sector, where the weak part is the quantity and not the shape. A test now asserts that the class a component declares is the class its own layer used, so the two cannot drift apart again.

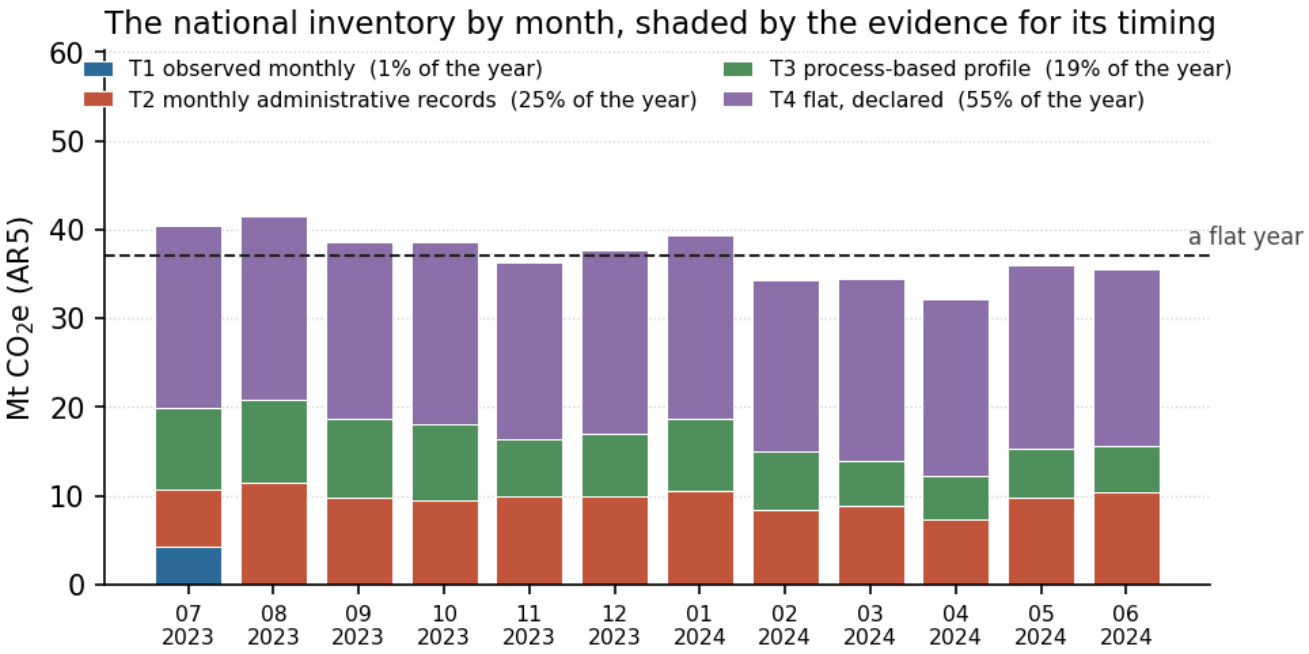


Figure 3. The national inventory resolved into the twelve months of FY2023–24, shaded by the evidence for the timing rather than for the quantity. The dashed line is one twelfth of the year. $54.65\,\%$ of the inventory sits in the flat TE4 band, so the visible curvature is produced by a minority of it—chiefly rice methane, fertiliser nitrogen and the thermal fleet's summer dispatch.

4.7 The gridded cube

The abstract's gridded claim is discharged by one product built in this repository: 199 of the 202 source streams, spread over the national 0.05° grid by the most native geography its evidence supports and over the twelve months by the layer above; the remaining 3 are held at ADM1 because no geometry exists beneath their provincial control, and closure is therefore raster-plus-unallocated rather than raster alone. The placed streams follow the identity $E(s,m,g,i) = E(s,g)\,p(s,m)\,w(s,i)$ with $\sum_m p = 1$ and $\sum_i w = 1$ enforced per stream. The conservation statement is precise about what it covers: NATIONAL mass is invariant across representations—annual, monthly, gridded, district and provincial views all close to the same ledger, held by test on every build—while subnational marginals may differ under documented physical-versus-administrative allocation rules. The provincial check records the split exactly: 920 stream-province cells exact by construction, 15 unallocated below ADM1—the industry streams whose provincial share is the only spatial claim they make—45 carrying a declared divergence (nine streams: domestic aviation, administratively in ICT and physically at Attock's coordinates, and the six railway non-

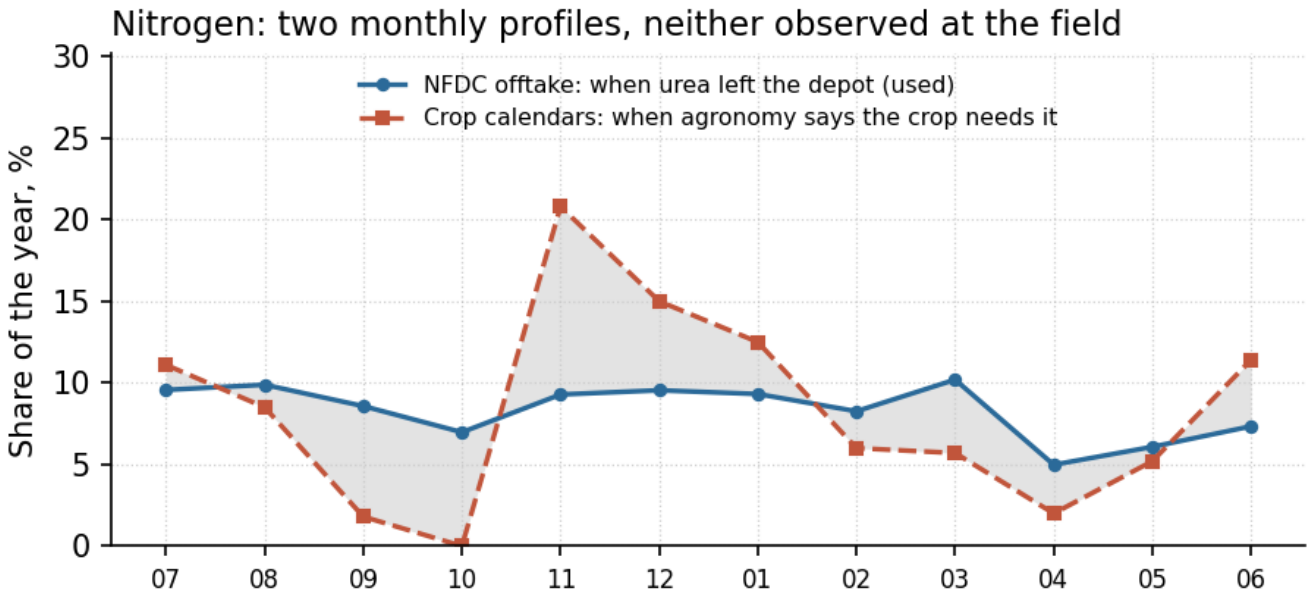


Figure 4. Two independent monthly profiles for the same nitrogen: NFDC's observed offtake (the year's own twelve fiscal months, all observed), which the inventory uses, against the agronomic crop calendars. They disagree by 51.5 percentage points summed over the year — the two-instrument gap that licenses the drawn timing dispersion for this category.

traction streams, whose border-yard cells bin across provincial boundaries), and 30 derived from the grid itself and therefore not an independent check. The cube can disagree with this paper's totals only by failing, never by drifting.

Six artifacts ship. The cube itself (`outputs/pakemis_grid_0p05_monthly.nc`) carries the gas×month layers in native gas mass—no CO_2e per cell, because a GWP is one multiplication and storing it would be invented precision—together with per-category annual surfaces. It no longer carries an `s4_co2e_share` layer. The layer once reported the per-cell footprint of streams spread on a proxy while carrying no native geometry; flagging placed mass is not the same as declining to place it, so later editions retained such streams at ADM1 instead. rc47 then gave industry its geography outright (kiln points, the clinker register, built-up), and what remains unallocated is the 3 cruise-aviation streams (0.06 % of the national total): fuel burned along corridors no held evidence can draw, booked at the uplift provinces and published per stream, province, gas and month in `pakemis_spatial_unallocated.csv`, with the closure identity carrying the term explicitly: `ADM1 == ADM2 + unallocated`. That identity is asserted for all nine fiscal periods and, until this release, its second term was published for one: the cube computed the unallocated mass every period and wrote only the reference year's, so a reader checking a historical year's arithmetic had the wrong year's number or none. `pakemis_spatial_unallocated_by_year.csv` carries the same rows with the period they belong to, and the mass it retains falls from 0.40 Mt in FY2017–18 to 0.25 Mt in FY2023–24 as facility geography reached more of the inventory. Nothing above ADM1 moved—the national total and all 1010 provincial marginals are unchanged—because only the placement *below* the province had been

invented. The ADM2×month view (`pakemis_adm2_monthly.csv`)—now also shipped sector-resolved as `pakemis_adm2_monthly_sector.csv`, module by shape by gas by month, a tested partition of the aggregate view; industry appears there carrying only its declared `unallocated-below-adm1` absence rows (an empty numeric cell, the reason in its own column), because the whole module is unplaced and a district pattern for it would be an invention—the provincial check (`pakemis_grid_provincial_check.csv`)—which records where the grid marginal is exact by construction, where it is derived, and where it *declares* divergence, as at Islamabad International, administratively in ICT and physically in Attock, where the cube follows the coordinates—and the fallback declarations (`pakemis_grid_fallbacks.csv`) complete the set. Substitutions are shipped as data rather than log lines, and so is every deliberate change of placement: the release's spatial re-bases are measured against the previous commit—shapes moved, gross and net mass, fallback mass before and after, provincial marginals asserted unchanged—and published as `pakemis_spatial_rebase_rc42.csv` and, for this release's re-bases, `pakemis_spatial_rebase_rc44.csv`. The second file exists because the first could not see this release: it reads three reference-year products, so a re-base that moved FY2017–18 and left FY2023–24 alone reported nothing at all. It now reads the nine-year district and provincial products as well, and distinguishes the provincial marginal it must hold fixed from the one a deliberate re-base is allowed to move.

The spatial layer is no longer a single year. The national ledger has carried nine fiscal years since rc22, while everything below the nation existed for FY2023-24 alone; the same dispatch is now run for every registered period and published annually as `pakemis_adm1_by_year.csv` (7864 rows, stream × province × period) and `pakemis_adm2_by_year.csv` (10 350 rows, shape × gas × view × period). Both close, per year and per gas: the national total equals the ADM1 marginal, and ADM1 equals ADM2 plus the mass explicitly unallocated below it, to 1×10^{-12} for the eight years the monthly panel controls and to 3.3×10^{-8} for the reference year, that residue being the stream ledger's eight printed decimals against the panel's twelve and nothing else. What these products do *not* claim is nine years of observed geography. Each year's national mass is evidenced for that year; the within-province pattern is the reference year's, rescaled to the year's own control, and it is graded as such per stream-year rather than in a footnote. The sharpest case is grid power, whose plant-fuel ledger begins in FY2019–20. The four years before it have no per-plant fuel evidence at all, so their grid streams ride the built-up substitute, and the difference is categorical rather than marginal: $17.66\,\mathrm{Mt}$ CO_2e is placed on that substitute in FY2015–16 against $0.62\,\mathrm{Mt}$ in the reference year. That second figure was $0.34\,\mathrm{Mt}$ until rc44, and the correction is the more interesting half of the comparison. A province's whole share used to be pooled across whichever of its plants happened to carry a coordinate, so fuel belonging to a plant without one was placed on its located siblings and no substitution was recorded: measured from the ledger, $3.4\,\%$ of grid gas, $0.0\,\%$ of furnace oil and $0.0\,\%$ of diesel sat on plants the register cannot place. Furnace oil read $17.0\,\%$ until rc60, and it did not fall by acquiring a coordinate: the whole remainder was one station, Lal Pir, which had held a GEM-exact point since rc46 and could not be matched to it because the plant ledger carried a column heading inside its name. The substituted mass in the reference year moves from $1.24\,\mathrm{Mt}$ to $0.94\,\mathrm{Mt}$ for the same reason. That fraction now rides the declared surface with an event, which is why thirteen streams move from `point_resolved` to `point_partial` in this release: they always were partial. The cube has always priced its substitutions, but it wrote them for the reference year alone and only where a province had no located plant at all, so a year that substituted one province of one stream and a year that substituted an entire sector looked

alike in everything published. `pakemis_grid_fallbacks_by_year.csv` publishes the record for all nine periods, per stream and province, with the mass each substitution moved.

How much of the nine-year spatial product rests on evidence from the year it describes is itself published, per stream-year, in `pakemis_spatial_weights_by_year.csv`—the spatial mirror of the temporal registry's `borrowed_from_year`. Of 1818 stream-periods, 29 carry no mass at all (the structural absences above) and the remaining 1789 are typed by how their geography was arrived at: 795 (44.4 %) were measured for their own year, 615 (34.4 %) ride a timeless structural surface, 306 (17.1 %) carry the reference year's pattern rescaled, 46 rode a substitute surface, and 27 are unallocated below ADM1 in every year. The first two of those were one category until this release, and separating them corrected the figure downwards: a stream placed on a 2021 land-cover snapshot was counted as measured for its own year in every one of the nine, and being the reference year does not make a 2021 raster an observation of FY2023–24.

The split between the second and third of those was itself wrong until this release, and the correction is worth stating precisely because of what it does *not* move. A surface the engine keys to no period at all has no time axis in any year, so it is timeless in all nine; the build applied that reading only to the reference year and typed the same surface a backcast in the other eight. 499 stream-years were affected, and their weight vectors hash identically to the reference year's, so nothing was backcast—there was no time-specific pattern to backcast from. The label asserted an operation that did not happen, and a reader counting how much of the map rests on a borrowed *shape* counted all 499 in every year but one. What the repair moves is the *kind* of non-evidence, not its quantity: the 44.4 % measured for their own year is unchanged, and the timeless and backcast classes together still account for 921 stream-periods. A timeless surface and a borrowed one are both silent about the year they are placed in, and the distinction matters for what would fix them—a backcast needs that year's evidence read, a timeless surface needs evidence that has a year axis at all. The registry keeps two axes apart on purpose. Whether a year's geography was *borrowed* is silent and must be derived; whether it was *substituted* is loud and is read from the events the build emitted. This pass is the argument for the distinction: the grid fleet was for a time borrowed from the reference year with nothing recording it, and a register reading only the emitted record would have called those years well placed. Field burning is the one sector whose within-province geography is measured in every year, because FIRMS holds all nine fiscal windows and its proxy is built for the period asked for.

The months below the nation follow the years: `pakemis_adm2_monthly_by_year.csv` carries all 108 of them at district grain (122 616 rows), and the reference year's rows in it are byte-identical to the monthly view that already shipped. The nine gridded cubes are published too, which an earlier release declined to do. The objection then was that at five hundredths of a degree the within-province pattern is, for two thirds of stream-years, the reference year's—precision the evidence does not have, in the format most likely to be read as though it did. That objection was right and it is answered rather than overridden: every layer now ships beside the per-stream, per-period evidence in `pakemis_spatial_weights_by_year.csv`, which says whether a period's spatial pattern is its own or the reference year's carried back, so a reader can see which years a map actually resolves. A mostly-borrowed raster with an honest provenance column is useful; the same raster without one is what was refused.

The file is pakemis_grid_0p05_annual_ref_fy2015_16_fy2023_24.nc: nine fiscal periods by three spatial views by three gases on the frozen pk_0p05_epsg4326_v1 lattice, 9.5 MB. Its time coordinate is the midpoint of each July-to-June interval with explicit bounds, because these are nine fiscal years and not ten calendar ones—a calendar-year product could be made directly only for CY2016–CY2023. It carries cell_area and pakistan_fraction as variables, so a cell total in Gg can be turned into a flux per square kilometre of land without reconstructing either. Per period and gas the cells plus the retained mass of pakemis_spatial_unallocated_by_year.csv reproduce the national total to within the precision of the stored single-precision raster. Whether a period's map is its own is now measured rather than graded. Each stream-year's normalised weight vector is hashed where the allocator uses it, and the digest ships in pakemis_grid_stream_surfaces_by_year.csv beside the surface the mass was placed on. The registry's provenance column was previously derived from what the engine *can* read for a surface and district panel—the right rule, and still a statement about the build rather than a measurement of it, which is how a fleet once satisfied one half of that rule while failing the other for an entire pass unnoticed. The two agree: all 306 stream-years labelled a backcast do carry the reference year's pattern, and of the 795 labelled as resting on their own year's evidence, none does. The hash also reaches past the label, which is how the mislabelling above was found rather than argued: 805 non-reference stream-years carry the reference year's pattern, 306 of them typed a backcast and 499 typed timeless, and it was the second group—identical patterns under two different names—that showed the label was describing the period rather than the surface. The check runs in both directions, because a grading that can only be wrong one way is a grading nobody checks. Monthly cells and the ledger configuration remain the reference year's alone: neither a historical month's seasonal geography nor a historical ledger configuration is gridded, and neither is invented.

One frame carries all nine periods, and the products say so in their own headers: the canonical 152-unit set on geoBoundaries-gbOpen-PAK-ADM2-2019. Eleven of those units were carved out of larger districts during the window, so an earlier year's row for such a district is the mass that would fall inside *today's* boundary rather than a figure any administration reported at the time. That is the only choice under which the nine years are comparable with one another, but it is a choice; the contemporaneous administrative geography is not held, and the frame's own date column carries source labels rather than dates, so the gap is registered as an acquisition request instead of estimated.

The nine years reach the query surface, which is where they were always needed: this pass began because a nine-year ledger was attached to a one-year renderer, and products that never enter the semantic release leave that exactly as it was. pakemis_adm2_sector_by_year.csv carries the district view with its module axis for every period, and the release publishes it as fact_adm2_year beside the reference year's fact_adm2_month. Two tables rather than one, deliberately: making the month table cover nine years at module grain would store a borrowed within-province pattern a hundred and eight times per district instead of nine, and a single table whose module coverage was complete for one year and absent for eight would answer a comparison across years wrongly while answering the reference year correctly. The split puts that asymmetry in the schema, where a consumer meets it. A district query for an earlier year is now served rather than refused—the refusal it used to raise, "spatial products exist for FY2023-24 only", had become false—while a *monthly* district request outside the reference year is still refused, now for the reason that is actually true. The provincial level follows the same arc one release later: fact_adm1_year publishes the nine-year ADM1 product (pakemis_adm1_by_year.csv) at stream grain with exact CO_2e

—ADM1 is the level the closure identity holds exactly at, so it carries no bounded cells—retiring the last stale spatial refusal, which had provinces answering monthly for the reference year and not at all for the other eight. This is a reconstruction of where past emissions most likely sat, and its value is that a reader can see, per row, which part of it was measured.

The allocation now also prices its own alternatives where an admissible second reading exists. `data/stream_spatial_alternatives.csv` registers, per allocation basis, whether another defensible proxy exists, and `outputs/pakemis_spatial_brackets.csv` publishes the spread: road diesel under the retired registration proxy disagrees with the fuel-sales allocation by up to 17.47 percentage points of a provincial share (the Karachi port effect), the neighbouring panel years drift from the reference year's own provincial shape—no longer borrowed since the direct editions' area pages were ingested—by at most 3.93 points, and the within-province alternatives (built-up against population, plant points against the fallback surface) are declared and deferred, because province marginals are share-constrained and only a grid-level product can price them. The 3 no-native-geometry cruise streams carry an explicit no-alternative admission instead of a bracket. Since the asset round the brackets also descend to district level (`outputs/pakemis_spatial_brackets_adm2.csv`): the captive-power point-versus-fallback alternative moves up to 29.18 percentage points of a district share—information no province number implies—while the residential raster choice is *proved* invariant at district level, because the district panels fix each shape's mass and the raster moves only cells within shapes. No probability attaches to any spread: these are disagreements between defensible readings, not error against an observation—no independent provincial fuel record exists to be wrong against.

The cube is the reference configuration: scenario REF_2023_24, whose residential dung axis is the household survey reading. Configuration L is gridded through the same dispatch as its own file (`pakemis_grid_0p05_monthly_ledger.nc`), differing in the two dung-cake streams' masses and in no other cell, and its delta is carried in the file's global attributes. The spatial-evidence classes behind the allocation are tabulated per module in Table S4. The allocation basis is declared per stream and the spatial class is now *derived* from it rather than declared beside it: 71.79 % of the inventory reaches the grid from an administrative statistic on a stated proxy, and the cube is exactly as good as those declarations—a placement of the ledger, not an observation of the atmosphere. 16.88 % now reaches it through facility registers the allocation actually consumes: grid stations at their own plant-ledger fuel, cement works at their company's census clinker, captive plants at their metered gas—each with a built-up fallback that is a declared ledger event, and each graded S2 rather than S1 because coverage sits below the 95-percent line the derivation demands. The remaining 0.05 % does not reach the grid at all: cruise aviation has no defensible placement below its uplift provinces (no origin–destination matrix is held), so that mass is retained at province level and published per stream, province, gas and month in `pakemis_spatial_unallocated.csv` — graded NA rather than given a letter it does not earn, with the closure identity carrying the term explicitly (`ADM1 == ADM2 + unallocated`). The administrative share was 96.59 % as recently as the previous release—the facility registers existed but nothing consumed them—and the derived class is what forced that honesty: an earlier hand-written letter had said S2 over 91.2 % of electricity generation that the cube then placed on a raster.

4.8 Four Enforced Accounting Closures

Four quantities in this inventory are estimated once and consumed twice across an accounting boundary. In each case the conventional treatment is for each side to estimate its own share and for a methods document to assert that the two agree. Here each is a single number computed once and consumed twice, and the equality is asserted with == rather than with a tolerance, because a tolerance would permit exactly the drift the closure exists to prevent.

The boundary is not the same kind of boundary in all four cases, and the distinction matters for what the technique transfers to. Two are *transfers between modules*: a quantity leaves one sector's books and must arrive on another's. Two are *partitions within one module*: a single total is divided between categories that are reported separately, and the parts must exhaust the whole. Calling all four cross-sector, as an earlier edition of this paper did, described the architecture more grandly than it works.

The four are the closures enforced with ==; they are not the whole of the inventory's cross-boundary accounting. The release publishes 8 ledgers in one schema (`data/cross_sector_ledger_map.csv`, projected into the validation package's cross-sector registry with every value read from the sector ledger it points to), each with its control, its transfers and its unresolved rows: the four here, and the delivered-gas ledger across the metered modules—whose net-of-leakage alternative is priced in `outputs/national_gas_closure_variant.csv`, one row per configuration with the burned-and-leaked volume and the leaked share behind it—the fertiliser feedstock-to-urea ledger, the livestock nitrogen-excretion ledger and the crop-residue ledger, whose closures are asserted in their own modules' tests. An earlier edition counted only the four.

Closure	Kind	Physical transfer
Dung-cake nitrogen, agriculture to residential	Cross-module transfer	Agriculture → Residential
Household-generator petrol, carved out of the road control	Cross-module transfer	Transport → Residential
Waste generation into disposal, uncollected and burned	Within-module fate closure	Waste categories
Road-class diesel into on-road and mixed off-road	Within-module category partition	Transport categories

Table 5. The four closures by the kind of boundary each crosses. The enforcement is identical in all four cases—one computation, two consumers, exact equality—which is why the broader name *cross-boundary accounting* is the transferable one. Arrows give the direction of the *physical* transfer: the module whose books the quantity leaves to the module on whose books it arrives. For the generator-petrol row the software dependency runs the other way—the residential module computes the estimate and the transport module subtracts it from its road control (Section 4.8.2); for the dung row the two directions coincide.

All four were known defects before they were closures. Each is documented as a limitation in an earlier version of this inventory—an orphaned flow, a double-counted fuel, an unpartitioned total—which is the point: the closures are not new physics but the conversion of four written-down warnings into four enforced properties.

4.8.1 Dung-cake nitrogen: agriculture to residential

Pakistani households burn cattle and buffalo dung as fuel. The agriculture sector has always accounted for this on the supply side, removing $28\,\%$ of cattle and $30\,\%$ of buffalo excreted nitrogen from the manure-management balance and reporting it with the notation key IE—included elsewhere—against the Energy sector. Until the residential sector existed, nothing was elsewhere. The nitrogen left agriculture's books and arrived nowhere: $1.260\,\mathrm{Mt}$ of excreted nitrogen a year, correctly removed from one balance and never entered in another.

The residential sector now receives it, and receives it as mass rather than as an estimate. The transfer runs through the same herd, the same species categories and the same burned-fuel shares that drive the agricultural removal, converted from nitrogen to volatile solids using IPCC's VS rates for the same animal categories and region that supply the nitrogen rates. The result is $38.77\,\mathrm{Mt}$ of *volatile solids*, which corresponds to $58.74\,\mathrm{Mt}$ of air-dry dung cake under the declared moisture and ash basis. Volatile solids rather than total dung mass is the quantity the closure runs on, because VS is the organic fraction and dung ash carries no energy—but the two are not interchangeable, and an earlier edition called the VS figure "dry matter burned as household fuel", which reads as the mass a household actually handles and is 1.5 times smaller than it.

The transfer fixes the diverted mass and not its destination. Of what the residential ledger receives, the reference burns in households the dung the household survey observes, $10.93\,\mathrm{Mt}$ of air-dry cake, and the other $47.81\,\mathrm{Mt}$ of the diversion has no observed destination: it is a typed unresolved row outside the total, with no emission applied, and Section 4.9.5 prices its candidate destinations. Every release through rc66 burned the whole diversion in households; that reading is retained beside `REF_2023_24` as configuration L, a destination assumption for the diverted dung rather than a reading of household use.

The identity is exact by construction: the diverted mass is a derivation from agriculture's own flow, and household combustion plus the unresolved remainder equals it. A change to the manure-management shares moves the diversion and the remainder together.

The biogenic CO_2 from this fuel is reported as a memo item outside the national total, following IPCC's biomass convention and Pakistan's own stated practice in BUR-1 (Ministry of Climate Change, Government of Pakistan, 2022). Its CH_4 and N_2O are counted: incomplete combustion is a real net source regardless of the carbon's origin.

4.8.2 Household generator petrol: carved out of the road control total

Pakistan's petroleum statistics report motor gasoline sold to a Road customer class. Not all of it moves a vehicle. Households running standby generators against load-shedding buy their petrol from the same forecourts and it is booked to the same class, so a road inventory anchored on that control silently includes stationary domestic generation, and a residential inventory that also estimates generator fuel counts it twice.

The residential sector derives the quantity—$56.33\,\mathrm{kt}$ of petrol in FY2023–24—from HIES generator-fuel expenditure divided by the administered pump price, and the transport sector subtracts exactly that figure from its road petrol control by calling the residential module rather than by holding a copy of the number. The two cannot drift apart, and the direction is the

one that matters: the fuel is not deleted, it is reassigned from 1A3b to 1A4b, so the national total is unchanged and only the attribution moves.

This closure replaced an earlier one. Until the transport sector was re-anchored on the petroleum statistics' own customer classes, it took a single national diesel supply figure as its control and subtracted the railway fuel account from it. That subtraction is now unnecessary—Railways is its own customer-class row, outside the Road row by construction—so the railway account was freed to do something better, and appears in Section 6 as an independent cross-source check rather than as an internal correction. A closure that becomes structurally impossible is a closure that has been superseded by better source data, which is the outcome to want.

4.8.3 One waste stream, partitioned twice

Municipal solid waste generation is a single national quantity—$31.56\,\mathrm{Mt}$ a year—that must be divided between the waste that reaches a disposal site and the waste that does not. Collection is $50.3\,\%$, so $15.88\,\mathrm{Mt}$ enters the solid-waste disposal model as the current year's addition to the first-order-decay stock, and $15.67\,\mathrm{Mt}$ remains uncollected, of which $60\,\%$ is openly burned.

The partition is two-stage, and the second stage does not close into two parts. Generation splits into collected and uncollected, $G = D + U$; the uncollected splits again into what is burned and what is not, $U = B + O$, and O—$6.27\,\mathrm{Mt}$ dumped to land and waterways—is a declared unresolved fate rather than a category. Reporting disposal and burning as though they exhausted the stream would imply an inventory of *everything* generated, which this is not.

The two categories are computed from one generation figure and the partition is asserted to be exhaustive: disposed plus uncollected equals generated, exactly. The failure this prevents is the natural one. Open burning and solid-waste disposal are usually estimated from separate literature—a per-capita burning rate here, a landfill tonnage there—and nothing then forces their sum to be a quantity of waste the country actually generates. Under this construction, an argument about the collection rate moves emissions between two categories without changing how much waste exists.

4.8.4 One diesel total, partitioned between on-road and off-road

The same road-class diesel row supports tractors, agricultural plant and—through the source data's residual bucket—construction machinery, which IPCC reports under 1A4c ii and 1A2 gvii and not under road transport. The transport sector's fleet model normalises vehicle-kilometre shares over all vehicle classes *including* the off-road class, deliberately: renormalising after dropping off-road would hand the on-road classes $100\,\%$ of the diesel while off-road separately claimed its own share, counting the same tonne twice.

Mixed off-road machinery is therefore a reported category of $1.73\,\mathrm{Mt}$ rather than the memo item it was, its name declaring the 1A4c/1A2 mixture, and the memo is asserted to be empty so that the same diesel cannot appear in both places. Since the August 2026 re-anchor the road control is OCAC's Road customer-class row, so the bulk non-road diesel claimants (industry, utilities, government, railways) are excluded by the source's own partition, and only the retail-channel agricultural and captive claims remain inside—carried as unresolved status rows of the transport fuel ledger.

4.8.5 The gas ledger, and the double-claim it prices

The natural-gas closure is the sharpest of the four, because its finding is a double-claim rather than a partition: the post-meter leakage factor rides delivered gas that four combustion claims burn *in full*. OGRA's Table 3.31 partitions every delivered molecule by end use, and `outputs/national_gas_closure.csv` joins each class to the quantity the consuming module actually burns—the same publication's number for residential, general industry, cement, captive power and CNG (the build aborts if any pair drifts past rounding), and a printed different-meter reconciliation for grid power, whose NEPRA TOE activity is another publication's meter. On the $18\,685$ million m^3 leak base, about $114\,\mathrm{kt}$ of methane is claimed as leakage against delivered gas—about $1.0\,\%$ of the same whole-gas mass. Until this release the reference booked that gas as combusted *in full*, keeping both claims and saying so; the physical-atmosphere identity is now enforced in the booking itself: the four combustion claims book their gas rows net of the leaked whole-gas share, all three gases, because gas that leaves upstream of the burner tip produces none of them. The leaked molecules stay in the inventory where they physically go, as 1.B.2 fugitive methane—the netting removes a double-claim, not emissions. The administrative fuel-accounting view, in which the full metered delivery is booked as combusted for reconciliation against fuel-sales statistics, survives as the named joint configuration *gross_delivered_gas_accounting* ($0.288\,\mathrm{Mt}$), applied to the gas rows the four modules book net. Two defects in the share's earlier derivation nearly cancelled and are corrected in the variant file: the leaked share divided methane mass by whole-gas mass, and the CO_2 base was imputed from delivered volumes rather than read from the modules. The sharpest inconsistency the ledger surfaced is removed from the reference in this release: through rc66 fertilizer feedstock leaked at a combustion-derived factor while being deliberately outside every combustion module. The factor is defined on gas consumed in combustion equipment, so the ledger now carries the fertilizer class as two rows. Feedstock—$83.5\,\%$ of the class on NFDC's metered feed and fuel gas, and $25.9\,\%$ of the leak base as it stood—is outside the base; process fuel, which is burned, stays inside it, its carbon booked by 2.B.1 on production. The earlier membership is retained and not defended as the fugitives scenario *post_meter_including_feedstock*, which adds $1.195\,\mathrm{Mt}$; the $0.288\,\mathrm{Mt}$ above contains no fertilizer term and is not the size of this correction. The ledger does not close the fertilizer stream's carbon: a gas-carbon potential of $13.2\,\mathrm{Mt}$ to $14.7\,\mathrm{Mt}$ CO_2 at the utility heating-value band stands against $8.08\,\mathrm{Mt}$ gross booked by 2.B.1 on ammonia, published in `outputs/pakemis_gas_flow_crosswalk.csv` as an open difference and assigned to no cause. The identity also holds per draw on the combustion side: the residential, CNG, industrial and captive delivered volumes share one activity draw through the cross-module dependency group `delivered_gas_ogra_3_31`, while the leak side's draws are separately injected and that decorrelation is declared rather than hidden—the grid volume rides its own NEPRA meter, reconciled in the ledger, never the shared draw.

4.8.6 What closure does and does not buy

None of these four operations makes the inventory more accurate in the sense of moving it closer to a measured truth; there is no measurement to move towards. What they buy is that a particular class of error becomes impossible rather than merely unobserved. A tonne of diesel cannot be burnt twice. A kilogram of excreted nitrogen cannot both leave the manure balance

and fail to arrive in the dung ledger, as household fuel or as its typed unresolved remainder. A tonne of municipal waste cannot be both landfilled and openly burned, nor can it be neither.

It is worth being precise about the residual risk, because closure invites overconfidence. An exhaustive partition guarantees that the parts sum to the whole; it says nothing about whether the split point is right. The waste collection rate of $50.3\,\%$ carries real uncertainty, and moving it moves emissions between two categories with quite different methane yields. The off-road class inherits its source workbook's residual bucket—the catch-all "Other vehicles" class maps into it wholesale—so its magnitude is a genuine carve-out of the fuel total while its composition is soft. These are stated as limitations in Section 7, and the closures are what make them visible as questions about a split rather than invisible as disagreements between two independent estimates.

4.9 Uncertainty, and What Correlation Does to It

Uncertainty is reported three ways: IPCC Approach 1 error propagation per sector, a pooled Approach 2 Monte-Carlo simulation over every component in the inventory, and the same simulation run again with the shared emission factors correlated. The third is the one with a finding in it.

4.9.1 Approach 1 and Approach 2 diverge, and the divergence is the finding

Approach 1 combines each component's relative uncertainty in quadrature, treating correlated components as summing linearly within their group before the squaring. Approach 2 accumulates $100\,000$ realisations of the national total—and it is a MIXTURE, not one family: the waste and fugitives modules inject their own sector engines' persisted draws (waste's ledger-true parameter rerun, whose fate identities hold in every realisation; fugitives' bounded split-triangular), while every other module draws mean-preserving lognormals. The draw files, their seeds and their SHA-256 hashes are in the release manifest and `outputs/national_uncertainty_sources.csv`; the published run uses ten times the sector default because the paper draws a conclusion from a difference between two runs, and at $10\,000$ draws the reported half-widths carry a Monte Carlo standard error of the same order as that difference.

The two no longer agree closely, and the gap is the finding: $10.9\,\%$ from Approach 1 against $11.72\,\%$ from Approach 2 at the national level—quadrature is the analytic limit of the simulation only under symmetric, independent, moderate uncertainties, and the injected engines are deliberately none of those. The honest national statement is asymmetric, $-10.3\,\%/+13.1\,\%$ about the median (`rel_down_pct_95`/`rel_up_pct_95` in the published table), and a single $\pm$ figure is printed only as a summary. The interval has two legitimate reference points and they are not interchangeable: the same 2.5th–97.5th percentile band ($400.30\,\mathrm{Mt}$ to $504.97\,\mathrm{Mt}$) read against the *deterministic reference* $444.96\,\mathrm{Mt}$ is $-10.0\,\%/+13.5\,\%$, because the deterministic total sits below the Monte-Carlo median ($446.52\,\mathrm{Mt}$) and mean ($448.14\,\mathrm{Mt}$)—the injected engines are right-skewed, so re-running the accounting at parameter means does not land on the distribution's centre. The recommended point estimate is the deterministic reference $444.96\,\mathrm{Mt}$ for accounting and comparison against other inventories, and the median for any probabilistic statement; a reader attaching the about-the-median percentages to the deterministic total is off by roughly a point in each direction, which is why both readings are printed.

And this interval is the narrowest statement the release makes, which is the wrong one to borrow. It is parametric and annual and national: it prices what the factors and the activity data could be, and prices nothing about *when* or *where* the mass was put. Both of those carry their own uncertainty, and neither is probabilistic—a timing or placement bracket is a disagreement between two defensible readings of the same evidence, and no probability attaches to a spread. So they are not summed into σ. `outputs/pakemis_joint_uncertainty.csv` publishes the three components separately, per category and per the *resolution of the figure being read*, together with an envelope—a bound, not a distribution—that widens the parametric interval by the structural half-widths. A bracket only bites on the figure it can move: re-allocating a category across districts does not change its national total, and re-profiling it across months does not change its annual one. Nationally the bound widens from $-10.37\,\%/+13.14\,\%$ annually to $-23.3\,\%/+26.10\,\%$ for a district in a month. The four resolutions form a lattice rather than a chain, because adding a month is not comparable to adding a district. Timing is priced for seven categories of twenty (six on measured dispersion, drawn in the ensemble) and placement for six, and every row names what its envelope therefore omits; filling those cells with a weak alternative would make the bound look rigorous and be worse.

Within a category the parametric error is perfectly correlated—one emission factor moves every district and every month of it together—so intervals add *linearly* across districts and months and never in quadrature, and the gridded products deliberately carry no per-cell interval that would invite the opposite. Across categories the simulation has already measured the dependence, and adding the category intervals instead inflates the national bound to $-22.0\,\%/+45.3\,\%$: not a conservative reading of the evidence but a different and wrong one, asserting a coupling the correlation model denies. `docs/uncertainty_contract.md` states both rules and `tests/test_joint_uncertainty.py` proves each by worked example.

None of this is validation, and the two are registered separately. A wide interval that contains the truth and a narrow one that does not are indistinguishable from inside the model, so `outputs/pakemis_validation_catalog.csv` records what has been compared against what: 102 entries across the three properties an inventory can get wrong independently—50 magnitude, 20 timing, 32 geography—because a national total agreeing with EDGAR does not validate a map and a satellite hotspot at a facility does not validate a total. Every entry declares how independent its evidence was, and the answer is usually not very: 20 carry *large* independence, 45 only *partial*, and 8 *none* at all, meaning the comparison shares the activity data it is testing (a further 10 are by-construction checks to which independence does not apply). 11 rows are declared absences—a module and property with no independent check, named rather than left blank, because a cell with nothing in it reads as a property that was never at issue. Six absences in the register this catalogue replaced were found to be false: the benchmarks had been scoring them all along and nobody had joined the two registers, so the register understated what the project had actually checked.

Release 1.0.0rc33 unified that register with the frozen benchmark tasks into this one catalogue (the older register itself was retired in 1.0.0rc35), in which every validation asset declares its grain, its producer, and how it is wired into the build; a validation product regenerated by the build itself must now be at least as new as the build's first output or the build fails naming it, which on first contact exposed one quality-cited check that nothing in the build produced and one whose producer depends on a model held outside this repository. The catalogue's verdicts, previously six incompatible vocabularies, are projected onto one closed set at read time. Its summary, `outputs/pakemis_validation_scorecard.csv`, tabulates the full property-

by-grain space—magnitude, timing and geography at twelve grains from national to worksheet cell— and 14 of its 36 cells are covered today; the remaining cells are written as declared absences, so the scorecard is simultaneously the record of what has been validated and the roadmap of what has not.

Three checks added this release each came back *narrower* than they were built to be, and the narrowing is recorded rather than rounded off. ERA5, on an independent reanalysis, tracks the NASA POWER temperature behind the residential profile almost exactly in shape—correlation 0.969 to 0.989 across five cities and 54 months—and disagrees on level by a systematic 3.2 °C to 6.1 °C. That offset is definitional: the held grid carries one value per cell per day and no metadata saying which value, so it validates the seasonal *timing* and cannot validate the degree-day series itself, degree days being a function of level (`outputs/era5_degree_day_crosscheck.csv`). APCMA's cement despatches, compiled by the industry association rather than PBS, reproduce the QIM production shape to a mean 0.22 percentage points of the year over 60 months (`apcma_qim_cement_crosscheck.csv`)—a real corroboration of the shape cement now rides, on magnitudes that are deliberately not compared because despatches are sales. Its feedstock corroborates it from a third direction: PBS limestone production tracks the QIM cement shape to a mean 1.20 percentage points of the year over their six overlapping years (`limestone_qim_cement_crosscheck.csv`), once one flagged source anomaly is set aside—November 2016 carries 36.7 % of its year's limestone in a single month, a defect in the mines-department series that is published and flagged rather than dropped. That parse is gated on three of the publication's own controls—each row's printed total, each block's annual total, and agreement between the two pages that publish every year—and the cells failing one are kept unrepaired with their raw text (`apcma_dispatch_quarantine.csv`, `apcma_dispatch_parse_summary.csv`). They include a published digit-grouping error, an annual export total printed as 202 989 where its own twelve months sum to 2 020 569, and small restatements between the two editions of a year. And FAO's rice calendar agrees with three of four provincial harvest windows (`fao_crop_calendar_crosscheck.csv`), which changed no provenance class, because the methane profile rides the *flooded* window and harvest is not one of its endpoints.

The divergence localises the assumption doing the work. The completed fugitives ledger gives ±37.7 % under Approach 1 and ±19.0 % under the injected Approach 2—NARROWER, because the sector engine's bounded split-triangular draws respect the published range endpoints where the old lognormal surrogate's tails did not. Its Monte-Carlo median is 24.15 Mt, its mean 24.73 Mt—the sector engine's own quantities, reaching the national row unconverted for the first time.

4.9.2 The correlation question

Six of the ten sector modules report CO_2 under an IPCC 1.A code, and they draw it through the same default carbon-content factors. Transport (both its road and its off-road categories), electricity generation, industrial fuel combustion, residential fuel, railways and domestic aviation all take their fuel CO_2 factors from the same table entries with the same published uncertainty ranges. The second case groups those components by the physical factor each one uses: every row computed with the gas/diesel-oil carbon content draws one realisation of it, every row computed with the natural-gas content draws another, and the two are independent of each other. **It is a declared covariance sensitivity, not an estimate of the true dependency structure**—see the

limitation stated at the end of this subsection. Process CO_2 from cement, fugitive CO_2 from venting and flaring, open-burning CO_2 in waste, and agriculture's urea and liming are untouched, because none of them is fuel combustion.

An earlier edition of this section described a single correlation group spanning all six modules, which is what the first implementation did. That is not a defensible physical claim: it asserts that diesel, petrol, natural gas, furnace oil, coal and jet kerosene are all wrong together, in the same direction, by the same relative amount. It also inflated the correlated band, and the inflation was most of what the paper had reported as a headline finding.

The membership is derived rather than declared, and that is a repair. Industrial fuel combustion was outside this group for the whole of the release in which it was added: the sector list was a hand-written literal that no test compared against the inventory, so a module burning natural gas and six petroleum products through exactly these table entries was silently treated as uncorrelated with the rest. The group is now computed from the components' own IPCC codes—a sector qualifies if it reports CO_2 under 1.A—and a test asserts the declared list equals the derived one, so the next fuel-burning module cannot be omitted the way this one was.

Treating those draws as independent asserts that the true diesel carbon content could be at the low end of its range for railways and at the high end for road transport simultaneously. It cannot. There is one carbon content of the diesel that Pakistan burns, and if the IPCC default is wrong about it, it is wrong in the same direction everywhere at once.

Almost no published national inventory states which assumption it made. Approach 2 as described in the Guidelines permits correlation but does not require its declaration, and the Approach 1 formula makes independence the path of least resistance. We ran both and report the gap as a result in its own right.

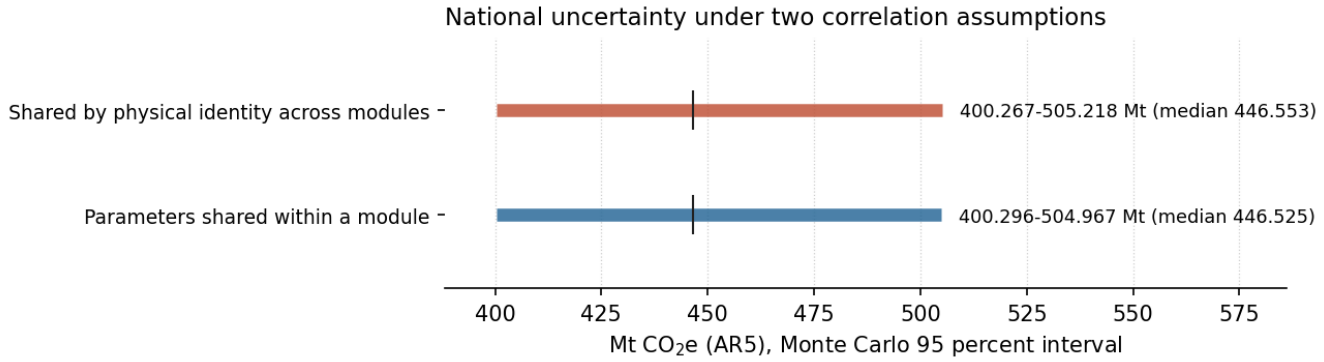


Figure 5. 95 percent Monte-Carlo intervals for the national total (2.5th to 97.5th percentile, median ticked) under the two correlation assumptions. The two bars are nearly identical, which is the finding. Sharing a physical factor prevents the contribution of that factor from averaging independently across the rows that use it, while preserving each row's other activity, conversion and residual uncertainties.

4.9.3 Reading the gap

Three things follow, and the first is a correction to what this paper previously reported.

	Level	Independent	Correlated
National mean (Mt CO_2e)	national	448.14	448.15
National median	national	446.52	446.55
2.5th percentile	national	400.30	400.27
97.5th percentile	national	504.97	505.22
National ±%	national	**11.72**	**11.75**
−/+% about median	national	10.3/13.1	10.4/13.1
−/+% about det. ref.	national	10.0/13.5	10.0/13.5
Transport module ±%	module	12.93	12.93
Road transport ±%	category	13.26	13.24
Power module ±%	module	8.34	8.34
Residential ±%	module	30.45	30.41
Railways ±%	module	15.27	15.13
Agriculture ±%	module	21.48	21.48

Table 6. Monte-Carlo uncertainty (Approach 2, 100 000 iterations, seed 42), with the IPCC fuel-carbon factors drawn independently and then with **one draw per physical factor**—diesel, motor gasoline, natural gas, residual fuel oil, coal, jet kerosene, other kerosene and LPG. Every row states its aggregation LEVEL, because the previous edition reported 11.54 in this table and 11.96 in the prose beneath it and both were called "transport": one is the module, which includes mixed off-road machinery, and the other is the road-transport category. The power module barely moves, and that is a disclosure rather than a null result. Its grid rows carry the TOE-to-terajoule conversion *and* their fuels' carbon contents as separate parameters, so they participate in both; what does not move is the balance between them, since the conversion is shared inside the module under either scope. An earlier implementation gave each row one group and overwrote this one, which made the correlated case markedly NARROWER than the independent one—and *positive* shared-factor dependence should not reduce the variance of the affected non-negative sum relative to otherwise identical independent draws. That is the form of the claim: correlation in general can reduce a sum's variance, when it is negative. Every group's draws are anchored to a representative member row (common random numbers), so the two columns share their randomness and differ *only* in dependency structure; each remaining movement sits inside three batch-means standard errors of zero, and the largest anywhere in the table is railways' 0.18 points, where cross-module sharing of the diesel carbon content is a genuine structural difference. The overwrite was forty.

At the national level the effect is below the reporting precision: ±11.72 % either way. An earlier edition put every fossil-CO_2 component of every fuel-burning module into ONE correlation group and reported the national band widening by a fifth. That grouping is not a physical claim. It asserts that the IPCC carbon contents for diesel, petrol, natural gas, coal, furnace oil and jet kerosene are all wrong together, in the same direction, by the same relative amount. Grouped per factor—which is what the shared-default argument actually supports—the eight factor groups are independent of one another, their variances add in quadrature, and the national effect largely cancels. The finding as previously stated was an artefact of the grouping.

One coupling was, however, genuinely missing rather than over-asserted, and closing it is what brought the two scenarios onto the same reported figure. Urea CO_2 is *derived* from the audited national nitrogen offtake—the same series that drives the synthetic-fertiliser soil pathways—yet the scope registry named that one series twice, once per receiving category, so a draw moved managed soils and urea application independently. No physical reading supports that: if the national offtake is wrong, it is wrong for both. Scoping it as one parameter across both categories is that change, and it widens the agriculture band by about a hundredth of a percentage point—the direction the supplement predicted, at a magnitude far below what any reader should act on.

At parameter level the effect becomes negligible and is indistinguishable from Monte Carlo sampling noise at the reported precision. The transport module reads $\pm 12.93\,\%$ under both correlation structures and the road-transport category from $\pm 13.26\,\%$ to $\pm 13.24\,\%$—a movement smaller than the batch-means standard error published beside its band. The earlier edition reported the road category widening from 11.96 to $18.46\,\%$ on this assumption, and that was an artefact of the model rather than a property of the inventory: a component could carry only ONE correlation group, so putting a row in the fuel-carbon group made its *entire* variance move with the factor. The carbon content of diesel carries about $2\,\%$; a road class row carries thirty to sixty. Correlating two points of a fifty-point variance across rows cannot widen their sum, and once the row's activity, conversion and factor are carried separately it does not.

The road-transport band is wider than the previous edition published—13.3 against $12.0\,\%$—and that is the same correction seen from the other side. The OCAC Road control is genuinely one quantity behind every class row, and the old model had no way to say so while the fuel factor occupied the row's only correlation slot.

Once uncertainty is decomposed by parameter, cross-module sharing of the same physical fuel-carbon factors has a negligible effect on both national and transport intervals. The earlier large effect was created by moving the whole component variance with a factor representing only a small portion of that variance. The argument for publishing the correlation model beside every band survives the correction: the two scenarios answer different questions, and only reporting both shows that the answer no longer depends on which one is asked.

The correlated case is a *bound*, not a better estimate. Perfect correlation is as strong an assumption as perfect independence, and the truth is between them: the six modules do share an IPCC default, but they burn physically different fuel batches with genuinely different carbon contents, and their activity uncertainties remain independent. We report both endpoints and choose neither, on the same principle applied to the waste sector's disputed methane correction factor. The honest statement is that the national uncertainty is $-10.3\,\%/+13.1\,\%$ about the median (symmetric summary $11.8\,\%$; $-10.0\,\%/+13.5\,\%$ about the deterministic reference) under either assumption, and that what the assumption is worth at national scale is now smaller than the Monte Carlo noise on the estimate.

What this exercise is, and what it is not. It is a *declared covariance sensitivity*: two named dependency assumptions, run and reported, with the difference between them attributable to the assumption. It is not an estimate of the true covariance structure of the underlying activities and factors, and it should not be read as a confidence bound on that structure.

What the two scenarios vary is *where sharing stops*. A component's draw is the product of one multiplier per named parameter that enters it and a residual multiplier private to the row, so a power gas row draws the TOE-to-terajoule conversion, the IPCC carbon content of natural gas and its own metered activity as three separate quantities. Under the within-module scope every parameter is qualified by its module, so the carbon content of diesel is treated as a different quantity in railways and in road transport—which is what an inventory compiled sector by sector implicitly assumes. Under the cross-module scope it is one quantity everywhere.

An earlier edition of this machinery gave each component a single correlation group. That forced a choice between a row's conversion and its factor, and excluded the power module from the fuel-carbon scenario altogether because its rows were already spoken for by the TOE group. That limitation is gone. The one that remains is coverage rather than architecture: a

parameter can be shared only once it has been named, and the registry does not yet name every quantity behind every row. Section 4.9.4 reports how far it reaches.

4.9.4 Parameter coverage, and what the variance is attributed to

Sharing parameters requires naming them, so the natural question is how much of the inventory the registry reaches. The answer should be read as a statement about the registry rather than about the physics.

Of the 202 component rows, 202 carry at least one named parameter and 200 carry more than one—the case the previous machinery could not express at all. The registry holds 672 parameters; 141 dependency groups carry variance in the published run, and 92 of those enter more than one row, which is what makes correlation possible at all. The largest single addition is agriculture's factor table (`outputs/agriculture_uncertainty_inputs_2024.csv`): the sector composed its IPCC factor uncertainties inside each row's declared width, where the registry could not see them, and naming them—the same values, the same widths, no new numbers—is what brought them inside it. Each row's marginal is decomposed as a sum over its named parameters plus a residual: where the declared row width covers its parameters, the residual closes the identity exactly and the marginal is the declared one; where the named parameters imply *more* variance than the row declares, the parameter graph is authoritative—the marginal is derived from the parameters under a per-row disposition in `data/uncertainty_reconciliation.csv`, and the declared width is published beside it as a comparator. The audit register carries 59 dispositioned rows: the original twenty-three were rows a previous edition clamped silently, and the audit found most of them one resolver defect (a median summarising a multi-member group with another row's width); after its repair 37 rows remain derived at the reference year, each carrying a named cause—a declared width that omits terms the sector's own input table carries, or a sector width propagated through a nonlinear model the linear quadrature legitimately disagrees with. Seven of the dispositions are the naming's own find: HISTORICAL panel years compose the FAOSTAT projection's activity widths while the registry holds the census year's, so those rows derive from the graph in the panel years and reconcile exactly at the reference—the vintage difference published, not clamped. The twelve newest are the manure and managed-soils N_2O rows: once the herd and crop draws were scoped to the streams they physically drive, their declared widths—the emission factor's alone—were found to omit the activity the engine multiplies in, and the graph now carries them at 107 % to 124 % against the 101 % to 114 % they declared.

Which quantities drive the national uncertainty is read from the released ensemble itself, not from a decomposition fitted to it. `outputs/pakemis_uncertainty_drivers.csv` gives each category's and each module's share of the variance of the national total across the 10 000 released draws—the covariance of the category with the total over the variance of the total, so the shares sum to one—its share of the excess in the top and bottom 2.5 per cent of draws, and the national interval with that one category held at its deterministic value. Enteric fermentation carries 59.8 % of the variance, managed soils 14.0 %, domestic wastewater 9.0 %, manure management 5.6 % and rice cultivation 4.0 %; by module, agriculture carries 83.5 % and waste 11.1 %. The upper tail has the same order: of the excess in the top 2.5 per cent of draws, enteric fermentation is 60.1 %, managed soils 14.7 % and domestic wastewater 10.5 %. With enteric fermentation held fixed the national interval narrows from 399.8 Mt to 505.4 Mt to 420.3 Mt to 483.9 Mt.

The product carries one parameter row. The wastewater effluent factor, frozen at its default in a rerun of the waste engine on the same draws, carries $4.1\,\%$ of the national variance and $4.8\,\%$ of the upper tail. Parameter-level importance is not otherwise claimed. Earlier editions ranked named parameters by Sobol indices estimated on a surrogate of the simulation; that decomposition, its estimator sweep and its convergence table are retired, with every figure drawn from them.

4.9.5 What the interval leaves out

The interval above is *conditional on one structural configuration*, and that configuration is larger than any one axis of it. The residential dung basis is a single axis—which reading of the dung evidence the residential sector takes, the household survey in the reference—while the $444.96\,\mathrm{Mt}$ total rests on 23 such choices, of which it is one. Configuration L, the alternative on that axis, is a first-class key rather than a delta: the annual product carries its residential rows under `configuration_id`, beside the reference's typed dung residual, its spatial products are gridded through the same dispatch as the reference: `pakemis_grid_0p05_monthly_ledger.nc`, `pakemis_adm1_ledger_2024.csv` and `pakemis_adm2_ledger_2024.csv` at stream × gas × month, differing from the reference in the two dung-cake streams' masses and in no other cell, and `pakemis_ensemble_ledger_summary.csv` prices it on the same `draw_id` as the reference ensemble, so the configuration delta is a per-draw difference rather than two unrelated intervals. They are recorded as a scenario registry, `REF_2023_24` in `data/scenarios.csv`, and the release manifest carries the whole object rather than the letter.

Figure 6 prices each published alternative and draws the one unpriced axis as a labelled empty slot; `outputs/national_axis_register.csv` is the definitive one-row-per-axis table (family, reference setting, deltas, pricing status, and the reason where unpriced). The coal energy basis in power is worth $1.6\,\mathrm{Mt}$, the sewer methane correction factor $9.1\,\mathrm{Mt}$, and the fugitive practice assumption $8.6\,\mathrm{Mt}$. Each of those is *smaller* than the parametric interval, which is about $52\,\mathrm{Mt}$ at the 95th percentile—and *none of them is inside it*. That is the point, and it is not the same point as being larger: the interval is conditional on the reference configuration and does not vary these axes at all, so their effect is not represented in it at any width. A reader given only $\pm 11.7\,\%$ has no way to see that they exist.

The same pricing ships in a second, policy-facing form: `outputs/national_configurations.csv` names nineteen single-axis configurations—the reference, seventeen alternatives a policy reader would ask for by name (the aviation scope fork ships as three of them: military attribution, general-aviation activity and the bridge's unresolved residual are separate evidence questions, priced separately; the post-meter feedstock scope returns fertilizer feedstock to the leak base, the membership of every release through rc66, retained and not defended, and the newest is the power coal calorific basis, which asks the coal-energy question a second way with no heat rate anywhere in it and so, unlike the member beside it, does not divide the generation back out of an energy that contains it), and one reporting recode whose delta is zero by construction—each the scenario with exactly one axis moved to its published alternative reading, resolved through the same machinery as the bridge so the two products cannot disagree. Eight further rows are declared exceptions to the one-axis rule, each labelled as the exception it is: the fugitives envelope floor (seven axes at their minimal legs at once), the cross-sector gas deduction (a net-of-leakage adjustment no single module's scenario table can carry), the central practice reading (a third reading of the fugitive-practice control, which the one-alternative-per-axis registry schema cannot hold), and the five scope screenings whose

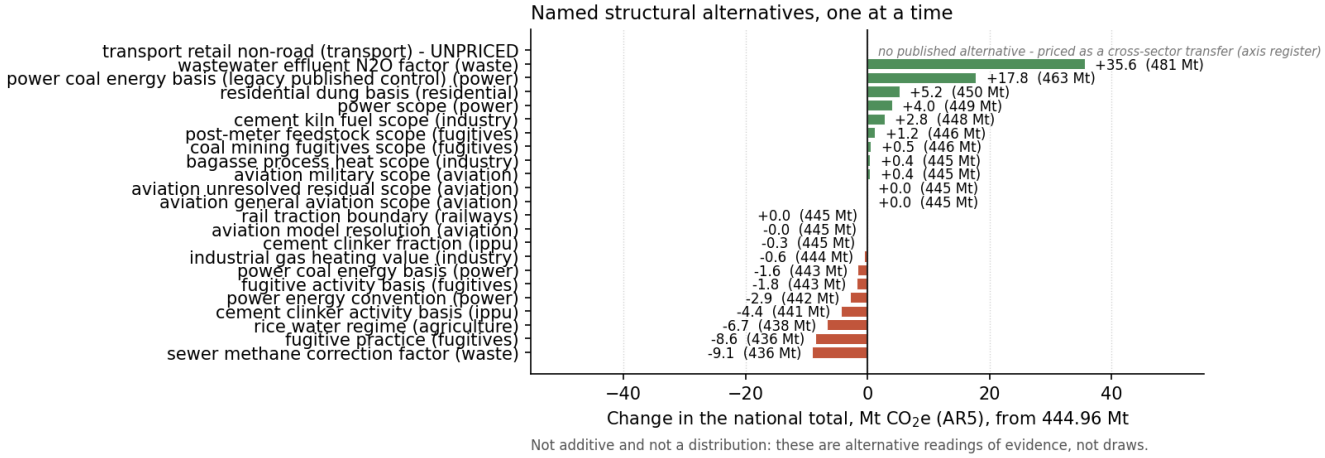


Figure 6. The deterministic effect of each named structural alternative on the national total, one at a time, from `outputs/national_structural_bridge.csv`. **The bars are not additive and are not a distribution.** Two of the power axes describe the same coal energy in different ways and cannot both be taken; none carries a probability, because assigning one would be the fiat this inventory declines. The transport retail non-road axis has no single published alternative and is priced as a cross-sector transfer in the transport supplement instead.

categories sit outside every module's own scenario table—the F-gas consumption bound, the commercial/institutional floor, nitric acid unabated and at its last verified abatement, and the ammonia gas-balance ceiling. The reference now books the stations' own metered coal at IPCC rank defaults, so the members of this family describe the accounts it replaced. If the coal energy is instead taken at the calorific values the plants' own generation implies—the heat-rate route, whose energy contains the generation it is later divided by—the total is 443.4 Mt (*power_plant_physical_coal*); on NEPRA Table 3's published conversion, the reference for every release before the coal re-base and retained as a named control because it implies net fleet efficiencies outside 28–47 per cent in seven of nine years—it is 462.7 Mt (*power_table3_legacy_control*); if the Tier 1 clinker model at the 0.75 fraction were taken in place of the producer census, it is 440.5 Mt (*cement_tier1_model_low*); with the K-Electric system's fuel added to the power boundary it is 449.0 Mt (*power_with_kelectric*), a coverage alternative priced beside the reference and not added to it: whether the national fuel statistic already contains K-Electric's fuel is not established, and if it does, adding it would count that fuel twice.

The file also carries a structural grouping: the power alternatives and the reference are labelled ONE FAMILY (`family_id power_grid_control`), because they are competing definitions of the same grid control—38.45, 40.04, 44.05 and 57.82 Mt at the sector level—rather than independent axes. They are not alternatives of one kind, and the file's `family_kind` column says which each is: 38.45 is a fuel-property alternative on the reference's own fleet, 44.05 a coverage alternative that adds the K-Electric system's fuel, and 57.82 a legacy published account retained for traceability. A range may be formed only within one kind.

The selection rule changed with this release, and the previous one rested on an argument that does not hold. Every release until now took the regulator's published national statistic, and the declared reason was that Pakistan's UNFCCC filing

agreed with it on solid fuels to 2 per cent, which was held to be standing counter-evidence against the plant-implied coal case. That agreement is not evidence: the filing is compiled from the same energy balance the regulator reports, so the two agree by provenance, and `data/validation_catalog.csv` has carried that row at independence class `none` throughout. The rule now rests on a test neither account can satisfy by construction. Dividing each candidate coal energy into the regulator's own printed generation gives an implied net fleet efficiency; against the 28–47 per cent envelope the published conversion lands inside in TWO of nine panel years and the generating stations' own metered coal at IPCC rank default calorific values lands inside in SEVEN, with the remaining two straddling the envelope rather than falling outside it. Those two are FY2016–17 and FY2017–18, the years whose commissioning coal is bounded rather than printed, and the straddle is a property of that interval: it is bounded by the intensity range the same two stations run at over the ledger years, which includes a sharply curtailed year, and a commissioning plant is a part-load plant, so the low-intensity end of that range is the relevant one rather than an outlier to discard. A narrower band would place the whole interval inside the envelope, and would do so by excluding the most analogous observation available. Neither side of that test touches a heat rate, so neither restates the regulator's own print—which the third candidate, the heat-rate route at $38.45\,\mathrm{Mt}$, does by construction and is why the re-base did not go there.

The reference is therefore $40.04\,\mathrm{Mt}$, and the conversion it replaced is retained as `power_table3_legacy_control` at $57.82\,\mathrm{Mt}$ rather than deleted, so a reader holding a total published before this release can find the account it came from. It is not equally probable with the reference and the registry says so, which is also why it is excluded from the priority ranking of unresolved choices. The per-year moves, and the mass and calorific terms whose product is each year's ratio, are published in `outputs/pakemis_power_coal_rebase_ledger.csv`; the four candidate accounts with their efficiency verdicts are in `outputs/pakemis_power_coal_configurations.csv`. Every quote of the grid figure, here and in the results section, is conditional on that rule with the alternatives named beside it; the structural bridge carries the same `family_id` tag on its power axes.

One constraint on those candidate explanations is new here and narrows all of them at once. Each supposes a constant difference between the regulator's two tables—a calorific convention, a gross basis, a scope, a stock change—and a constant difference has a constant sign. Across the five years the plant ledger spans it does not. `outputs/pakemis_power_coal_energy_adjudication.csv` divides each energy figure by the coal generation the regulator itself reports, a test that uses no tonnage and so cannot be undone by an error in one: Table 3's energy implies a fleet net thermal efficiency of $57.5\,\%$ in FY2020–21 and $18.9\,\%$ in FY2023–24. No thermal fleet spans that range, and a plant set that is too wide makes energy too high in every year rather than too low in one. Over the same years the plant-level route implies $32.4\,\%$ to $45.7\,\%$, which is a fleet.

The break is dated, and it is not a revision between editions: the 2024 edition prints FY2018–19 to FY2021–22 side by side in the same table, and the 2025 edition reproduces those columns exactly. Across two adjacent columns of that one table the coal tonne-of-oil-equivalent figure rises $87.5\,\%$, from $4\,123\,008$ to $7\,731\,514$ TOE, while the same report's per-station table has coal mass falling $7.3\,\%$ and coal generation falling $3.5\,\%$.

It *is* a conversion, and over a panel longer than the plant ledger it is a constant one. Table 3 states that it takes these figures from the HDIP *Pakistan Energy Yearbook*, and HDIP's own power-sector coal tonnage is held here as well, republished in the

Economic Survey's statistical supplement. Against *that* tonnage Table 3's energy works out at 0.447 tonnes of oil equivalent per tonne in every year from FY2015–16 to FY2020–21—six consecutive years spanning $0.000\,072$ TOE per tonne, or $0.016\,\%$—and 0.604, 0.511, 0.478 thereafter. `outputs/pakemis_power_reference_audit.csv` carries the span year by year, and `outputs/pakemis_power_fuel_reconciliation.csv` carries the plant, fuel and year account the constant is measured against: coal mass, rank, generation and the energy they imply, for all nine years rather than the five the per-station table reaches, with the edition and table each number was read from and an interval wherever no edition prints a mass. That constant is $18.73\,\mathrm{TJ\,Gg^{-1}}$: one generic coal factor, applied whatever the fleet happened to be burning. It breaks in FY2021–22, and it breaks *upward*—to 25.27, then 21.40 and 20.02—over exactly the years the fleet's own physical calorific value falls from 20.34 to $13.01\,\mathrm{TJ\,Gg^{-1}}$ as Thar lignite scales. A measured energy cannot move the wrong way against its own fuel. An earlier version of this section read the first two of those six years and concluded the opposite; the panel simply began too late to see the constant. What *also* moves is the ratio of the two tonnage accounts: HDIP's national series against the regulator's own per-station quantities runs 0.88, 0.70, 1.05, 1.11, 1.21, crossing unity between the two years in question. A scope difference cannot cross unity. A consumption-against-receipts difference could, in either direction, and nothing held settles which.

A physical screen then narrows them, though it does not choose between them. Taking each tonnage account at the fleet's own rank composition—IPCC default calorific values, no heat rates—and dividing the electricity out of it, the per-station account gives $34.4\,\%$ to $38.7\,\%$ in every one of the five years: a fleet, inside the band throughout. HDIP's national series gives $49.0\,\%$ falling to $29.1\,\%$, outside at both ends. Of three accounts of the same coal, one is physically admissible across the panel and two are not. That screen is a plausibility filter on a candidate energy level and *not* independent corroboration of the account it favours: the efficiency it reports is the electricity divided by exactly the energy that route constructs, which is an identity rather than a comparison, and the audit above recomputes it year by year to show so. It can reject an energy level as unphysical; it cannot adjudicate between two tonnage accounts that imply the same energy.

`outputs/pakemis_power_coal_configurations.csv` states the four candidate coal accounts year by year with what each depends on, and the column that matters is whether the calorific value was derived from generation. One of them was: the $38.45\,\mathrm{Mt}$ alternative this paper names beside the reference takes heat-rate-implied calorific values, so its energy contains the generation it is later divided by and its implied efficiency restates the regulator's own print rather than testing it. The rank-default route, which uses no heat rate and is the only candidate independent of the electricity it is judged against, was published nowhere before that file. Its ratio to the published energy runs from 0.47 to 2.00 across the panel and crosses unity twice, and decomposes exactly into a tonnage term that stays within 0.74 to 0.99 and a calorific term that runs 0.63 to 1.38: the sign change is the conversion, which is why no single correction factor exists. That the composition is genuinely moving underneath all three is not in doubt: the fleet's lignite share by mass goes from $27.6\,\%$ in FY2020–21 to $39.3\,\%$ in FY2021–22 and reaches $92.0\,\%$ by the reference year, which is why a calorific value that rises over the same span is the wrong shape whatever tonnage it is divided by. A third route, taking the same tonnages by rank at IPCC default calorific values and no heat rate at all, falls with the fleet as the plant route does.

None of this overturns the selection. The official reporting filing agrees with the regulator's coal energy to within $2\,\%$, which is the standing counter-evidence the rule already cites, and this inventory continues to take the published national statistic.

What the constraint establishes is narrower and, for a reader deciding how much weight the $40.04\,\mathrm{Mt}$ figure carries—against the 38.45, 44.05 and 57.82 alternatives named beside it throughout—more useful: whichever reading is right, it is not one that holds constant across the panel, so the open question is not only *which* table to believe but *when* its basis changed. No probability attaches to any row—the schema forbids the column—and the rows do not compose: each is one axis off the reference, not a corner of a scenario space.

One axis carries a finer question than its name, and it gets its own table. Under the reference the household survey observes $10.93\,\mathrm{Mt}$ of dung cake against the ledger's $58.74\,\mathrm{Mt}$ diversion, and mass conservation cannot say where the other $47.81\,\mathrm{Mt}$ went—conservation proves the mass exists, not its destination. But the candidate destinations are not interchangeable: each has different CH_4, N_2O and biogenic-CO_2 consequences, so the survey reading alone under-describes the national reading. Table 7 prices each destination declared in `data/scenario_variants.csv`, with no probabilities. The non-household-combustion variant lands on the L total exactly, because the residual burning anywhere on the same fuel chain emits what L counts—a reclassification, not a new quantity—and that equality is asserted by test rather than observed. The survey reading carries its own uncertainty, and the reference prices it: the dung stream is drawn at $35\,\%$ activity uncertainty, the width firewood carries.

Destination of the 47.81 Mt residual	**Δ vs reference**	**National**
Declared unresolved stock (the reference)	0.00	444.96
Non-household combustion, same fuel chain	5.25	450.20
Return to manure management	4.49	449.44
Field application (soils N_2O)	3.49	448.44
Diversion fraction revised down	4.49	449.44

Table 7. National consequences of each destination for the reference's unresolved dung residual, Mt CO_2e (AR5), from `outputs/national_dung_variants.csv`. Every variant conserves the diverted mass exactly (asserted at write time); none carries a probability. Return-to-manure and revised-diversion price identically because the arithmetic is the same—what differs is which reading of the evidence is being revised, and the published note says so. The residual is worth between 3.5 and $5.2\,\mathrm{Mt}$ depending on where it went, which is why conservation alone cannot close the scenario.

4.9.6 One ensemble, joinable: the common draw_id

Until this release, every uncertainty product answered its own question in its own file and nothing shared a draw: the annual Monte Carlo published percentiles, the monthly layer carried deterministic brackets, the spatial products carried none by declared design. A reader could not ask "in the realisations where the national total is high, which months and which provinces are high?"—and the two closure identities the deterministic products enforce at 10^{-9} were not even *expressible* per draw. The common ensemble makes them expressible and then asserts them inside every one of its $10\,000$ draws: `outputs/mc_draws/pakemis_ensemble_{annual,monthly,spatial}.parquet`, joined by a single `draw_id`, with $\sum_{m=1}^{12} E(d,c,m) = E_{\mathrm{annual}}(d,c)$ and $\sum_{\mathrm{prov}} E(d,c,p) + E_{\mathrm{retained}}(d,c) = E_{\mathrm{annual}}(d,c)$ holding to 10^{-9} in every draw d (the second is the gridded-plus-unallocated identity, carried as a placement axis). Every stream in the national pool is identity-seeded, so draw d of the ensemble is *bit-identically* draw d of the published $100\,000$-draw run—a prefix given a

name, not a second simulation—and a test asserts the bit-identity. The summary surface (`outputs/pakemis_ensemble_summary.csv`) publishes each grain's quantiles with batch standard errors beside the published run's, which is the convergence statement for the prefix.

What is drawn and what is not follows the evidence, per category and stated in a column. Monthly profile weights are *drawn* only where a measured dispersion exists—the hindcast WAPE streams, road transport and railways, whose weights take multiplicative noise sized by the measured error and renormalise to sum to one inside every draw—and are degenerate everywhere else (`profile_uncertainty_status`); the power fuel-mix variant remains a conditional configuration and is never mixed in. Spatial shares are fixed per draw—allocation is declared, not sampled—so the spatial layer's per-draw variation is the parametric mass reaching each province, and the priced allocation alternatives ship as conditional quantile blocks over the same draw_ids (`outputs/pakemis_ensemble_spatial_conditionals.csv`), never as a mixture. A month-by-cell ensemble would be roughly 118 gigabytes and is deliberately absent: the ensemble is marginal by design, category grain for months and province grain for space, and the deterministic weights whose sums-to-one the cube gates already enforce carry each draw the rest of the way. The five kinds of uncertainty this release separates—parameter, profile, placement, structure, completeness—each have a carrier product and an aggregation rule, stated in `docs/uncertainty_contract.md`; the completeness kind rides the joint lattice as a bound (`completeness_bound_Mt`) beside the interval, never a distribution inside it.

Structure gains a time axis the same honest way. The twenty-two registered axes were priced against the reference year only; `outputs/pakemis_structural_trajectories.csv` now re-prices each axis in each of the nine panel years where its engine honestly can—92 axis-years priced on their own years' activity—and refuses the rest with the missing evidence named per row (93 refusals; the producer census that is own-year by construction, the scenario runners that price the reference year only). A refusal is a row, never a silence, and no probability attaches to any point: these are deterministic forks re-read per year, not a trajectory ensemble, and the longitudinal product's refusal to cross structural axes into a probabilistic ensemble stands.

4.9.7 Where the uncertainty actually lives

The national band is much narrower than most of its sectors'—$10.9\,\%$ against waste's $51.1\,\%$ and fugitives' $37.7\,\%$—which is the ordinary behaviour of partly independent errors in a sum. Fugitives now contribute $5.3\,\%$ of emissions and $3.3\,\%$ of Approach-1 variance. Its discrete practice and coverage scenarios remain outside the pooled Monte Carlo, so the national probability interval should not be read as absorbing that structural envelope.

The same caveat now applies to electricity generation, whose uncertainty was rebuilt around its evidence classes: its parametric band fell to $\pm 8.3\,\%$ (per-fuel activity widths, correlated through the shared TOE conversion) precisely because its one-sided questions—the coal calorific conversion (worth $1.6\,\mathrm{Mt}$), the TOE gross/net basis, and K-Electric coverage—moved into named structural cases with conditional intervals and no probabilities. The national interval absorbs none of them, and the power sector's scenario table is where a reader finds what they are worth.

4.9.8 What to measure next depends on the question

The driver shares above answer one of three different priority questions this inventory publishes, and they are not interchangeable. Which *category* would most narrow the conditional interval is a parametric question, and its answer is enteric fermentation, read from the ensemble at category grain with no ranking of named parameters. The other two are published together (`outputs/national_priority_rankings.csv`). Which *unresolved model choice* most moves the deterministic total is a structural question the interval cannot see, and its answer is the power coal energy basis at $9.1\,\mathrm{Mt}$. Which *missing category* most changes completeness is a coverage question neither of the first two can express, and its answer is industrial solid fuel: EDGAR carries $19.4\,\mathrm{Mt}$ for 1.A.2 that this inventory does not estimate. Reading any one of the three as the measurement priority list silently answers the other two questions with it.

4.9.9 Nine years, and whether their differences are resolved

Everything above conditions on the reference year. The nine-year panel is the product a reader is most likely to difference, and until now it carried no interval at all—so the only bound available for a year-on-year change was the national $-10.4\,\%$ to $16.0\,\%$, which is a statement about a *level* and does not apply to a *change*. `outputs/pakemis_longitudinal_uncertainty.csv` gives every fiscal year a band and `outputs/pakemis_trend_uncertainty.csv` gives every consecutive pair and the full window one, differenced draw by draw so that shared error cancels as the real quantity does.

How much of it cancels is not something the evidence settles, so three correlation scenarios are published rather than one. At one extreme every component's error is a single draw shared by all nine years; at the other every error is redrawn wherever the underlying number moved. The reported scenario is neither: it is typed from the parameter graph, sharing across years what names a factor, a physical constant or a held parameter, and redrawing what names a quantity measured in a particular year. Whether a quantity "moved" is decided by equality in-process and enters a stream's seed as the first fiscal year that carries the value, never as the value's own digits: a one-ulp platform difference in a computed number therefore cannot re-seed a stream, and the first 200 draws of every stream ship as `outputs/pakemis_longitudinal_prefix.csv` so that a rerun elsewhere is compared with what this release drew rather than trusted.

The resolution statement, read from the product's own rows with scenario and scope named on every number. Under the reported (typed) scenario, 2 of the 8 consecutive year-on-year changes in the panel are resolved at $95\,\%$. The decadal window IS: since the FY2015–16 recovery the panel has one window rather than two—the common-scope span and the full panel coincide at ten comparable sectors—and the FY2015–16 to FY2023–24 change of $51.12\,\mathrm{Mt}$ resolves at $15.94\,\mathrm{Mt}$ to $89.57\,\mathrm{Mt}$ under the typed scenario and at $40.08\,\mathrm{Mt}$ to $64.04\,\mathrm{Mt}$ under time correlation, but NOT under the deliberately-widest reading, full time independence ($-10.01\,\mathrm{Mt}$ to $114.07\,\mathrm{Mt}$), which spans zero.

That is a retreat from what this paragraph said before the coal re-base, and the retreat is the honest reading. It said that no end of the bracket left the decadal change spanning zero any more. That was true of the larger change the previous edition reported and is false of this one under the same intervals. Booking public-power coal on the generating stations' own metered fuel rather than on the aggregate conversion lowers the reference year by $17.78\,\mathrm{Mt}$ and the FY2015–16 start year by

only $0.19\,\mathrm{Mt}$, so almost the whole move lands on the decadal difference. The inventory can therefore support a decadal trend statement under two of the three correlation readings it publishes, cannot support one under the third, and still cannot resolve a single year-on-year change. A claim that strengthened when the evidence did not is worth stating as a correction rather than quietly re-scoping.

An earlier edition of this section reported one resolved change, and it was an artefact of a defect in the product beneath it. The difference table carried a `comparable_sectors` column reporting the sectors two years have in common, while the arithmetic subtracted every sector present in each year—so any pair reaching back to FY2015–16, where transport and industry are absent, subtracted eight sectors from ten and published it under a column saying eight. The one change that appeared resolved was FY2015–16 to FY2016–17 at $77.92\,\mathrm{Mt}$, of which $61.08\,\mathrm{Mt}$ was two sectors entering the record; restricted to the scope its own column claimed it fell to $16.83\,\mathrm{Mt}$ and stopped resolving, and on the recovered ten-sector panel the first step is $25.60\,\mathrm{Mt}$ and still not resolved. The full-window figure fell the same way, from $131.27\,\mathrm{Mt}$ to $80.33\,\mathrm{Mt}$—$39\,\%$ of the headline was the record filling in. Both the point estimate and the draws behind the interval were restricted to the scope the column claimed, so the band was corrected and not only the number—and since the FY2015–16 recovery the restriction itself is history: the published full-panel change is $51.12\,\mathrm{Mt}$ over all ten sectors, like for like.

The reason was measurable, and this release acted on it. The previous edition found $81\,\%$ of the national variance row-private residual—variance the parameter graph attributed to no named quantity—and stated that naming parameters, not collecting more years, was the work that would narrow a trend band. Naming agriculture's factors did exactly that to the variance composition: $76.56\,\%$ of the reference year's variance now sits in parameters that persist across years, $20.96\,\%$ in year-resampling activity, and $2.48\,\%$ in residual. The shares are covariance-aware: members of one dependency group draw one factor, so their first-order terms add before they are squared (the previous edition summed the squares and printed $10.67\,\%$ for the resampling share, half its value). Persistent error is precisely what a difference cancels, and the cancellation is what resolves the windows stated above while leaving every consecutive change unresolved—the year steps are small against the fifth of the variance that resamples each year and the residual that no parameter names.

5 Data records

Four product families are defined, and this release implements them to the maturity reported in Table 8—which is not the same as four products shipping. Stating which is which is the point of this section: a reader must be able to tell a mass-conserving reconstruction from a declared allocation without reading the code. The annual family is the reference configuration and the fiscal-year panel; the monthly family is the monthly estimate on every annual cell; the spatial family is the spatial allocation on administrative units and on the grid, with its unallocated remainder; the uncertainty family is the parameter-level ensemble and the structural bridge. Every product is an evidence-graded record: its rows carry the temporal and spatial evidence class of the streams they aggregate, and the coverage matrix of Section 5.1 states, per module, which geometry the release holds and which classes dominate.

Two schema rules hold across all of them. Every published row states its reporting period as explicit dates rather than an integer year, and the fiscal-start sectors' filenames carry the period too. And every number a manuscript prints is checked against the product it came from on every commit; a printed quantity with nothing behind it is a build failure, not an editorial matter.

5.1 The product-coverage matrix

Table 9 answers, per module: what native geometry the release holds, the dominant spatial-evidence class the cube allocates on, which spatial products the sector publishes itself, and the dominant temporal class of its monthly resolution. It is generated from `outputs/pakemis_product_coverage.csv`, itself derived from the source-stream ledger, so a sector whose coverage changes cannot leave this table behind. The ADM2 monthly view now ships sector-resolved as `outputs/pakemis_adm2_monthly_sector.csv` (module × shape × gas × month, a tested partition of the aggregate view); nesting agriculture's raster inside its district table remains a declared deferral, carried here rather than implied to exist.

6 Technical validation

Validation here means two different things, and the section keeps them apart. The first is internal: the annual, category and gas totals close exactly; every annual cell equals the sum of its twelve months; the national total equals the spatial allocation plus its unallocated remainder; and the ledgers between sectors balance. Those identities are asserted on every build and are reported with the generation stages that produce them. The second is external, and is the substance of this section: each sector is compared against whichever independent estimates exist for it, the disagreements are explained or defended rather than absorbed, and the key-category analysis says which categories carry the level and which carry the variance—so that a reader knows where the evidence-graded product is strong and where a proxy is doing the work.

Every sector is compared against whichever external estimates exist for it: EDGAR (Crippa et al., 2025) for all nine non-agricultural sectors, FAOSTAT (FAO, 2025) and Climate TRACE (Climate TRACE, 2026) for agriculture, and Pakistan's own BUR-1 (Ministry of Climate Change, Government of Pakistan, 2022) where its categories are separable. The comparison is generated on every run rather than compiled by hand, and 26 of its 27 rows resolve against an engine category; the one that does not is the red-flagged CRT ammonia comparator, which reports native CO_2 and is never divided into a CO_2e value. Each row's register entry declares its GWP basis, and the engine side is resolved in that basis: Climate TRACE publishes under AR6, so those seven rows compare AR6 against AR6, where an earlier edition compared them against AR5 values and manufactured a few percent of disagreement in the comparison itself. Resolution does not imply completeness: the industrial row measures the metered gas component against EDGAR's all-fuel category, so its low ratio remains visible.

6.1 The aggregate reading

At the whole-inventory level this work is $87\,\%$ of EDGAR's Pakistani sectors summed. That number is a *magnitude ratio* and is reported as one, because on its own it conflates two questions that different work answers. Source domains this inventory

Table 8. The release's core products and their declared status. **This table is GENERATED** from `outputs/pakemis_product_catalog.csv` rather than maintained by hand, because the hand-maintained version drifted: it described the uncertainty family as reference-year only, and denied that any time-correlated ensemble was published, while two longitudinal products had been shipping for releases and the main paper described their three scenarios. A product can no longer be in the release and absent from this table.

Product	Status	What it is, and what it is not
Annual: reference year	mature	Ten sectors on one reporting period, exact mass closure and executable cross-boundary ledgers. `national_all_sectors.csv`.
Annual: panel	mature	Sector, category and gas for all 90 sector-years - the rectangle is complete since the FY2015-16 recovery; category scope stays partial and the completeness matrix says where. TWO views; only the harmonized common scope may be differenced. `pakemis_annual_fy2015_16_fy2023_24.csv`.
Annual: pass2 scorecard	new	The Pass-2 exit gate scored from shipped products: two outcomes met, one met with named exclusions, and the monthly outcome refused by type with its arithmetic printed. A verdict register, never an estimate. `pakemis_pass2_scorecard.csv`.
Annual: completeness mass	new	Every absent stream joined to the bound the shipped arithmetic supports - a measured completeness band with contained claimants excluded from its sums. A register of what is missing and what it would weigh, never a distribution and never part of any total. `pakemis_completeness_mass.csv`.
Annual: atomic	mature	The component grain beneath the panel: every priced stream, with its evidence class, for all nine periods. `pakemis_atomic_annual.csv`.
Monthly: panel	allocation, not observation	108 months, July 2015 to June 2024, every non-absent component resolved into twelve months closing exactly to its annual figure. 60.8 per cent of the published monthly mass sits on a flat profile, against 54.6 per cent in the reference year alone. Read a flat month as an allocation, never as an observation. `pakemis_monthly_fy2015_16_fy2023_24.csv`.
Monthly: coverage	complete	Exactly 108 rows: what each month's timing rests on, and which sectors a month does not have at all. `pakemis_monthly_coverage_summary.csv`.
Spatial: rice geography crosscheck	new	The declared MapSPAM rice surface measured against the own-year GCD-Rice 2023 30 m classification, as within-province ADM2 shares with per-unit verdicts. A measurement of a disagreement, never a surface switch - the fork is priced as the agriculture_rice_gcd2023_adm2 bracket. `pakemis_rice_geography_crosscheck.csv`.
Spatial: stream support ledger	new	One row per (period, stream): the mass beside the support that placed it - geometry, surface, panel, measured located share, priced fallback and ADM1-retained masses, grade, pattern hash and the monthly-support refusal stated per row. Rectangular with typed absence. The audit's three-file hand-join, done once and pinned; never a second allocation. `pakemis_stream_support_ledger.csv`.
Spatial: adm2 frame 2022	new	The OCHA COD-AB 2022 ADM2 frame (160 units, CC BY-IGO) crosswalked to the canonical 152-unit set: twelve of the fourteen collision groups plus Larkana and Tor Ghar RESOLVE, Keamari and Surab stay part_of. The complete, tested preparation for the v1.1 re-rasterisation - and deliberately NOT that re-rasterisation: every shipped shape_id stays on the frozen 2019 frame this release. `pakemis_adm2_frame_2022.csv`.
Spatial: grid 0p1 edgar aligned	new	The annual 0.05-degree cube aggregated 2x2 at the ODD parity that tiles EDGAR's 0.1-degree lattice exactly (interior 133x169; three half-covered boundary strips declared in coverage_fraction), conservation asserted per layer. A derived benchmark-comparison surface - the frozen 0.05 lattice remains the product of record. `pakemis_grid_0p1_edgar_aligned.nc`.
Spatial: agri reference panel divergence	new	The reference-year crop split measured proxy-vs-panel (rc47 Wave 1): within-province shares per (category, shape) under the MapSPAM/cropland proxy product and the PBS district crop panel the allocation reads. The record of what putting FY2023-24 on the panel changed, and of why field burning declined both panels for its own-year FIRMS raster - a measured disagreement, never a second allocation. `agri_reference_panel_divergence.csv`.
Spatial: cube	allocation, partly observed	The 0.05 degree MONTHLY cube in native gas mass, with district and provincial views, for the reference year. The monthly grain is the reference year's alone; the nine fiscal years are gridded ANNUALLY in pakemis_grid_0p05_annual_ref_fy2015_16_fy2023_24.nc and tabulated at ADM1 and ADM2 in pakemis_adm1_by_year.csv and pakemis_adm2_by_year.csv. `pakemis_grid_0p05_monthly.nc`.
Spatial: cube annual	allocation, evidence-graded per stream and period	Nine fiscal periods at 0.05 degrees, three spatial views, three gases, in native mass. Each period is generated from its own atomic streams, NOT by rescaling the reference year's map. Cells plus pakemis_spatial_unallocated_by_year.csv reproduce each period's national control. The views PARTITION the placed mass and must not be added to one another's totals. Whether a period's within-province pattern is its own or borrowed is stated per stream in pakemis_spatial_weights_by_year.csv. There is no historical configuration S and no historical monthly grain. `pakemis_grid_0p05_annual_ref_fy2015_16_fy2023_24.nc`.
Spatial: by year	allocation, geography borrowed outside the reference year	Nine fiscal years at ADM1 and ADM2, annually. Each year's national mass is evidenced for that year; the within-province geography is the reference year's, rescaled - so this is a reconstruction of where past emissions most likely sat, not nine years of observed district data. It is also the only ADM1 table configuration L has: pakemis_grid_provincial_check.csv is a share check. `pakemis_adm1_by_year.csv`.
Spatial: unallocated	declared, not gridded	The mass with no defensible placement below ADM1, retained rather than spread. National closure is raster plus this. `pakemis_spatial_unallocated.csv`.
Uncertainty: reference year	mature	A 100,000-draw national Monte Carlo consuming two sectors' own engines, plus twenty structural axes priced as named configurations. `national_uncertainty_montecarlo.csv`.
Uncertainty: common ensemble	joinable, reference year	A 10,000-draw prefix of the national Monte Carlo carried to months and provinces on one draw_id, with both closure identities holding inside every draw. Profile weights are drawn only where a measured dispersion exists; allocation is conditional, never mixed. `pakemis_ensemble_summary.csv`.
Uncertainty: longitudinal	published, and negative	Nine annual bands under three temporal-correlation assumptions, typed persistent against redrawn. It exists; an earlier edition of this table said it did not. `pakemis_longitudinal_uncertainty.csv`.
Uncertainty: trend	published, and negative	Draw-wise year-on-year and window differences, each pair restricted to the sectors both years report. No year-on-year change is resolved at 95 per cent; the multi-year windows now are, under the typed

Module	Geometry held	Spatial class	Own products	Monthly
Agriculture	diffuse	S3	district+grid	TE4
Electricity generation	facilities	S1	provincial	TE2
Road transport	diffuse	S3	provincial	TE2
Residential fuel	diffuse	S3	provincial+district+grid	TE3
Industrial combustion	diffuse, facilities, points	S2	provincial	TE3
Cement (IPPU)	diffuse, facilities	S2	provincial	TE2
Waste	diffuse, points	S2	provincial+district	TE4
Fugitive oil and gas	diffuse, facilities, points	S3	provincial	TE4
Railways	network, points	S2	provincial+district+grid	TE2
Domestic aviation	none (S4), facilities	NA	provincial	TE2

Table 9. Product coverage by module. Geometry held lists the geometry-id kinds in the source-stream ledger (none (S4) marks streams with no native geometry: they are retained at ADM1 and enter no grid cell, and closure is raster plus unallocated); spatial and monthly columns give the emissions-dominant class.

Comparison	Source	Period	Bench.	Engine	Ratio
Comparisons on matched boundaries					
Electricity (1A1a)	EDGAR	CY2024	56.80	40.04	0.70
Residential (households)	EDGAR	CY2024	31.42	24.22	0.77
Waste (year-matched)	EDGAR	CY2023	26.57	37.45	1.41
Agriculture total	FAOSTAT	CY2023	233.30	223.45	0.96
Agriculture total	EDGAR	CY2024	261.20	223.45	0.86
Agriculture total	Climate TRACE	CY2024	261.50	218.07	0.83
Managed soils	FAOSTAT	CY2023	58.50	28.53	0.49
Rice cultivation	Climate TRACE	CY2024	7.80	15.96	2.05
Partial-scope diagnostics — ratio withheld or scope-noted					
Fugitives (1B2 vs all 1B)[e]	EDGAR	CY2024	17.54	23.50	1.34
Transport module vs EDGAR modes	EDGAR	CY2024	48.76	41.93	—
Coverage comparisons — not agreement					
All sectors	EDGAR	CY2024	514.33	444.96	0.87
Industrial combustion (1A2)	EDGAR	CY2024	43.98	28.13	0.64
IPPU: cement term vs all processes	EDGAR	CY2024	27.38	21.82	—

Table 10. The 1.A.1.a grid figure is the `power_grid_control` family reference; the family's other members read 38.45 Mt (fuel property), 44.05 Mt (coverage) and 57.82 Mt (legacy account), and none is a sensitivity. [e] The 1.B.2 reference sits one membership step below the top of its own 9.601–24.694 Mt structural envelope, fertiliser feedstock being outside the post-meter leakage base; the priced envelope members, including the 17.68 Mt geometric-mean central reading, sit beside it in the configurations file. External comparison, Mt CO_2e, **grouped by what the comparison can support**. A previous edition printed all three groups as one undifferentiated ratio column, which invited a scope difference to be read as an error—and one row was worse than that: it was labelled "Road transport" while carrying the transport *module*, road plus mixed off-road, against an EDGAR row for a different set of modes in a different year. Every row now carries its period, and the register behind it (`data/benchmarks.csv`) carries both boundaries and a `ratio_permitted` flag; a dash means the boundaries do not support a ratio, not that one could not be computed. The inventory period is FY2023–24 throughout, so a row whose benchmark period is a calendar year is comparing across a partial-year offset as well.

does not estimate at all pull it down; domains it does estimate and where the two disagree pull it in either direction—waste sits at 141 % of EDGAR and agriculture at 86 %—and netting those against each other produces a single figure that reads as completeness and is not one. Three numbers are therefore reported separately (`outputs/national_benchmark_scope_coverage.csv`):

1. **Magnitude ratio,** 87 %—444.96 against 514.33 Mt. How big, not how complete.
2. **Benchmark-scope representation,** 11 % **fully and** 89 % **partially**—the share of EDGAR's source domain, weighted by EDGAR's own magnitudes, in which *every* IPCC category mapped to the domain is estimated, included elsewhere or does not occur. Only public electricity meets that. Nothing is wholly absent: every domain carries at least one estimated category, which the magnitude ratio cannot show. The test is deliberately strict—one unestimated category makes a domain partial, however small it is—so the unestimated categories are named per domain rather than absorbed: liming and other carbon-containing fertilisers in agriculture, domestic navigation in transport, industrial solid fuels in 1.A.2, commercial buildings, four IPPU categories, and industrial wastewater, incineration and biological treatment in waste. *An earlier edition of this paper claimed five domains estimated "whole", which the completeness matrix in the same repository contradicted.* The classification is derived from the notation keys now, through `data/edgar_domain_map.csv`, rather than asserted.
3. **Inventory completeness**—the IPCC notation-key matrix over all 138 category–gas combinations in Section 2.5, which is the only one of the three that is independent of a benchmark's own scope.

The bridge from this inventory to EDGAR's total has three terms, and mixing them is the error to avoid. Categories absent here, valued on EDGAR's own basis, are a *coverage* gap: industrial combustion (15.85 Mt of its 43.98 Mt, after the brick and cement coal activation) and F-gases (11.54 Mt) are the large ones. Differences where both systems estimate the same category are a *method or activity* disagreement, and they run in both directions: agriculture is 38 Mt below EDGAR and waste is above it. A previous edition listed the agriculture difference beside the two absences as though it were a third gap, which it is not—nothing is missing there, the two estimates disagree. Adding only the coverage term at EDGAR's own values would put this inventory slightly *above* EDGAR overall, which is the right sanity check on the estimated categories: they are not systematically low, the inventory remains incomplete in specific places, and the completeness table (Section 2.5) names exactly where.

The resolved sector comparisons fall broadly within a factor of two, which for independent Tier 1 estimates built from different activity data is close agreement. The flagged component rows are examined rather than adjusted toward their benchmarks.

6.2 Railways against Pakistan's own submission: a factor of ten

BUR-1 reports railways at 4.06 Mt CO_2e for FY2017–18, and the comparison this paper makes is *year-matched*: run on the operator's metered fuel for that same fiscal year, by the identical method, this inventory reconstructs 0.4663 Mt. BUR-1 is 8.7 times it. Quoting the current year against a 2017–18 filing would fold six years of activity change into a ratio that is about method, which is why the matched year leads; both years are set out together so neither can be read as the other.

Quantity	FY2017–18	FY2023–24
Reported diesel, thousand litres	156 396	132 588
Estimated diesel mass, kt	132	112
This inventory, Mt CO_2e	0.466	0.394
BUR-1, Mt CO_2e	4.06	—
Submission over this inventory	8.7	—

Table 11. The railway comparison, matched. *A previous edition mixed three years in one paragraph*: it quoted the matched-year estimate beside FY2017's 144 631 thousand litres, which is a different year again, and beside the current year's mass. The matched-year estimate is built on the 156 396 thousand litres the operator reports for FY2017–18. The difference between the two years is fuel, not method: FY2017–18 diesel is 18.0 % above FY2023–24, not a factor of ten in either direction.

The disagreement resolves on arithmetic rather than on judgement. Producing 4.06 Mt of CO_2 from diesel requires roughly 1280 kt of fuel. Pakistan Railways—a single state operator that meters and publishes its own fuel purchases—reports buying 112 kt in FY2023–24 and 132 kt in FY2017–18, so BUR-1's implied fuel is roughly ten times either. The traffic comparison is a *physical-scale diagnostic* rather than a matched-year test, and is labelled as one because it pairs the current year's train-kilometres with the older submission: the metered purchase works out at 4.0 L per train-kilometre over the operator's own 33.1 million train-km, which is where a diesel locomotive belongs, while BUR-1's implied fuel would need about 46 L per train-kilometre, which no locomotive does. The conclusion does not rest on it—the fuel-mass route above is matched on both years.

Three independent quantities—the operator's fuel account, the published traffic statistics and the carbon content of diesel—all support the lower figure, and both the fuel bound and the per-train-kilometre intensity are pinned by tests so the claim is rechecked on every run. We record the disagreement openly, on the principle that a national inventory which silently diverges from its country's own submission is worse than one that diverges and says so.

Three further facts from the same register (`railways_official_comparison_2024.csv`) place that disagreement, and none of them flatters this inventory by construction.

It is not confined to the superseded narrative filing. Pakistan's first Common Reporting Tables, the current submission and the one this paper prefers wherever a cell exists, book 56 609 TJ under 1.A.3.c for calendar 2022. Against this inventory's value for the two fiscal years that calendar year straddles, the filing is 11.6 times it—worse than BUR-1's 8.7, not better.

It runs the other way against Pakistan's oldest filing. The Initial National Communication implies 0.5575 Mt for FY1993–94, which is 11.9 % *below* this inventory's independent reconstruction of the same period from the operator's own fuel history. That figure is derived from a stated share rather than read from a printed row, and the register labels it a magnitude check rather than a point match; it is reported here because it is the one official comparison this inventory does not win, and because of what it implies about timing. Between the only two filings that report railways separately the official figure rose by a factor of 7.3 while this inventory's reconstruction *fell* by 26 %. The order-of-magnitude disagreement is therefore not a standing feature of Pakistan's reporting: it entered the official record somewhere between 2003 and 2022. It cannot be dated more closely than that, because the two filings in between report only the 1.A.3 aggregate.

And it exceeds anything the railway has ever emitted. The operator's fuel history, reconstructed back to 1950, peaks at $2.3049\,\mathrm{Mt}$ CO_2e per year in 1965–70, when Pakistan Railways burned $48\,720$ tonnes of coal and $542\,078$ tonnes of furnace oil annually at the end of the steam era. BUR-1's $4.06\,\mathrm{Mt}$ is 1.8 times that peak. A diesel-only railway a quarter the size of the steam network cannot emit twice what the steam network did, and no assumption about factors or boundaries recovers the difference—only the activity can, which is where the filing and the operator's metered purchases part company.

The railway fuel account also supplies this inventory's only genuinely *independent* source-boundary check, and it is worth separating from the closures of Section 4.8 because it is the opposite kind of evidence. A closure is an identity: one quantity computed once and consumed twice, which cannot disagree because it is the same number. This is two unrelated reporting chains describing the same physical fuel. Pakistan Railways meters and publishes its own purchases in its Year Book; the Oil Companies Advisory Council independently reports a Railways customer-class row in its national petroleum statistics. Neither is derived from the other. Year-matched for FY2021–22 they differ by $3.80\,\%$—$112\,037\,\mathrm{t}$ against $107\,932\,\mathrm{t}$—which is about as close as two independent national fuel accounts get.

That check exists because a closure stopped being necessary. The transport sector once subtracted the railway fuel account from a single national diesel supply figure; since it was re-anchored on the petroleum statistics' own customer classes, Railways sits outside the Road row by construction and there is nothing left to subtract. The quantity that used to enforce a boundary now tests one instead.

6.3 Waste against EDGAR: an unresolved methane correction factor

The waste sector is $141\,\%$ of EDGAR's year-matched Pakistani waste total. This row has changed sides during the project: before domestic wastewater was built it read $47\,\%$, and the gap was a measure of what 4D would add. With 4D in, the sector overshoots.

The likeliest single cause is the methane correction factor for sewers. IPCC gives 0.5 for stagnant sewers and 0.1 for wastewater discharged to a flowing river. At 0.5, domestic wastewater is $24.46\,\mathrm{Mt}$ and the waste sector overshoots EDGAR. At 0.1 it is $15.39\,\mathrm{Mt}$ and the sector lands close.

The evidence points both ways, and genuinely so. The physical description of Pakistani urban drains—open, slow-moving, heavily organically loaded—supports 0.5. The only independent comparator supports 0.1. EDGAR is itself a model and does not settle the question.

We publish both readings rather than resolving the tension by fiat, and report the 0.5 case as the headline because it is the value the physical description supports. This is the same treatment given to the correlated-uncertainty case in Section 4.9: where an assumption moves a result by a factor approaching two and the evidence does not choose, the defensible output is both numbers and a statement of what would settle it. Here, what would settle it is a dissolved-methane measurement campaign on urban drains in two or three Pakistani cities.

6.4 The other flagged rows

Fugitives at 1.34 **of EDGAR** is a matched-year but only partly matched-scope comparison. The 2019 segment account includes production, gathering, processing, transmission, distribution, refining and post-meter sources; EDGAR labels its aggregate "fuel exploitation". The difference lies inside the engine's declared structural and parametric ranges, but no adjustment is made toward EDGAR because both products depend on modelled factors rather than Pakistani component measurements.

Managed soils at 0.49 **of FAOSTAT** and **rice at** 2.05 **of Climate TRACE** are agriculture-sector questions treated in the companion manuscript (Ahmad, 2026a). Both survive scrutiny, and the managed-soils one has been measured rather than asserted since this release. `outputs/pakemis_faostat_nitrogen_flows.csv` compares the two compilations at the nitrogen flows rather than at the total, and the flows agree. FAOSTAT's four managed-soils items map one-to-one onto this inventory's four pools, so none of the difference is scope. Synthetic fertiliser nitrogen agrees to 0.9 %—3 775 000 against 3 808 775 t N—and the six livestock species both compilations carry agree to 0.9 % in total, with the residual running the wrong way for the headline: this inventory counts 1.36 times FAOSTAT's sheep and 1.30 times its camels, so more excreted nitrogen, not less.

The factor of two is therefore in the emission factors and in the partition of nitrogen, not in the nitrogen. Two-fifths of it is the factor vintage: FAOSTAT applies the 2006 Guidelines' aggregate EF1 of 0.010 where this inventory applies the 2019 Refinement's 0.005 for dry climates, and taking FAOSTAT's vintage as a named scenario moves managed soils from 28.53 Mt to 40.43 Mt, closing 11.90 Mt of a 29.97 Mt gap. The remaining 18.07 Mt is neither the nitrogen nor that vintage, and it is not diffuse either. FAOSTAT's four items, acquired separately for this release, put two-thirds of the whole gap in one pool: pasture deposition is 7.71 Mt here against 27.29 Mt, a difference of 19.58 Mt and 65.3 % of the total. Synthetic fertiliser contributes 8.77 Mt (29.3 %) and crop residues 2.70 Mt, and the mechanism in all three is the one already named: the 2019 Refinement disaggregated these direct factors by climate and Pakistan sits at the dry end. Beside EF1's 0.005 against 0.010, the pasture factor for cattle is 0.002 here against the 2006 default's 0.02—a tenfold cut, applied to the 40.9 % of this inventory's 7.7834 Mt of excreted nitrogen that its ledger routes to grazing.

Taken together the two direct factors account for almost the whole difference. The pasture factor had no scenario beside it until this release, which was an asymmetry rather than a judgement: the same climate choice on EF1 has carried three alternatives for releases—the wet value, the 2006 aggregate and an empirical adjudicator—while the larger lever carried none. It has one now, and applying both 2006 direct factors together moves managed soils to 54.72 Mt, closing 26.19 Mt or 87 % of the gap. The two scenarios are disjoint by construction: EF1 scales synthetic fertiliser, organic manure and crop residues, the pasture factor scales the two deposition rows, and no subitem belongs to both. What remains is 3.78 Mt, in the indirect factors—EF4 and EF5 changed between the vintages too, and these scenarios do not touch them—and in the nitrogen partition.

One pool runs the other way and the factor vintage cannot explain it. Manure applied to soils is 1.16 times FAOSTAT's despite the lower factor, so this inventory routes more excreted nitrogen to field application than FAOSTAT does. That is a disagreement about where the nitrogen goes rather than about what happens to it, and it is the only one of the four.

Whether the dry classification is right for grazing land is a separate question from whether it is right for cropland, and the evidence behind it is weaker. The dry EF1 is recorded as *best-supported for irrigated arid cropland*, derived largely from irrigated dry fields—which is what Pakistan's fertilised area is. Pasture, range and paddock land is neither irrigated nor cropland, so that basis does not transfer to it, and no comparable Pakistani evidence stands behind the pasture choice. The factor stays at the Refinement's dry value, because that is the Refinement's own classification for an arid country, and the alternative is now priced beside it rather than left implicit. Climate TRACE's rice estimate, the other agriculture question here, uses a cultivated-area basis that diverges from the national crop statistics.

IPPU at 0.93 of EDGAR is nearer like-for-like than the label it used to carry, and what remains unmatched is now named rather than assumed. EDGAR's "Processes" row covers cement, lime, soda ash, ammonia, and iron and steel. Of those five this inventory estimates three—cement at $21.82\,\mathrm{Mt}$, ammonia at $3.24\,\mathrm{Mt}$, and iron and steel at $0.39\,\mathrm{Mt}$—and adds lubricant use, $0.08\,\mathrm{Mt}$, which EDGAR's row does not name. Lime is NE for want of activity data; soda ash is NA rather than NE, its Solvay-process carbon allocated to Energy under IPCC Volume 3 §3.8.3.1 (Section 2.5). The $1.84\,\mathrm{Mt}$ by which EDGAR's row exceeds this module is therefore of the order of one absent category, not of four, and the comparison is recorded as a coverage reading rather than as agreement.

Cement alone is 0.80 of the whole-processes row, the right order for a country whose IPPU is dominated by cement. The cement term's own configuration range brackets matched-period clinker trade and the clinker-fraction choice: the Tier 1 diagnostic at the 0.75 fraction gives $17.39\,\mathrm{Mt}$, while the 0.95-fraction trade-corrected configuration gives $21.50\,\mathrm{Mt}$; the reference itself is the producer census's observed clinker, $21.82\,\mathrm{Mt}$, above both. The earlier 48.011-Mt cement control was resolved as FY2021–22, not an alternative FY2023–24 series.

6.5 The national submission, year-matched: the gap is completeness, not estimation

The comparators above all require an approximation. EDGAR and Climate TRACE publish calendar years against this inventory's fiscal one; Climate TRACE publishes under AR6 against an AR5 account; FAOSTAT does both. One comparator requires neither, and it is the most authoritative one available: Pakistan's First Biennial Transparency Report (Government of Pakistan, 2024) publishes a full IPCC-category inventory on AR5 global warming potentials, and it states its own period convention—"the fiscal year 1993–94 (referred to as 1994)", "Pakistan's 2020–21 inventory (referred to as 2021)". Its inventory year is a Pakistani fiscal year. It therefore faces FY2020–21 directly, with no blend and no metric conversion, and this release carries that year.

Four of this project's own ingests had read that document and three of them recorded it as calendar 2021; only the aviation ingest read it as fiscal. The correction is why this comparison is possible at all, and it is recorded in the register rather than in a footnote.

A join on the IPCC code would be wrong, and silently so. The submission books rice cultivation at 3.D and managed soils at 3.C; this inventory books rice at 3.C and soils at 3.D. The codes are *swapped* between the two accounts, so a code join compares rice against soil nitrous oxide and returns two plausible ratios. Every row of `outputs/pakemis_btr_comparison.csv` is therefore an explicit crosswalk entry with a declared tier, and a ratio is emitted only where the boundaries support one.

Excluding the submission's land-use block—34.82 Mt, outside this inventory by declared scope—the two national totals are 483.45 and 486.65 Mt, a ratio of 0.993. That agreement is not the result worth reporting, because it averages category ratios spanning more than an order of magnitude: 6.91 on oil and gas fugitives, 0.44 on managed soils, 1.19 on enteric fermentation, 0.43 on lubricants. What the headline does support is a decomposition the single ratio hides. Restricted to the categories both accounts actually estimate, the totals are 482.49 and 480.61 Mt—a ratio of 1.004, this inventory above the submission. The 3.20 Mt national difference is the net of two movements in opposite directions. 6.04 Mt is seven categories this inventory declares NE and the submission books—solid-fuel fugitives, lime, glass, other carbonate uses, other industrial processes, waste incineration and industrial wastewater—against 0.382 Mt this inventory estimates where the submission books nothing or zero, which is iron and steel: a net 5.658 Mt of pure completeness difference in the submission's favour. On the categories both accounts estimate, this inventory is 1.88 Mt above the submission, which offsets part of it. Those seven rows are the same rows that sit in `outputs/pakemis_completeness_mass.csv` with no bound, and they now have an official magnitude.

The sign of that national difference reversed with the coal re-base and has reversed back in this release, and earlier editions of this paragraph read it each way. The first described the gap as this inventory's *shortfall* and concluded that the difference against Pakistan's own account was a completeness difference rather than an estimation one. The matched comparison year is inventory year 2022, which the re-base raised by 11.07 Mt—Table 3's flat conversion had been reading that year's coal low—so the shortfall became an excess and the clean conclusion with it. This release lowers the same year again—household dung on the survey, wastewater nitrous oxide on the 2019 Refinement with each year's own sanitation shares, fertiliser feedstock outside the post-meter leak base—and the excess is a shortfall once more. Both components were always there; what changes is which one dominates. The completeness difference is unaffected and still 5.658 Mt; the estimation difference is again the smaller term, and the submission's own coal energy descends from the same HDIP series the re-base moved away from, so the two accounts are not independent on exactly the category that first reversed the sign.

Two of the apparent order-of-magnitude disagreements are one disagreement. On landfill methane this inventory is 0.49 of the submission; on open burning of waste it is 17.74 times it. Summed over the solid stream the ratio is 0.68, so what the accounts mainly disagree about is the *fate split* of the same municipal waste—how much is collected to a disposal site against how much burns where it falls. That is a question about collection statistics rather than about an emission factor, and neither individual ratio should be quoted as a level error.

The managed-soils row is the one that should change a reader's mind, because it is the second independent account to say the same thing. The submission puts direct and indirect managed-soil nitrous oxide at 58.05 Mt against this inventory's 25.77 Mt; FAOSTAT, built from different inputs, puts the same quantity at 58.50 against 28.53 Mt for the reference year. Two accounts, one official and one international, land at about twice this inventory, and the cause this paper already publishes—the 2019 Refinement's dry-climate EF_1 against the 2006 aggregate value—predicts exactly that. A second comparator agreeing with the first is worth more than either alone, and it moves the dry-climate choice from a defensible assumption to the single most consequential open question in the agricultural account.

The comparison runs both ways, which a completeness argument has to. The submission reports the metal industry as zero, where this inventory estimates iron and steel; and it reports substitutes for ozone-depleting substances as zero, where this inventory declares the category NE and carries an external bound beside it. A typed absence with a bound states more than a reported zero.

One entry in that list has to be withdrawn, and the withdrawal is the more useful result. The submission carries no biomass-burning *line*, and reading that as an absence—as an earlier version of this crosswalk did—makes this inventory look like the more complete account. It is not. The submission folds residue burning into its land block: cropland's non-CO_2 half is $3.16\,\mathrm{Mt}$ beside $2.50\,\mathrm{Mt}$ of land-use CO_2 in the same row, and the agriculture Common Reporting Table carries crop residue burning as a line of its own—the same quantity filed twice by one submission, which is also why a land layer built here cannot simply adopt both. Against the folded figure this inventory's field burning is 0.18, not a surplus. The exclusion of the land block is therefore not as clean as scope makes it sound: $3.47\,\mathrm{Mt}$ of its $34.82\,\mathrm{Mt}$ is CH_4 and N_2O, which for a land account is fire, and about $3.5\,\mathrm{Mt}$ of what the scope exclusion withholds is comparable in kind to emissions this inventory does estimate. It is also the hazard to avoid in reverse: a land layer here that adopted the submission's cropland figure beside this inventory's existing residue burning would count the burning twice.

6.6 Provincial SO_2: where national agreement hides a fivefold gap

Every comparison above is national, and a national ratio is the one statistic a spatial inventory can pass without being right anywhere. The provincial test is `outputs/pakemis_air_provincial_external_comparison.csv`, which places this inventory's ADM1 allocation beside EDGAR v8.1's gridded SO_2 summed by area fraction into the same boundaries, on a fiscal blend of the two calendar years each fiscal year spans.

At FY2021–22 the national ratio is 0.92—as close to agreement as anything in Table 10—and it is not informative. Underneath it Punjab is 1.25, Khyber Pakhtunkhwa 1.67, Balochistan 2.05 and Sindh 0.18: a spread of an order of magnitude, cancelling to within four percent of unity. A compilation can be wrong in every province and right in the sum, and this one very nearly is.

The disagreement is not diffuse. Decomposing by stream puts it almost entirely in electricity generation, where this inventory books $167.0\,\mathrm{Gg}$ against EDGAR's $409.3\,\mathrm{Gg}$, and within that in Sindh, where it books $4.5\,\mathrm{Gg}$ against $196.9\,\mathrm{Gg}$. That is not a defect in the allocation but a declared absence appearing where it was declared: power-sector SO_2 *as actually emitted* is NE in this release, pending a control-device inventory, and what is booked is the oil-and-diesel admission register alone—no coal. The coal fleet that EDGAR is seeing is in Sindh.

Read against the release's own screening the comparison resolves rather than merely disagrees. Adding the uncontrolled coal screening to the booked register—the two are disjoint by construction, the screening's oil rows covering only the remainder the register does not reach—gives $362.2\,\mathrm{Gg}$; assuming flue-gas desulphurisation at the screening's controlled case gives $202.0\,\mathrm{Gg}$. EDGAR's $409.3\,\mathrm{Gg}$ falls inside that bracket, at $129\,\%$ of its width. An independent compilation therefore places Pakistani coal generation near the uncontrolled end of this inventory's own declared range, which is the most that can be said before the control-device inventory exists, and is more than the national ratio could say at all.

One caveat belongs with these provincial numbers rather than after them. The coordinate ingest resolves every located station's province by dropping its coordinate into the 0.05° grid and reading the cell, and a cell is about $5.5\,\mathrm{km}$ across. `outputs/pakemis_power_province_boundary_audit.csv` tests all 77 located stations against their curated records and against their sources' own province fields: 7 are misplaced, every one of them within $3.4\,\mathrm{km}$ of a district line. The clearest is the Hub site, where two stations $1.11\,\mathrm{km}$ apart already sit in different provinces in this repository—the oil station recorded in Balochistan, the coal station in Sindh because its ingest copies the raster rather than the Global Energy Monitor field beside it, which says Balochistan. At FY2021–22 the coal station alone burned $25.5\,\%$ of the coal generated in Pakistan, some $6.4\,\mathrm{Mt}$ of CO_2; across the ledger years the misplaced stations carry between $1.5\,\%$ and $10.3\,\%$ of plant-ledger generation, the low end being the FY2023–24 reference, where the Hub coal station barely ran.

A finer grid would not settle this, which is why the audit corrects nothing. For four of the seven the simplified district polygons disagree with the curated record as well—ray casting places the Hub oil station in Karachi, both Guddu units in Dera Ghazi Khan and Tapal Energy in Lasbela—so the error is in the boundary geometry at these sites and not only in the rasterisation of it. Province and district are also one decision here: the district raster is built inside the province raster, and the nine-year products assert $\mathrm{ADM1} = \mathrm{ADM2} + \text{outside} + \text{unallocated}$, so relabelling a station's province while its mass stays binned in the old district would break the identity that the spatial layer exists to hold. Correcting these seven is a declared provincial re-base with a geometry decision inside it; the audit measures what it would move, and a guard holds the list at seven so an eighth cannot appear in silence.

The audit now publishes *which kind* of conflict each station has, because the seven are three problems and not one. For 2 the frame disagrees with itself—the 0.05° province raster and the district polygon name different provinces, so no relabelling settles it. 1 sits in a hole: no district polygon contains the Hub coal station at all, $0.13\,\mathrm{km}$ from the nearest vertex. For the remaining 4 the curated and source records agree and the frame dissents. 0 of the seven is a station-record error, which is the result that fixes the remedy: every conflict is in the geometry, so a boundary revision addresses all of them and a station relabelling addresses none. The classification is a declared vocabulary rather than a vote, because the raster reads the province file and the containing test reads the district file—a disagreement between them can be a file difference or a resolution difference, and nothing held says which, so counting them as two witnesses would invent an authority neither has.

6.7 Key Categories: Level Against Variance

A key category is one with a significant influence on a national inventory's total. IPCC Approach 1 sorts categories by their share of the absolute total and flags them cumulatively to a $95\,\%$ threshold. Thirteen of this inventory's 24 IPCC categories are key by that criterion.

We report a second ranking alongside it. A category's contribution to national variance is $(E_x U_x)^2$ as a share of the sum over categories, from the same Approach-1 quadrature that produces the national band. The level ranking says where emissions occur; the variance ranking says where reduced uncertainty would most narrow the national interval.

IPCC category	Code	Module	Mt CO_2e	Level (%)	Var. (%)
Enteric fermentation	3.A.1	agriculture	151.66	34.1	63.05
Road transportation	1.A.3.b	transport	40.19	9.0	0.95
Public electricity and heat	1.A.1.a	power	40.04	9.0	0.80
Manufacturing and construction	1.A.2	industry+power	28.13	6.3	0.79
Domestic wastewater	4.D.1	waste	24.46	5.5	13.03
Residential	1.A.4.b	residential	24.22	5.4	1.34
Fugitive emissions, oil and gas[e]	1.B.2	fugitives	23.50	5.3	3.33
Cement production	2.A.1	ippu	21.82	4.9	0.18
Manure management	3.A.2	agriculture	19.33	4.3	1.89
Rice cultivation	3.C.7	agriculture	16.55	3.7	4.31
Direct N2O emissions from managed soils	3.C.4	agriculture	16.43	3.7	3.95
Indirect N2O emissions from managed soils	3.C.5	agriculture	12.09	2.7	3.69
Solid waste disposal	4.A	waste	9.17	2.1	2.27
Not key by level:					
Urea application	3.C.3	agriculture	4.82	1.1	0.01
Open burning of waste	4.C.2	waste	3.83	0.9	0.22
Ammonia production	2.B.1	ippu	3.24	0.7	0.00
Indirect N2O emissions from manure management	3.C.6	agriculture	1.90	0.4	0.18
Agriculture/forestry/fishing[†]	1.A.4.c	transport	1.73	0.4	0.01
Biomass burning in croplands	3.C.1.b	agriculture	0.67	0.1	0.00
Iron and steel production	2.C.1	ippu	0.39	0.1	0.00
Railways	1.A.3.c	railways	0.39	0.1	0.00
Civil aviation	1.A.3.a	aviation	0.30	0.1	0.00
Lubricant use	2.D.1	ippu	0.08	0.0	0.00
Railway stationary/off-road[‡]	1.A.2/1.A.4.c	railways	0.005	0.0	0.00

Table 12. The 1.A.1.a grid figure is the `power_grid_control` family reference; the family's other members read 38.45 Mt (fuel property), 44.05 Mt (coverage) and 57.82 Mt (legacy account), and none is a sensitivity. [e] The 1.B.2 reference sits one membership step below the top of its own 9.601–24.694 Mt structural envelope, fertiliser feedstock being outside the post-meter leakage base; the priced envelope members, including the 17.68 Mt geometric-mean central reading, sit beside it in the configurations file. National key-category analysis, IPCC Vol 1 Ch 4 Approach 1, screened over **IPCC categories**. The first 11 cross the cumulative 95 % level threshold. *A previous edition screened the engine's own categories instead*, which split 1.A.2 between the power and industry modules and gave this table 25 rows against the category table's 19; the threshold, the ranking and every variance share were computed on that second hierarchy. The engine categories are now source streams beneath their code, in `outputs/national_key_categories.csv`. [†]1.A.4.c's placement is unresolved: the source's residual class mixes agricultural machinery with construction and industrial plant that belongs under 1.A.2, and the split is not published. [‡]The rail re-base's receiving category: the measured non-traction share of the railway fuel account (workshops, power houses, pumps, track machinery), whose own 1.A.2-versus-1.A.4.c split the Year Book does not publish. The variance column is a different ranking, and it assumes categories are independent—see Section 4.9. Level assessment only.

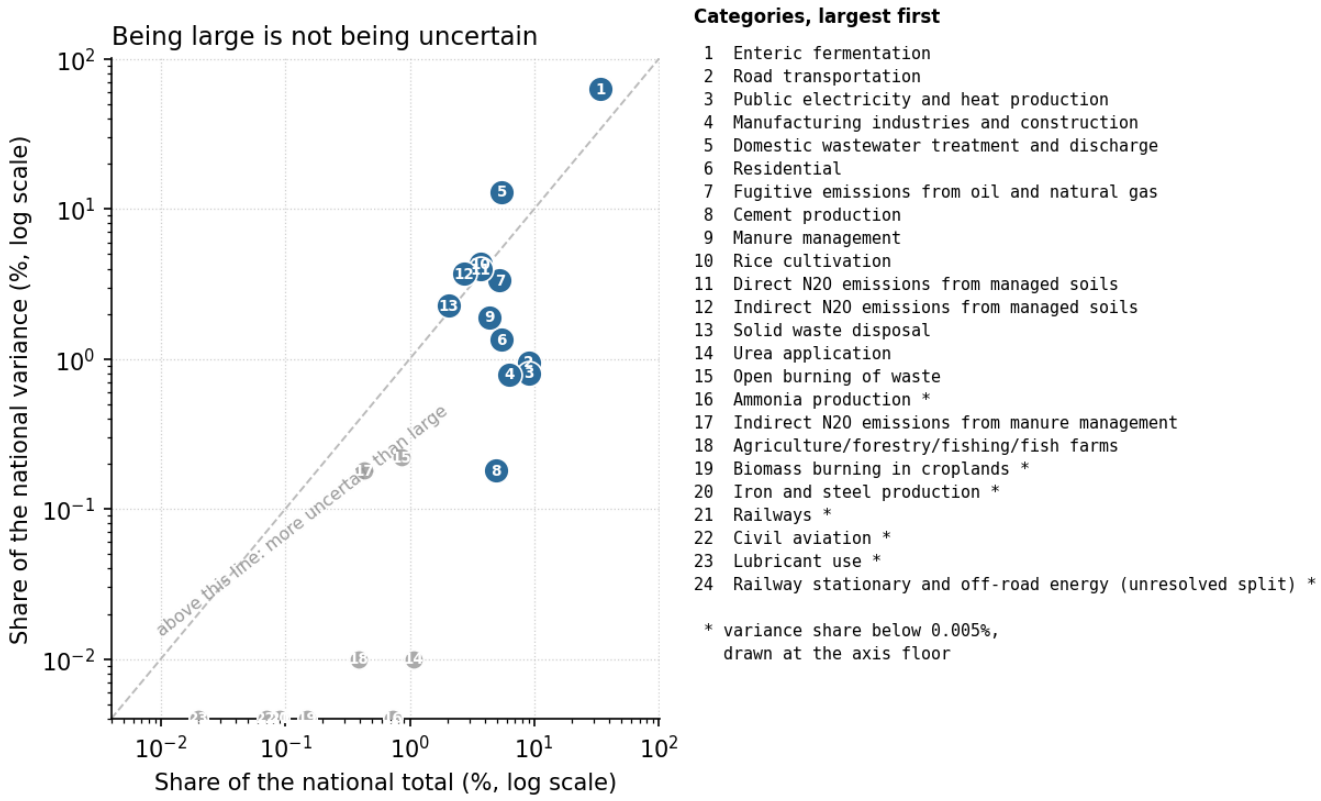


Figure 7. Level share against variance share for all 24 IPCC categories, on **log axes**. The diagonal still marks equal level and variance share, so the reading is unchanged: above it a category contributes more to national uncertainty than to magnitude. A previous edition drew this linearly, which put enteric fermentation at one corner and compressed eleven categories into a blot at the other—and the low corner is where the finding is.

6.7.1 What the two rankings imply

Road transport contributes 9.0 % of emissions but 0.9 % of variance. Its fuel control and carbon factor constrain national mass, so more fleet detail would improve composition and air-pollutant analysis more than the national greenhouse-gas interval.

Domestic wastewater contributes 5.5 % of emissions and 13.0 % of variance. It is the largest upward inversion after enteric fermentation, and its uncertainty has two named drivers: the methane correction factor for Pakistani sewers, now the larger term, and the effluent N_2O factor's bounded 150-fold published range, which the 2019 Refinement shortened from the 2006 Guidelines' 500-fold. Dissolved-methane sampling on urban drains and an effluent-pathway N_2O measurement would test the two directly, in that order.

Oil and gas fugitives are now key by level. The completed 2019 segment account is 5.3 % of emissions and 3.3 % of Approach-1 variance. This replaces the earlier production-only result in which a very small category carried a four-order-of-

magnitude factor range. The remaining priority is no longer simply utility throughput: it is technology and facility evidence for pipeline material, pressure, LDAR, compressors, processing and wells.

Enteric fermentation remains dominant on both rankings, at $34.1\,\%$ of emissions and $63.0\,\%$ of variance. The variance ranking therefore supports work on both livestock and wastewater; it does not imply that one small category alone controls national uncertainty.

6.7.2 The ten-sector trend, and what its window can and cannot say

IPCC's trend assessment requires a consistent series between one base and reference year. All ten sectors now satisfy that requirement from FY2016–17 onward and are published as the constant-scope view (Section 4.5), so the *ten-sector* trend this release previously declared out of reach is the one it supports. Cement and industrial combustion were the two holding it back; both were absent for want of documents that turned out to be held, and both now report every year of the window. The window now reaches FY2015–16 as well: the product splits that blocked transport and industry there were recovered from the publicly served Pakistan Oil Report 2015–2016 under a four-way identity gate, so the full nine-year panel is a like-for-like base for the trend. The agriculture-only trend assessment carries a heavier qualification than the others and it should be read with it: crop and nitrogen activity is now reconstructed on Pakistani fiscal-year tables in every year, but enteric fermentation and manure management still ride a calendar-year livestock projection because no fiscal enumeration of the herd exists, so the sector's history is fiscal in the categories that could be moved and a declared calendar projection in the $77.37\,\%$ that could not (Section 4.5.2).

7 Usage notes and limitations

What the dataset supports depends on the scale at which it is read. At national scale it is a reconstruction of the estimated scope on a named reference configuration, with a parametric interval and a priced set of structural alternatives; at monthly scale it is a monthly estimate whose evidence class is stated per stream and is flat for a declared share of the mass; at administrative and grid scale it is a spatial allocation on proxies and on registered point sources, and must not be read as observed facility emissions. The limitations below are labelled so that a claim elsewhere in the paper can be bounded by one of them, and the discussion that follows them says what the construction bought and what it did not.

The total is a single-year national total, and two of its ten modules pay for that with a borrowed shape. All ten sectors report FY2023–24, stated as explicit dates on every generated output. Getting there cost two declared compromises rather than a caveat: transport takes its magnitude and its provincial shares from the reference period's own direct OCAC editions but borrows its fleet/class composition from the registration panel's last year (FY2022–23), and waste's district population is projected one year past its census anchor to the fiscal midpoint. Neither moves a national tonne—a share set sums to one—but both are weaker evidence than the quantity they sit beside, and the component rows say so.

Roughly $47\,\mathrm{Mt}$ of this inventory's incompleteness can be given a number, and that figure is neither a floor nor a total. Industrial fuel combustion is the largest residual, and it halved in this release: EDGAR puts the category near $44\,\mathrm{Mt}$,

and beside the 5.55 Mt of captive self-generation, 5.61 Mt of industrial process gas and 3.40 Mt of industrial liquid fuels, the inventory now estimates 2.78 Mt of brick-kiln and 10.80 Mt of cement-sector coal from the HDIP coal-by-consumer series, leaving about 15.85 Mt outstanding rather than the whole category.

That 15.85 Mt is a coverage difference against one comparator and is not a lower bound on absent mass. Run over all nine years, the same comparison puts this inventory *above* EDGAR in four of them, by as much as 6.63 Mt in FY2020–21, and a quantity that changes sign cannot bound what is missing. The gap opens in FY2022–23 and FY2023–24 alone, the two years in which this inventory's own 1.A.2 falls 25.25 Mt below its FY2021–22 level—brick-kiln coal, cement-sector coal and industrial furnace oil together—while EDGAR's stays flat, so the difference prices a discontinuity in our activity series at least as much as it prices categories nobody estimated. EDGAR is in any case IEA-balance-derived and fed upstream by the same HDIP record this inventory reads, which makes it a diagnostic rather than an independent measurement of what we are missing.

F-gases (11.54 Mt) are absent entirely. Both figures come from an external estimate, which is what lets them be written down at all, but they are not the same kind of quantity: the F-gas figure prices a category this inventory does not estimate, while the 15.85 Mt is a difference between two estimates of a category both inventories estimate. The rest cannot be valued even that far. Chemical IPPU—ammonia, nitric acid, soda ash, lime, iron and steel—has no independent estimate here at all, and Pakistan's own filing puts the chemical industries near 19.4 Mt; industrial wastewater, waste incineration and biological treatment are likewise unquantified, as is mobile air-conditioning. Commercial and institutional combustion (1.A.4.a)—absent from this register altogether until an audit of the commercial-class rows the repository already held—is now declared and screened at roughly 2.4 Mt across its two quantifiable streams; domestic navigation and the coal-mining surface share carry published screening estimates outside the total. *The* 47 Mt *is therefore what can be counted, not what is missing.* Section 2.5 names all of them. This is a statement about coverage, not accuracy, and it means the national figure should be read as "ten sectors totalling 445 Mt under reference scenario `REF_2023_24`" rather than as Pakistan's emissions. One named structural case crosses sectors: the residential dung stream is reported there under two configurations, the household-survey reading and the agriculture-ledger supply derivation, which diverts 4.9 times what the 2024–25 survey round measured. This rollup carries the survey reading, which is the reference. The diverted dung it does not burn in households, 47.81 Mt of air-dry cake, is unresolved and outside the total, and configuration L, the destination assumption that households burn all of it, would raise the `REF_2023_24` total by 5.25 Mt.

LULUCF enters no total, on either side. No land-use, land-use-change or forestry flux enters the inventory total or any benchmark comparison. An exploratory Tier 1 land module ships with the release and publishes, outside every total, a provisional net sink of -4.01 Mt CO_2e (forest land -3.52 and Indus-delta mangrove blue carbon -0.49; cropland and grassland soil carbon not estimated) in `outputs/lulucf_totals.csv`, discussed in Supplement S1 of the agriculture paper. For a country with substantial mangrove and rangeland carbon the gap in the reported inventory is real, and it is a different kind of gap from the others: LULUCF can be a sink, so its absence does not simply bias the total low.

That provisional sink rests on an assumption Pakistan's own submission contradicts, and the disagreement is now priced in `outputs/pakemis_forest_flux_adjudication.csv`. The module sets deforestation to 0 ha a year, justified by FAO's net forest conversion being about zero in recent years; Pakistan's Common Reporting Tables put forest-land gross losses

at 14 868.348 against gross gains of 7113.083 kilotonnes of carbon, a ratio of 2.09 that nets to 28.436 Mt of CO_2 and reproduces the submission's own printed figure. So the two accounts that read forest land as a *sink* are not two: the exploratory module takes the decisive quantity from FAOSTAT, whose net forest conversion is exactly 0.000 kha from 2020 against 26.104 kha in 2015 while its Pakistani forest area rises at exactly 20.786 kha a year—a step constant to three decimals over four years, which is interpolation between Forest Resources Assessment anchors and not an annual observation. The gap between the submission and FAOSTAT is 32.55 Mt of CO_2 and it is a disagreement about direction, so no forest component can be built from either until the area trend is settled. The parameter is left at zero rather than changed, because moving it moves this printed figure and that is a declared decision rather than an audit edit.

The fugitive estimate is conditional on technology and coverage, and the reference sits high in its own envelope. The 2019 segment ledger replaces the capped 2006 range treatment and gives a parametric interval conditional on higher/default practices. Lower-practice, throughput-basis, incomplete-coverage and leak-base membership configurations form a separate structural envelope of 9.601 Mt to 24.694 Mt, and the reference (23.499 Mt) sits one membership step below the envelope's *top*—conservative default-practice factors with full coverage, and fertiliser feedstock outside the post-meter leakage base—not at an evidential centre. The top is the feedstock-inside membership of every release through rc66, retained and not defended. The envelope now ships as a named configuration family (`fugitives_structural_envelope`: the reference, the lower-practice member at 14.894 Mt, and the joint floor at 9.601 Mt) with its selection rule declared, so a national quote of the reference never travels without the envelope beside it. No Pakistani component-measurement campaign yet selects among those configurations.

The series is complete, and its residual weaknesses are in basis rather than coverage. All ten sectors are reconstructed consistently across FY2015–16 to FY2023–24: the trend-safe HARMONIZED COMMON SCOPE view and the full-scope view coincide since the FY2015–16 product splits were recovered from the public record (Section 4.5). The scope caveat that used to sit here—differencing across the first year measured sectors entering the record—is retired with its premise. What the completeness of the RECTANGLE does not change is the basis beneath it. Agriculture sits inside the constant scope on a *fiscal* reconstruction, and what remains outside that reconstruction is narrower than what used to. Earlier releases ran every pre-anchor year on FAOSTAT calendar-year activity—crops, fertiliser and livestock stocks alike—and called the result a hybrid diagnostic series rather than a trend-safe one. That description is superseded: crop and nitrogen activity now comes from the Economic Survey's own fiscal-year tables in all nine years, the series marks agriculture `reconstructed-fiscal`, and rebuilding it corrected a misalignment of a full calendar year in three of the four crops the old history carried (Section 4.5.2). **One half of the sector is still not on a fiscal basis, and it is the larger half.** Enteric fermentation and manure management—77.37 % of agriculture's reference-year mass—are driven by livestock stocks, and Pakistan publishes no fiscal-year *enumeration* of the national herd: the Economic Survey's fiscal head counts are, by their own footnote, the inter-census projection FAOSTAT already carries, so adopting them would buy a label and no observation. Those two categories keep a calendar driver and an inter-census projection rather than an enumeration, and the six-month phase error that leaves is measured at under 2 % of each category in every year rather than asserted. The step is priced rather than smoothed. The nine-year ten-sector panel is complete: the FY2015–16 product splits for road transport and industrial combustion—its last absent cells—were recovered at rc20, and

no sector-year in the panel is NE. It never did require more clinker trade returns: the monthly HS 252310 panel now runs July 2014 to June 2024 for both flows, and every year of the series resolves to an observed trade term—which is what opened FY2021–22 and FY2022–23, the two years that also held activity; the consequence for key-category analysis is in Section 6.7. The underlying cause is partly source availability and partly source *format*: five Pakistan Railways Year Book editions are scanned images with no extractable text layer. FY2018 and FY2019 were missing from the railway series for that reason and are not any more—both were recovered from the historical tables a later readable edition reprints, and the series is unbroken from FY2014 to FY2025.

Several partitions are exhaustive but soft at the split point. Section 4.8 guarantees that the parts sum to the whole; it does not guarantee the split. The waste collection rate of $50.3\,\%$ moves emissions between two categories with different methane yields. The off-road machinery class inherits its source workbook's residual bucket—the catch-all "Other vehicles" class maps into it wholesale and is then modelled as $100\,\%$ diesel—so the carve-out's magnitude is sound while its composition is not, and the cropland proxy that allocates it provincially is right for the tractors and wrong for the remainder. The transport sector's vehicle-class split is declared wrong for the same reason and is not used for any reported number that depends on it.

Household firewood is no longer the largest single activity uncertainty, and the reason is worth recording. It is the largest household fuel by energy— $38.7\,\%$ of household input, against dung cake's $13.6\,\%$—and this inventory used to carry it at $60\,\%$ activity uncertainty because the only national quantity available was FAOSTAT's, which carries FAO's own estimate flag in every year since 2009 and does not move.

It is now a measurement. The HIES 2024–25 round gives a design-weighted national household firewood quantity, and two independent anchors bracket it: the 2018–19 round reproduces the 2024–25 total to within six percent, and FAOSTAT's own series, converted on a moisture-matched basis rather than the oven-dry-with-as-received pairing that had been compared against it, sits seven percent above. The stream is therefore reported at $35\,\%$, which carries the survey's stratum-clustered sampling error plus the systematic risks a household survey cannot see—one-month consumption recall, out-of-home and informal-enterprise use booked to households, and seasonal enumeration.

FAOSTAT has not been discarded; it has been demoted from anchor to named structural case, `firewood_faostat`, which puts the sector at $23.89\,\mathrm{Mt}$ against the reported $24.22\,\mathrm{Mt}$. Dropping firewood entirely gives $20.81\,\mathrm{Mt}$. Both are published on every run, because an excluded number is invisible in a way that a flagged one is not, and because a reader is entitled to see what the re-anchoring did rather than only its result.

Air pollutants are partial, but less partial than this section used to claim. Seven sectors—agriculture, residential, transport, waste, power, railways and aviation—now build an air-pollutant layer and write machine-readable outputs. A previous edition recorded power-sector and residential-gas pollutants as not estimated, blocked on EMEP/EEA factors the agency no longer serves in a retrievable form, and that diagnosis was wrong in both cases: the residential sector holds the small-combustion chapter and reports gas, LPG, kerosene and coal, and the power sector reads AP-42 directly, including a fuel sulphur content that published Thar analyses do bound. What remains genuinely not estimated is narrower and sharper: what the power fleet *actually* emits, since no national control-device inventory has been obtained here and the reported figures are uncontrolled upper bounds; the equipment mix behind industrial combustion, whose boiler, furnace, engine and turbine shares are declared rather

than surveyed and are worth a factor of thirty on diesel NO_x; within the IPPU and fugitive layers, the classes no admissible factor or series can price—urea finishing, nitric acid, lime, glass, venting, post-meter losses—each typed absent with its activity written beside it; and the transport processes—cold start, evaporative NMVOC, non-exhaust particulate—named in that sector's own register. The completeness table is the authority on which is which.

Emission factors are overwhelmingly Tier 1 defaults. Apart from the agriculture sector's Tier 2 elements, documented separately (Ahmad, 2026a), this inventory applies IPCC defaults to Pakistani activity data. That is the appropriate tier given the available information, and it means the estimates carry the defaults' assumptions about technology mix, combustion conditions and management practice—assumptions derived largely from countries whose infrastructure differs from Pakistan's.

The monthly layer rests on physics more than on observation, and on neither for a third of it. Every source stream is resolved into twelve months and every month row declares how. Stating that as one number—$54.65\,\%$ of the reference year on a flat profile—is true and misleading, because it does not distinguish a flat month that is the physical answer from a flat month that is nobody having read the records. Across the nine-year product the honest statement is three figures, generated from the released rows in `outputs/pakemis_temporal_decomposition.csv`: $37.3\,\%$ of monthly mass has its within-year variation carried by an observed monthly series; $41.0\,\%$ is continuous or stock-driven, where flat—properly, days-weighted—is the correct answer and acquiring a monthly series would not change the shape; and $21.8\,\%$ genuinely varies within the year and is carried flat because no monthly evidence was obtained. These three figures and the TE-class flat share are different classifications of the same mass—TE4 contains both physically appropriate flatness and unobserved variability—and the exact bridge between them is published as `outputs/pakemis_flatness_crosswalk.csv`, joined at the month-row grain the coverage summary grades, so it reproduces the TE-class flat share exactly and carries no grain difference.

The historical monthly product is graded below A−, explicitly. The release sets itself a gate for the monthly layer—no more than $15\,\%$ of nine-year monthly mass may be time-variable and carried flat for an A−—and the nine-year product does not meet it: $21.76\,\%$ of its mass varies within the year and is carried on the calendar alone. The reference-year product is a different object and is stated separately: FY2023–24 alone carries $9.57\,\%$, inside the gate. So the grade is a statement about the eight historical years, not about the reference year, and it is a row of the exit scorecard (`outputs/pakemis_pass2_scorecard.csv`, `2b-historical-temporal-grade`) rather than a sentence alone. One acquisition decides it: road transport's months before July 2021 are flat because OCAC's monthly editions before that date are not held, and the pre-July-2021 editions requested under `MON-OCAC-01` carry $283.2\,\mathrm{Mt}$ of nine-year mass; with them the figure would be $14.68\,\%$, inside the gate.

That gap is a property of the source rather than of the effort spent on it, which is worth stating plainly because the sentence above would otherwise read as a download nobody has got round to. OCAC's own archive lists monthly sectoral-sales editions from the current month back to July 2021 and stops; before that date it publishes fiscal-year compilations only, back to FY2016-17, and this inventory already holds those as its annual controls. The monthly series does not exist in public form for the six years at issue, so no acquisition of a published file closes the gap—only a request to the compiler does. That request is drafted and unsent, and an arrival is an ingest for a later release, not a promise made by this one.

The largest acquisition of this release moves less than it should, and where it stops is the finding. NFDC's consolidated volume publishes fertiliser offtake by product and month for FY2012-13 to FY2022-23 on this inventory's own July–June fiscal year, so the eight years before the reference period need no borrowed months at all; the parse is gated on the three controls the table prints for itself—each row's total, each product's `TOTAL` row, and the page grand total—and validated against the independently transcribed Annex-7 at their three-month overlap, where the two agree to two tonnes on four hundred thousand (`outputs/nfdc_monthly_parse_summary.csv`, `nfdc_monthly_quarantine.csv`). Urea application reaches $100\,\%$ observed. Managed soils then moved for a reason that had nothing to do with acquisition. The offtake shape may only ride the synthetic-fertiliser share, and the atomic annual table appeared to split managed soils into subitems for the reference year alone. It never lacked the split: `fiscal_series` had been computing all thirty-five components for every year and discarding them on the way out, while the annual product returned nothing for those years and recorded in its own documentation that agriculture's history was "carried by category and gas, not by stream"—a sentence true of the FAOSTAT reconstruction that preceded the fiscal seam and false of the engine that replaced it. Removing that declared absence let the offtake shape reach the synthetic share in all nine years: $95.09\,\mathrm{Mt}$ moved from unobserved to observed, no annual number changed, and the re-aggregated subitems reproduce the published category totals across all 72 year-category-gas cells to $5 \times 10^{-7}\,\mathrm{Mt}$, which is the series' own rounding. **The deposited and applied census nitrogen—pasture, organic manure, crop residues—stayed flat**, because it rides a livestock count and a harvest rather than a dealer's depot, and a test asserts it did not come along.

The PBS minerals panel, parsed two releases ago and deliberately left unread because each canonical mineral resolves in only five or six of the eight held editions, is now spent—per source-year, as review ruled it should be: observed years carry the year's own twelve months at TE2, missing years are days-weighted TE4, and no year wears another's shape. The upstream fugitive streams—gas production, gathering, processing and sour-gas processing on the monthly gas series, oil production on crude—moved $93.66\,\mathrm{Mt}$ of cumulative nine-year mass from unobserved to observed timing, $43.0\,\%$ of the sector in the reference year, without any annual figure moving; eight of nine years now carry their own months for both streams, three of them recovered from parse defects inside editions the repository already held. The demand-side streams—distribution, transmission, post-meter, LNG—deliberately did not: deliveries and send-out have their own season that field output does not observe, and a production shape on them would be the wrong-driver overclaim; they stay flat with the acquisition named (`outputs/pbs_minerals_parse_summary.csv` carries the per-mineral coverage, `pbs_minerals_quarantine.csv` the rows the source's own totals reject, and a test holds the production shape to the upstream streams alone).

"Observed" in that sentence means the panel's own twelve months, and for part of the panel it does not mean the country. `outputs/pakemis_minerals_panel_screening.csv` reports which province-years are whole: of 28 mineral-years, 13 are served as covered from an incomplete set of provinces, because the coverage test asks for twelve months rather than for four provinces. The consequence is bounded and it is not uniform. For natural gas it is small—Pakistani production has almost no season and the provinces' curves are alike, so recomputing a complete year's shape from only the provinces a broken year holds moves a monthly share by about a percentage point. For crude oil in FY2021–22 it is not small: the panel

holds Balochistan alone, and the profile served for that year puts $78.76\,\%$ of the year in a single month and labels it observed. No real month of Pakistani crude production is most of the year.

The provinces are not missing from the source, and that is the part worth stating plainly. PBS prints all four. The mineral's name cell is vertically centred across its five province rows, so a layout-preserving text extraction places the name on the *middle* line, and a parser that treats the name line as the start of the group loses the rows printed above it. The arithmetic identifies them beyond argument: for coal in FY2021–22 the Punjab and Sindh rows printed above the name line, taken with the KPK and Balochistan rows on and below it, equal the edition's printed total in every one of the twelve months exactly. The panel carries $42.1\,\%$ of that year's coal.

Worse, the parse's own assurance cannot see this. It reports how many mineral-years close their group control, and that control compares the rows the parser held in hand against the printed total—not the rows the panel ships. For coal in FY2021–22 it closes to the unit while certifying a quantity the panel does not contain. 9 canonical mineral-years disagree between a closing control and the shipped panel, in both directions: limestone FY2016–17 ships $6.9\,\%$ of what its control counted, and natural gas FY2020–21 ships $230.8\,\%$ of it, that one being the worksheet duplication described next. A control that passes while its subject is short is not a control, and the comparison it omitted is now computed and declared.

The same screening separates a second defect that points the other way. The spreadsheet editions are read across every worksheet, and some workbooks repeat their table on working sheets, so those years are counted once per sheet—FY2020–21 carries 31 rows in every month for gas and oil and 100 for coal and limestone, and its published gas quantity is 12.83 times the last year the panel records correctly. That is a wrong number in a shipped activity table and it costs the inventory nothing, because the duplication is uniform across the twelve months and every consumer normalises; the missing provinces cost the inventory something and leave the quantity looking plausible. Neither is repaired here: the first would move a published activity table and the second a published monthly shape, and both are decisions rather than audit edits. They are counted so the decision can be taken on a measurement.

Only that last third is a gap, and it is the only one of the three that acquiring data would close. A livestock census counts a stock once and carries no month, so equal twelfths—days-weighted—is the complete statement the activity data supports, and the layer is built so that such a row can never receive a shaped profile. The rest is a gap with a name. Road CNG is rationed by season and by load-shedding and is spread flat because no monthly metered series was obtained. Sugar-mill bagasse burns in a crushing season this release does not assemble. Crop-residue burning rode that same sentence for three releases and no longer belongs in it: the nine-year VIIRS fire record now times it (TE3, every month observed within the attribution ladder's terms). Waste's disposal stream likewise left the flat set—the continuous FOD state engine computes its months from the deposit history—while the sector's other streams stay flat with their reasons named. Fugitives is flat only on its demand side: the upstream production, gathering and processing streams ride PBS's monthly gas and crude series where a year holds them ($43.0\,\%$ of the sector's reference year at TE2), while distribution, transmission, LNG and post-meter streams stay deliberately flat, because field production is not a defensible timing proxy for downstream deliveries. None of this moves the annual total by construction, and all of it would move the months. The single largest improvement available to the temporal dimension

is acquisition—OGRA's monthly gas system reports, a crushing-season series, provincial harvest dates—rather than a better model of what is already held.

The archive deposit contains products, not the source documents. The inventory reads 217 registered sources, of which 153 are third-party publications archived in the repository—OGRA, NEPRA, PBS, NFDC and PCAA reports, and PBS household survey microdata. *None of them is in the deposit.* For 126 of those 217 registered sources the publisher states no redistribution terms at all, which is the ordinary condition of a government statistical publication and not a defect in the record; but a deposit is a publication, and this project cannot license what it does not own. Depositing the repository whole would have offered the 60 distinct third-party documents it holds under a CC BY grant it had no right to make.

What ships instead is `outputs/pakemis_source_register.csv`: every source named, its publisher and edition recorded, its archive hashed, and the instruction for obtaining it from that publisher. The two questions the register keeps apart are the reason this works—whether a document may be *read and derived from* is not the question of whether it may be *republished*, and only the second is at issue here. The derived quantities are facts and this project's compilations of them, and those are deposited: 1761 files under licences the project can actually grant, each hashed in `outputs/pakemis_deposit_manifest.csv` with the licence it ships under.

One file cannot have a single licence, and saying that it did was a real error rather than an imprecision. The facility registry carries a licence column qualifying each of its 12 680 rows, and 71 of them are OpenStreetMap derivatives under ODbL 1.0—a share-alike position this project cannot waive, and the same position the paragraph above asserts it cannot assume for anything it does not own. Both the register and its published twin were nevertheless typed `own-work` in the deposit manifest, with the reason "this project's own code, product or prose", so the archive would have offered those 71 database rows under a CC BY grant. They now ship as `mixed-per-row` with the per-row licence counts in the reason, which is the only honest whole-file statement about a file whose rows disagree. The distinction that keeps this a narrow repair rather than a sweeping rule is that two other shipped registers also carry non-uniform licence columns, and theirs describe *other* artefacts: a register's judgement about a third-party document is this project's own work however that document is licensed. Which of the two a file is, is now a declared list on both sides rather than a default.

The ODbL carve-outs are also named twice now, by published identity and by the path the archive actually ships them at, because the two differ after the data-layout migration and a reader searching the deposit for the published identity alone found nothing—from which they could reasonably have concluded the files were excluded. They are not excluded; they ship, correctly typed, under ODbL.

A citation names a document; a reader needs the table. The standing objection to this release is the finer one—given the report, where exactly is the number?—and until now nothing counted the answer. `outputs/pakemis_citation_pointer_audit.csv` does, and the denominator is most of the work. A registered source that was *derived* inside this repository has a derivation chain rather than a page number, and a source that is a dataset rather than a document has a file and a variable rather than a table; scoring either against a table pointer would inflate the gap with rows no citation could close. Of the 131 sources where a table pointer is the right notion, 115 have one. A pointer counts whether it is in the register row or in the produced table's own per-row source column, which is where this project's hand-verified ingests put it—checking only the

first is how the same gap previously read as 157 rather than 16. And *pointed* is not the same as *checkable*: 103 of the 112 point into a document this repository archives, where a reader can open it and confirm the figure, while 7 point into documents the project does not hold and 5 into files derived inside it. The distinction changes no verdict—a pointer into a document the reader must obtain is still a pointer—but the audit's strongest number should not imply more than it has. One pointer was checked to the cell while that column was added: the bagasse generation ingest against NEPRA's held State of Industry Report, where a plant reads 15.10 GW h, 29.83 GW h, 17.38 GW h, 25.02 GW h and 29.10 GW h in the source and the same five values in the ingest.

Both halves are ratcheted, in the same direction, because the easiest way to improve this share is not to cite anything but to widen the exemption. The 16 that remain are a long tail of one-offs rather than a systematic failure, and the count is deliberately conservative: a handful of dataset sources whose honest pointer is a file and a row filter are scored as unpointed because the audit's publisher markers do not recognise them, and widening those markers would improve the share by exemption rather than by citation. Three named groups were closed while measuring. The seven OCAC extractions had no pointer anywhere, and the documents were held all along—each annual edition is two pages and each monthly edition eighteen, nine areas of energy products followed by the same nine of non-energy. Their table titles also drift, from "Sectoral Sales / Consumption of POL Products" through FY2019–20 to a split pair from FY2020–21 onward, which is recorded with the pointers because an edition whose vocabulary changed is exactly where a reader following a citation would go wrong. The Pakistani BTR row was a third and cost nothing at all, since the document was already open for the comparator work. The PBS minerals family was the other: four of its eight editions are PDFs carrying one running-head table over one to ten pages, and four are workbooks with no table title at all, where the *sheet* is the pointer—and one of those, FY2020–21, carries an "Electricity generation" sheet beside "mineral production", which is why two different parses read the same workbook for two different tables. The acceptance note that prompted this named HDIP among the publishers lacking pointers; every HDIP row carries one, and did before the audit was written.

"Deposited" is not the same set as "cited", and this release measures the difference rather than asserting it away. `outputs/pakemis_citation_deposit_audit.csv` takes every output file the manuscripts name—352 of them—and asks whether the archive contains it. 181 are deposited and 171 are not.

That denominator was 111 before the notation repair described below, and the correction is a lesson about measurement rather than about the archive. The manuscripts name a file in two notations: the national sections and agriculture's appendix write the full path, under `outputs/`, while eight sector appendices write the bare filename alone in an itemised list—the same file, typeset two ways. The audit matched the first notation alone, which reaches 132 of the 352 files the manuscripts name; the other 220 are named only as a bare filename, and of those 102 are in the archive and 118 are not. The register's two halves had been disagreeing about the word *cited* without either being wrong: the artifact inventory counts a bare filename, because it searches for basenames, which is why those files carry the evidence role at all, and this audit did not count it as a citation. A reader cannot obtain a file any better for its name having been typeset with a directory in front of it. The ratchet is re-based on the wider set and the pre-repair subset keeps its own ceiling, so widening the measurement cannot hide a regression in the part that was already measured.

The repair also settled a question the blindness had been answering by default. When the bare notation came into view it exposed 22 *sector headline* products—a sector's own totals, stream split, provincial table, monthly layer and uncertainty summary—named in that sector's own paper and absent from the archive, which this project's own test calls indefensible at that tier. A further 22 were the same gap seen from the other side: fugitives' and ippu's appendices described their headline tables by function and named none of them; aviation's five appeared in neither of its appendix's lists at all; and industry, the one covered sector with no paper of its own, has its four named in this paper. All 44 are named and deposited in this release, and that tier now measures zero. What remains undeposited is 168 files typed *diagnostic*, two typed *intermediate* and one evaluation-lane table, where the choice per file is a real one. The deposit ships a curated list and the papers cite whichever table actually carries the number, and until the round that built this audit nothing compared the two. Every absent file exists in the repository, so no citation points at nothing; what they point at is something a reader holding only the archive cannot open. Most are typed *intermediate* or *diagnostic*—working files no reader-facing claim was supposed to rest on—which is the interesting part, because a results section citing a table is a reader-facing claim resting on it. For those the role and the citation disagree about what the file is, and the fix is one of two things per file: ship it, or cite the deposited product carrying the same numbers. Seen from the register's own side the disagreement is wider than the deposit question, and tracing it changed the register itself. 11 files are now typed as working files on which *no reader-facing claim rests* while being named in a manuscript, 8 of them in this one. That count was 84 before this release, and the way it fell is the substance.

The register has always required a canonical product to be cited; it never required a cited file not to be called a working file, and that asymmetry is what let the set grow. Asking the question in the other direction needed two repairs.

The first was to the measurement, not the tree. The scan that decides whether a manuscript names a file read the national paper's sections, every sector paper's sections, the README and the documentation—and not the sector *supplements*' own bodies, four of which are only partly assembled from section files. 17 artifacts were recorded as named by nobody while a supplement named them, among them every figure in the aviation supplement and two regulatory cell audits that exist precisely to be the evidence behind a claim. Repairing the scan *raised* this count, from 77 to 84, without a single citation being added: the figure earlier releases printed was an under-count of a set that was always larger. Run again against the present tree, the same reconstruction finds 12 rather than 17, and the difference is instructive: a long filename in a monospace label needs typesetting break hints to stay inside the page, and the scan that reads a printed name has to strip them or it does not recognise the name it is looking for. Five of the seventeen have since been named in a section as well, with break hints, and one normalisation was silently blind to them—so the printed 17 is what the repair found in the tree of the day, and the 12 is what the instrument reports now that every reader of a printed name shares one normalisation.

The second repair was to what the register does when it has no answer. Every one of the files in this set acquired its role from the register's final catch-all pattern rather than from a decision, and that pattern declares a working file on which no reader-facing claim rests—so the role a file received when nobody had classified it carried a substantive claim about it. More patterns were the obvious fix and the wrong one: these files fall through precisely because their names share no structure a pattern can reach, and a cover validated against the whole inventory needed 61 patterns for 74 files. The catch-all now defers to

the evidence instead. A file nothing else classifies, which a manuscript names, is evidence behind a claim; one nothing names is a working file. 74 files moved on that one change, with no new pattern to go stale.

The part no re-typing could reach closed last, and it closed completely. A claim in the verification registry reads a product by name and this paper prints the figure computed from it, so a file a claim reads is a file a reader-facing claim rests on, by construction—which is the one thing the working-file role promises is not true of it. That contradiction stood at 83 products when it was first measured. The count of products a registered claim reads while the register types them a working file is now 0, and the subset named in no manuscript at all is 0 with it. None of the residue was closed by re-typing, because the evidence role requires the file to be named somewhere: each was instead named in the sentence that already discussed it—the livestock nitrogen stage-flow ledger, its IPCC-against-EMEP stage comparison and the air-pollutant Monte Carlo intervals in the agriculture appendix, the national uncertainty decomposition beside its component-row count, the waste release assignment beside the candidate list it was always printed with, the two compressed-gas cross-check tables inside the clause that invoked their epistemic class, the industry captive screening beside the captive figure it produced, and the delivered-gas closure variant beside the ledger it prices. Both gates are exact now rather than ratchets, so a single new printed figure whose evidence a reader cannot reach fails the build.

Following the same thread into the archive found the sharper version of it. A deposit-manifest product row whose release tier is *build* was skipped from the data archive under the rule that no build artifact is deposited, while the builder's own description promised every product row—two selections with no criterion in common, since the manifest chooses products by name pattern and never consults the role. 13 hashed products sat in that gap, essentially the whole air-pollutant family including the district panel and the 1990–2024 series. Nine are resolved by the two repairs above and by typing three of them from the manifest's own literal naming; the remaining four are declared exclusions with their reasons, because for those the manifest's claim may be the incidental one and which register is right is a question about what the archive is for. The test that should have caught all of this was named for a partition and asserted one inclusion of the two. Four were closed here in both ways—three tables with no deposited equivalent added to the shipped set, and one monthly table re-pointed at the nine-year product of which it is a single year's slice—and the remainder is held under a ratchet that may only fall.

The cost is real and belongs here rather than in a footnote. A reader who wants to rebuild this inventory from nothing must re-acquire those publications themselves. The hashes are what make that a procedure rather than a guess: obtain the document, compare the digest, and you can be certain you are reading what this release read. Everything else needed—the engine, the derived activity tables, the products, the manuscripts—is in the deposit.

7.1 A version string is not a pin, and the spatial axis had no fingerprint

The release machinery described above protects a release's *identity*: the manifest hashes every product and the release cover prints the digest, so a reader can tell whether the artefacts in hand are the artefacts that were built. What it did not protect is the claim that a version *label* identifies a particular content, and one case shows why that distinction matters.

The API that serves these products fetches a release directory and verifies it against that directory's own semantic contract, so a served copy is always internally consistent. A provincial query answered below this repository's own figure for the same

province and year, and the cause was neither side's arithmetic: the served release was byte-identical to the commit it was fetched from, and this repository's tree, still carrying the same version string, had moved past it. The two trees agree on *every* national total to nine decimal places—the reported figure for the reference year is identical on both sides—while the provincial allocation differs. The change was a deliberate re-base giving furnace oil its own product geography, which eliminated a residual province and redistributed its mass; it conserved the national total at every period, exactly as intended.

That is the structural point, and it generalises beyond this one release and beyond geography. Every guard this project has on a release's numbers is a guard on a *period total*: the manifest's national figures, the pinned configurations, the claim matrix, the registered manuscript claims, the query goldens. A change that conserves the period total passes all of them, so an allocation *within* those totals could move without any check noticing—and two did. The provincial case above is one. The other is temporal: a set of cells in the national monthly table, all of them electricity and heat production, moved within fiscal years at exactly zero net, so the annual table's fingerprint is identical to the published vintage's while the within-year shape of power generation is not.

It is now measured. `data/catalog/published_release_vintages.csv` records each published release as a commit and two digests, every row re-hashed out of this repository's history, and `outputs/pakemis_release_axis_fingerprints.csv` carries, for each disaggregated table and each vintage, a coarse digest over period totals and a fine digest over period-by-axis totals—the axis being a geography for the spatial tables and the month for the monthly one. Of the 6 disaggregated tables fingerprinted, 0 currently show a moved fine fingerprint under an unmoved coarse one. The other two are controls: the air-pollutant provincial layer moved on both axes, which is precisely why the existing guards did register that change, and the national annual table—the grain those guards actually watch—is unmoved. Neither file alters a number, and a label whose content has moved is now a row rather than a discovery. The magnitudes of both divergences are in those two builders' own documentation rather than here, because each is measured against an externally served copy this repository does not hold—and a figure a reader cannot recompute from the archive does not belong in a sentence that the archive is supposed to support.

7.2 Discussion

7.2.1 What the architecture actually bought

The claim in Section 2 is testable against the project's own history, and it is worth being honest about which parts of it paid off.

The clearest return was on *late additions*. Fugitives and cement were built last, after the analysis layer was mature. Neither required any change to the uncertainty analysis, the key-category screen, the benchmark comparison, the provincial allocation or the figure generation; both appeared in every national output on first registration. In a spreadsheet inventory, adding a sector means extending every downstream sheet by hand, and the cost of a new sector rises with the number of sectors already present. Here it is flat.

The second return was on *invariants catching real errors*. Several of the 27 contract tests exist because they failed, and Table 13 is the audit rather than the anecdote: each row is a defect this project shipped, what it would have cost, how it was

found, and the guard that now makes it a build failure. The pattern across them is worth more than any single row. **Every one was invisible to review**, because nothing about the resulting numbers looked wrong—and the later ones were invisible to the numeric tests too, because they were claims about words, boundaries or layout rather than about digits.

The third, and least anticipated, was that *closures turned prose caveats into arithmetic*. The four enforced closures in Section 4.8 existed first as written warnings—"this biases the sector high", "there is no national figure to remove them by"—in the documentation of sectors that had done the honest thing and declared a known defect. Building the counterpart sector converted each warning into a number. That is a general pattern worth naming: a well-documented known defect is a specification for the work that fixes it.

What the architecture did *not* buy is accuracy. The engine's activity data are still national statistics of variable quality, its factors are still mostly IPCC Tier 1 defaults, and the largest sector rests on a livestock census. Structure prevents inconsistency; it does not manufacture information.

7.2.2 Two undeclared assumptions worth a table apiece

Two results in this paper share a shape, and the shape is the point.

Sharing parameters by physical identity across modules, rather than confining them within one, leaves the national uncertainty at $\pm 11.72\,\%$ and the road-transport category unmoved—a result that reverses what the previous edition reported, for reasons Section 4.9 sets out. The sewer methane correction factor moves domestic wastewater from $15.39\,\mathrm{Mt}$ to $24.46\,\mathrm{Mt}$—a factor of 1.6 on a category that is $5.5\,\%$ of the national inventory. In both cases the assumption is defensible in either direction, the evidence does not settle it, and the conventional treatment is to pick one and not mention it.

We publish both endpoints in both cases. That is not fence-sitting; it is the recognition that a single number carries a false claim about the state of knowledge, and that the correct output of an analysis whose answer depends on an unsettled assumption is the dependence itself. In both cases we also state what would settle it: for correlation, a Pakistani fuel carbon-content measurement programme; for the MCF, dissolved-methane sampling on urban drains in two or three cities.

7.2.3 Implications for Pakistani inventory work

Three practical conclusions follow, in descending order of confidence.

Resolve gas-system practices before refining the vehicle fleet. The 2019 segment ledger now includes utility throughput and pipeline length, making fugitives $5.3\,\%$ of national emissions and $3.3\,\%$ of variance. Road transport is $9.0\,\%$ and $0.9\,\%$. The binding fugitive evidence is therefore pipeline material and pressure, LDAR, compressor configuration, processing throughput and well status rather than another aggregate production factor.

Industrial fuel combustion is the binding constraint on completeness. At roughly $16\,\mathrm{Mt}$ still missing, the unestimated remainder alone would rank sixth among this inventory's modules—just below residential fuel—while the complete $43.98\,\mathrm{Mt}$ category EDGAR reports would rank third, above road transport. Its absence is most of why this inventory reports $87\,\%$ of EDGAR's aggregate magnitude. No amount of refinement elsewhere compensates for the unmeasured fuels within this partially estimated industrial category.

Defect	Consequence	Detection	Correction	Regression guard
Two sectors proposing one category name	silent double-count	contract test on registration	globally unique names	category-name invariant
Correlation group spanning two categories	variance apportionment ill-defined	contract test	groups within one category	correlation-group invariant
Wrong parallel factor table	category trace gases 8% out, CO_2 exact	factor audit	category-specific factor identity	factor-identity test
Ambiguous integer year labels	mixed-period national total	period audit	explicit start/end dates	period-schema test
Unread public activity table	$3.40\,\mathrm{Mt}$ coverage gap	repository audit	OCAC liquid customer class read	completeness test
Retired provincial power proxy	$8.8\,\mathrm{Mt}$ in the wrong province	spatial diagnostic	plant-wise generation	allocation test
Source documents never archived	provenance unverifiable	archive audit	source-document registry	document-manifest test
Figure sources named by year	two manuscripts shipped pre-harmonisation figures	external review	year resolved at read time, absence raises	figure-source test
Stale gas decomposition	a decomposition summing to $455.14\,\mathrm{Mt}$ against a printed 448.00	external review	generated category $\times$ gas ledger	gas-total invariant
Declared-debt list	printed quantities tracked but unverified	external review	the list deleted; no exemption without a reason	unconditional coverage test
Two category hierarchies	20 categories screened against 24 reported	external review	IPCC category made canonical	one-hierarchy test
Stale comparative language	"doubles" surviving a ratio of 1.61	external review	claims registry binding phrase to relation	semantic-claims test
One correlation group per row	a factor correlation reported as widening a category by more than half when it moves it by nothing	external review	uncertainty carried per parameter	decomposition-identity test
Content printed off the page	$104\,\mathrm{pt}$ of a table outside the margin	external review	tables narrowed	log-derived layout test
Superseded output vintage	a retired FY2019–20 fugitives file quoting $3.81\,\mathrm{Mt}$ beside the then-current 24.69 (the envelope's top)	rebuild audit	vintage retired	orphan and twin sweep over `outputs/`
Staleness guard reading it	the shipped-versus-engine check ran on three sectors of nine and reported passing	rebuild audit	the check reads each sector's declared categories	per-sector coverage asserted, not assumed
Orchestrator declaring outputs it did not write	four phases named files that existed nowhere and printed OK beside them	rebuild audit	the declared artefact must be written during the run	orchestrator output check
Run manifest listing a directory scrape	a sector's manifest claimed any leftover file as its own product	rebuild audit	the manifest records what the run wrote	write-tracked manifest
The build defined three times	the branch that gates merges built a smaller inventory than either local command	rebuild audit	the three lists reconciled	build-definition test
The release record	it printed $455.14\,\mathrm{Mt}$ and described a mixed-year inventory three rounds after both stopped being true	rebuild audit	repaired and anchored	the release record is a claims document

Table 13. Defects this inventory shipped, and what now prevents each. The right-hand column is the deliverable: a defect without a regression guard has been corrected, not closed. The detection column is the uncomfortable one—the middle entries were found by an external reader rather than by the suite, and each of those had passed a suite of more than a thousand tests, because the property that was wrong had never been made checkable. The last six were found by rebuilding the whole release from one commit and asking a narrower question than the suite asks: not whether each artefact is correct, but whether anything still writes it. Four of the six are the same defect wearing different clothes—a check that reports passing while testing nothing—which is the failure mode a growing test suite is least able to see in itself.

Methane is the metric-sensitive majority. At $54.6\,\%$ of the national total on AR5, Pakistan's inventory is dominated by a short-lived gas, and mitigation assessments will be unusually sensitive to the metric and time horizon chosen. Reporting both AR5 and AR6 at every level of aggregation, as here, is a minimum; a country in this position has a stronger case than most for also reporting on a short-horizon metric.

7.2.4 Co-emitted air pollutants: a companion product, not a result of this paper

Seven of the ten modules also compute an air-pollutant layer from the same activity quantities, through EMEP/EEA factors (EMEP/EEA, 2019, 2023) rather than from a separate study. That shared activity basis is the methodologically interesting part, and it supports one robust statement: on primary $PM_{2.5}$, household solid fuel is the largest estimated source in this inventory by a wide margin, while being a minor greenhouse-gas category—the two inventories rank sources close to inversely.

It is reported in the supplement rather than here, deliberately. The seven layers are not of comparable assurance: two are bounds or screening estimates, transport is exhaust-only, and industrial combustion and IPPU have no layer at all. A ranking drawn across evidence that uneven is easy to over-read, and it should not carry weight the greenhouse-gas product has earned and it has not. Supplementary Note S1 gives the layer-by-layer assurance matrix, the magnitudes and the figure, released as **PakEmis-Air beta**.

7.2.5 Generalisability

The architecture is not Pakistan-specific. The contract in Section 2 encodes properties of IPCC inventories generally—categories, gases, notation keys, allocation proxies, correlation groups—and the accounting closures are instances of a pattern (one national quantity consumed by two sectors) that recurs in every national inventory: refinery fuel between 1A1b and 1A2, waste between disposal and incineration, land between cropland and grassland.

The parallel-table trap of Section S3 is entirely general and, we would expect, common. Any inventory compiled by a small team under time pressure, using the 2006 Guidelines, is exposed to it, and the exposure is invisible to CO_2-based validation. Checking which of the four Chapter 2 tables and which of the Chapter 3 tables each fuel-using category actually drew from would be a cheap and, we suspect, productive audit of many published national inventories.

8 Conclusions

We have presented a ten-sector national greenhouse-gas inventory for Pakistan totalling $444.96\,\mathrm{Mt}$ CO_2e on AR5 under reference scenario `REF_2023_24`, built as a modular software system in which each sector satisfies a common contract enforced by 32 invariants, and in which the properties that span sectors are executable assertions rather than statements in a methods document.

Three results generalise beyond this country.

Accounting boundaries can be closed by construction. Four quantities that are conventionally estimated twice and reconciled by reading are here computed once and consumed twice, with equality asserted exactly: two are transfers between modules,

one closes a waste-fate partition inside one module, and one partitions a transport total between its reported categories. Each of the four existed first as a written-down known defect—an orphaned nitrogen flow, a double-counted diesel, two unpartitioned totals—which suggests that a well-documented limitation is a specification for the work that removes it.

The IPCC default tables are a systematic trap. In four of six fuel-burning sectors, the plausible wrong table leaves CO_2 exactly right while the trace gases move: the affected category by $8.0\,\%$ in the two cases where it is not CO_2-dominated, and the CH_4 row several-fold in the two where it is. Either way the error is invisible to every external validation available, because every comparator is CO_2-dominated. We would expect an audit of which table each fuel-using category actually drew from to be productive across many published inventories.

An uncertainty architecture can manufacture its own findings. An earlier edition of this paper reported that correlating shared emission factors widened the road-transport category from 11.96 to $18.46\,\%$, and read that as a property of the inventory. It was a property of the model: a component could carry only one correlation group, so putting a row into the fuel-carbon group moved its *entire* variance with a factor that accounts for about two points of it. Once each row's activity, conversion and factor are carried as separate parameters, the same question leaves the national uncertainty at $\pm 11.72\,\%$ either way, and road transport's at $\pm 13.3\,\%$. We report both readings and choose neither, as we do for the disputed sewer methane correction factor that moves domestic wastewater by a factor of 1.6. Both are conditional on the reference structural configuration, and Section 4.9.5 prices the named alternatives that no parametric interval contains. The lesson is not that shared factors matter more than was thought; it is that a dependency model too coarse to represent the question will answer it anyway.

For Pakistan specifically, the analysis produces one clear recommendation and one clear constraint. The recommendation is to resolve gas-system technology and facility activity: the 2019 segment ledger now includes utility throughput and pipelines, while material, pressure, LDAR, compressor, processing and well evidence still determine its structural range. The constraint is that part of industrial fuel combustion is still outside the estimate. The solid-fuel activation admitted the brick-kiln and cement-sector coal this sentence once called absent, so what stands against the roughly $44\,\mathrm{Mt}$ EDGAR reports for 1A2 is a smaller and now unidentified remainder rather than the non-gas fuels as a class, and no refinement of the estimated gas component substitutes for what is still missing.

Everything here is *computationally* reproducible from the committed activity tables, and the qualifier is the honest part. The engine, the activity data, the emission factors and every output file quoted are public; the test suite runs on every commit; each figure is regenerated from the engine's own outputs and a test fails if any is stale; and the numbers in this manuscript are checked against the generated CSVs they came from. What is *not* reproducible end-to-end is the step before that: 12 of 217 activity files cite at least one document this project does not hold, so their derivation from the raw source cannot be re-run by a reader in full, only their use downstream. The register names every one. Six of the 217 are this project's own compiled fact tables—the fugitives anchor activity, the two residential fuel-supply tables and the three IPPU company tables—which the sector data layout registered as sources in their own right rather than leaving them implicit behind the compatibility views generated from them. A seventh joined them with the power coal re-base: the coal-energy booking table, which declares which account the engine books and is derived from two products rather than transcribed from a document. The other 210 rows are

unchanged. An inventory whose claims can be re-derived by a reader is a different kind of object from one that can only be read, and building it that way cost less than the reconciliation work it replaced.

9 Data availability

The dataset described here is PakEmis 1.0.0rc67. The dataset DOI is assigned on deposit; until then the release is identified by version and manifest digest (`outputs/release_manifest.json`, whose SHA-256 the release cover of every technical note prints). The technical documentation—the sector technical notes and this supplement—is deposited with the data under the documentation record, whose identifier is assigned with the dataset's. The manifest records the licences as MIT for the code and CC-BY-4.0 for the data and text, with the OpenStreetMap-derived carve-outs it lists under ODbL 1.0. One evaluation-only input carries no licence declaration in the record and is excluded from the primary data record until its rights are established (the manifest's `licences.undetermined` note). Redistribution rights are a property of each source document rather than of this project—and the OpenStreetMap-derived geometry tables and rasters (landfill polygons, the rail network and yards, the road-density and building-fraction diagnostics, and 71 rows of the facility registry, each flagged in its `licence` column) are *derivative databases* under ODbL 1.0 rather than CC-BY: ODbL's own distinction, under which a rendered map is a produced work needing only attribution but an extracted coordinate table remains a database and keeps ODbL's share-alike terms. `LICENSE-data` lists every such file. `data/catalog/source_documents.csv` classifies all 217 activity files as archived (153), derived from an archived source (52) or citing at least one document this project has not kept (12), and a not-archived file must carry a gap-register row saying so; `data/catalog/cited_documents.csv` names, for each cited document, the file that holds it. The source documents themselves are not redistributed: the source register names each one, records its hash and says where its publisher provides it.

10 Code availability

The software is release 1.0.0rc67 of the `pkinv` package (manifest schema `pkinv.release/v1`). The software DOI is assigned on deposit, with the dataset's; until then the release is identified by version and manifest digest. The inventory engine, the activity data, the emission-factor tables and every output file quoted in this paper are in a single repository (https://github.com/RehanAhmad13/pakistan-emissions-sectors); the reproduction procedure is Supplement S4. Every number printed here is read from a generated CSV rather than transcribed, and a test suite asserts that the manuscript and the engine agree. This paper describes release `1.0.0rc67` of PakEmis, schema `pkinv.release/v1`. The release is identified by `outputs/release_manifest.json`, which records the commit, the reporting period, the baseline configuration, both national totals, and a SHA-256 for every activity file the run read; each technical note reproduces that block on its first page, so a printed document always says which release it documents. Reproduction is `pip install -e .` followed by `make sectors`, `make national` and `make papers`, and continuous integration runs exactly that sequence and then diffs the regenerated outputs against the committed ones.

Author contributions. RA designed the inventory system, acquired and transcribed the source documents, wrote the engine and its tests, produced the data records and wrote the paper.

Competing interests. The author declares no competing interests.

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